

The Star of Bethlehem: A Skeptical View

By Aaron Adair





The Star of Bethlehem: A Skeptical View
Copyright © 2013 Aaron Adair

Published by *Onus Books*

Printed by Lightning Source International

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form by any means, electronic, mechanical, photocopy, recording, or otherwise, without the prior permission of the publisher, except as provided for by UK copyright law.

Cover image, “Under the Milky Way” by Steve Jurvetson (2007), used under Creative Commons licensing.

Cover design: Onus Books

Revised Standard Version of the Bible, copyright 1952 [2nd edition, 1971] by the Division of Christian Education of the National Council of the Churches of Christ in the United States of America. Used by permission. All rights reserved.

Trade paperback ISBN: 978-0-9926000-1-3

OB 05e/09e

PRAISE FOR *THE STAR OF BETHLEHEM: A SKEPTICAL VIEW*:

“Aaron Adair’s investigation of The Star of Bethlehem doesn’t just debunk a beloved Christian legend – it shows what a rough and treacherous neighborhood is that problematic intersection where scripture and science meet. Extensively researched, Adair’s keen detective work uncovers a treasure-trove of both actual astronomical history and wild and wooly folklore; including the intrigues of Han Dynasty astrologers, the Antikythera Device - an ancient Greek clockwork computer, Hipparchus of Rhodes’ discovery that the entire universe was shifting on its axis (or was it?), imaginary worlds like Immanuel Velikovsky’s mysterious Planet X and pseudo-scholar Zecharia Sitchin’s pseudo-Sumerian planet Nibiru. A fascinating and readable feat of hardcore historical legwork and keen scientific analysis.”

—David Fitzgerald, author of *The Complete Heretic’s Guide to Western Religion: The Mormons*.

“Well researched, scientifically reasoned, elegantly concise, this book will long be required reading on the ‘Star of Bethlehem’. Full of fascinating historical facts, and better informed and more careful than any other book on the subject, this should be on the shelf of everyone interested in that legendary celestial event.”

—Richard Carrier, Ph.D., author of *Proving History: Bayes’s Theorem and the Quest for the Historical Jesus*.

"The Star of Bethlehem is a tightly-argued, well-reasoned exploration of a fascinating question. The explanations for the Star of Bethlehem are as varied as the people who proposed them, and Aaron Adair masterfully demonstrates why every effort to rationalize the Star thus far has failed—and offers the best explanation yet for what the Star really meant to the earliest Christian writers. *The Star of Bethlehem* is a concise and rigorous must-read for anyone interested in religion, history, and modern efforts to understand the past."

—Jason Colavito, author of *The Cult of Alien Gods*.

"Aaron Adair's book *The Star of Bethlehem: A Skeptical View* offers readers a fresh and interesting look into an old academic question. While the argument that the 'Star of Bethlehem' story is a myth isn't a new one, Aaron Adair—an astronomer and physicist at The Ohio State University—offers a look into the past through the eyes of a scientist, while not once ignoring the value of New Testament scholarship. This is a must-read, and perhaps the definitive, book on this subject."

—Thomas Verenna, co-editor of *'Is This Not the Carpenter?'* *The Question of the Historicity of the Figure of Jesus* and undergraduate student at Rutgers University.

*To Sarah, 'mein Schatz',
for being with me
when both near
and far away.*

About the author:

Aaron Adair is a recent PhD in physics from the Ohio State University, primarily working in the area of physics education research (PER). Previously, he studied physics, astronomy, and mathematics at Michigan State University. His scientific interests have included working as a show presenter at a planetarium, researching at the SETI Institute, and examining parts for the ATLAS particle detector at CERN. His current areas of research are technological innovations in the classroom, project-based learning curricula, the origins of physics misconceptions, the history of astronomy, and classical history and religions. He has published on the Star of Bethlehem previously in *Sky & Telescope* and *Zygon: Journal of Science & Religion*. Born and raised in Michigan, Aaron currently resides in Columbus, Ohio, but he is on the move, to where no one knows.

He blogs at gilgamesh42.wordpress.com.

Acknowledgements:

This volume has been in the making for years, but its creation was helped by many different friends, family, and teachers in many fields. I cannot thank all of them, but some deserve particular attention. The professor that more than anyone else brought me into the classical world was Debra Nails and her classes of Plato, Aristotle, and classical Athens. In order to engage in the texts, I needed to learn the Greek language, in which Carl Anderson and William Blake Tyrell have helped me, though I dare not claim proficiency as they can. In history, Linda Johnson showed me many of the methods of investigation, but a large debt also goes to Richard Carrier when it comes to studying ancient Christianity and with particular help related to this project. Robert M. Price has also been of great help, including publishing a paper on this subject in his (now defunct) journal. When I worked at a planetarium as a show presenter, which is also where I learned of these Star theories, I received help and support from my co-workers Don Batch, Shane Horvatin, John French, and Mary Gowans. When it came to these matters, I also had several times received direct help from David Hughes, who has one of the better books on the subject of the Star of Bethlehem, though his conclusions are very different than mine. My first publication on the Star was encouraged by my astronomy senior thesis advisor, Mark Voit, who also taught me much about modern research tools in cosmology. David Fitzgerald has been of great support in this project, and he got me in contact with the person that became my editor, Jonathan MS Pearce. It is because of his efforts that this volume is orders of magnitude better than it would have otherwise been. Lastly, I must thank Sarah Heim, who has had to hear the most about my research

and has given me what I needed to hear in return and so much more.

Contents

Foreword _____	1
Preface _____	3
<i>PART I: THE STORY</i> _____	11
Chapter 1: Matthew's Account and Timelines _____	11
Chapter 2: Methodological Difficulties _____	19
<i>PART II: THE HYPOTHESES</i> _____	31
Chapter 3: Comets and Meteors _____	31
Chapter 4: Novae and Supernova _____	43
Chapter 5: The Planets and their Positions _____	51
Part A: Conjunctions _____	53
Part B: Horoscopes _____	69
Chapter 6: What (Other) Stars do Spangle Heaven with Such Error? Last Ditch Efforts _____	83
<i>PART III: FATAL PROBLEMS</i> _____	91
Chapter 7: Failure of all Natural Hypotheses _____	91
Chapter 8: Historical Issues _____	107
APPENDIX: THE TEXT OF MATTHEW _____	129
GLOSSARY _____	135
BIBLIOGRAPHY _____	147

Foreword

Like many astronomers, I've always been intrigued by the Star of Bethlehem.

It's the best-known and most legendary star, bar none. People who cannot name a single luminary in the actual night sky readily acknowledge their familiarity with the Bethlehem Star, and its religious role. No wonder it has inspired vast scientific speculation. For more than a century, various natural explanations have gained and then lost fashion, periodically making headlines. Moreover, it has been a staple of holiday planetarium shows since the 1930s.

For me, personally, this celestial topic launched a quarter-century career. My "Night Watchman" and then "Strange Universe" pages have appeared in every issue of first *Discover*, and then *Astronomy* magazines. But the very first column, published in *Discover* in December 1989, was a two-page spread about the Star of Bethlehem.

Basically I summarized the various "explanations" shown to the public during planetariums' annual "Star of Wonder" shows, then noted that Planetarium Directors—I'd interviewed quite a few—were well aware that each was impossible. Nonetheless, the shows remain popular, and have become such a tradition in and of themselves that no one seems bothered by such make-believe science being annually offered to the public.

I did not have the research time, nor found it appropriate, to delve into various other aspects of The Star. How, for example, the nativity account in Matthew is so utterly different from the gospel of Luke. How astrologi-

cal explanations might (or might not) solve the puzzle. Or the hard evidence behind the idea that Matthew—a century after the fact—may have simply created the Star story by borrowing specific aspects from popular contemporary literature.

Aaron Adair does all this and much more, and does so in an immensely readable way. The result is not a rehash of previous research into this timeless subject. Rather, it is a wonderfully meticulous, thoroughly engaging "final word" that leaves the reader not with nagging questions or dangling what-ifs, but a strong sense that the case may finally be closed.

Dr. Adair dissects each science "explanation" with astronomical insights both accurate and inarguable. Refreshingly, he does so without in any way disparaging religion in general or Christianity in particular. He does not cross the hubris line as a few popular physicists do, by suggesting that as a scientist, he possesses a superior "take" on theology. Nowhere do we feel smugness, or regard his brushstrokes as too wide.

For anyone truly interested in that most famous Star in history, this book is essential reading.

Bob Berman
Willow, New York

Preface

“Finally, the clouds are parting. I may get to do some observations tonight!” he thought while shivering, his face cold save his beard which had reached new lengths. He had been monitoring the skies quite avidly for the last year or so. Perhaps he was so keen to get out and look, taking every opportunity he could find, because the skies were particularly poor for star watching where he was. But there were others like him who believed that the stars had been boding for something quite remarkable. In the previous year, there was a rather ominous conjunction of the planets Jupiter and Saturn, who in myth were not friendly, though they shared Titanic blood. Could this mean there was new blood to be spilled? Was there to be another war with the king’s neighbors? Or even a civil war?

The signs were becoming even starker as Jupiter and Saturn were now massing with a third wandering star, forming a triangle in the night sky, the likes of which had not been seen for some time. The only object, the Arabs thought, that could make the great conjunction and massing even more potent a portent was the sight of a comet. Astrologers about the land were most curious if something they could not predict with their charts was to happen.

And so this one astrologer was out in the chilly October night, perhaps finding a sign of things to come. He knew what to look for, or so he believed. After all, he had mastered and taught his craft for years, though his genius was not recognized by his bored students. But political figures respected him and sought his judg-

ments, which the astrologer was reluctant and nervous to give. Did he really trust his art? Was there any wheat in all so much chaff? Diligently he worked out new ways to advance his field, saving it from ridicule the best he could. If anyone could save this art, many believed it was him. He was recognized for his scientific prowess, and his mathematical abilities were first-rate. In recent years, he pored over data from ages past, trying to find patterns, and occasionally an abrupt "Eureka!" would stir him away from the rest of the world. His overwhelming insight from nine years earlier about the orbits of the planets still pushed him, and tonight was one of those nights that he would look into the heavens and be lost to all else.

As the clouds broke, the familiar stars were beginning to be seen. It was still too early to see the brightest star of all, Sirius, while the constellations of spring were well past their zenith. Winter was coming, and with it many another constellation that told of coming storms and conflicts. The north was the only constant, at peace, and yet there was the terrible dragon forever chasing about the North Star all day and night. Only when God finally ended it all would such evils be replaced in the new heavens and new earth. Or so he pondered as infinities floated about his mind. Along with that circle in the north, the astrologer could see the constellation that would point to Polaris, a collection of stars which some called a bear, and if he could just squint right he could see the double-star down its back and tail. A rider on a horse, some say, forever circling the north. Horses on bear tails; dragons spinning about; a thousand fantasies all trapped in the unchanging firmament of the north.

"Enough of this," he blurted to himself. What would those like Caspar think of him if he lost himself there?

He was not supposed to focus long on what was constant, as it was change he was interested in seeing, including the change in the wanderers, the planets. Perhaps even more had changed, if the rumors were true.

The gaps in the clouds had expanded to the south, the zodiac undressing before his eyes, a body of stars he had come to know intimately over the years. At first, to the east, the beautiful Andromeda was uncovered. When he squinted he could still see a bit of unmoving cloud at her hip. Curious, but it was not the place to find any of the planets. Tonight he needed to concentrate elsewhere. With the skies clearing, he could see the constellation of the serpent-bearer, what the Greeks called Ophiuchus. The Greeks also thought him a healer, an Asclepius. Was this pattern of stars telling us of the resurrection of the dead then, or was the serpent to poison the land? Tonight the snake handler would also be bearing planets and prophecies.

And there he saw it: amongst the massed planets was something new and bright. Between Saturn and Jupiter, who had just returned to the Fiery Trine of the astrologers, there emerged a new flame, a new star. Had the prophecy of the Arabs come true? The astrologer carefully wrote down his observations to later share with his peers what he deemed may come about. A great tumult was had amongst them because of the newness of the star. Some prophesied doom, others revolution, others still the End of Days. While religiously important to him, the key astronomer of his country was not so quick to interpret.

The cautiousness of the experienced scientist was not simply a quirk of personality, but he had read of other stories of new stars and what they could have

meant. He was a dabbler in books of history and religion, understanding both present and past with his craft. In his consumption and obsession, he consulted rare books concerning the powers that ruled a small territory on the Mediterranean, the land of Judea.

A new star. The lands of the Jews. Ancient prophecy. Writing to his benefactor, he began to pen his interpretation of the brilliant phenomenon. Perhaps at last he understood the truth...

This story may seem an interesting tale that would connect to one recognized by literally billions of people today. As many know, in December (and January for some) there is the celebration of Christmas, the time of the birth of one of the best-known religious figures, Jesus of Nazareth, called Lord Jesus Christ by the reverent. Part of the story of his birth includes the coming of men from the East who had seen some star in the sky that told them of the birth of the King of the Jews. And many have wondered what these men, called Magi (sometimes translated 'wise men'), could have seen that would have inspired them to travel from their native land to the hamlet of Bethlehem in order to see and pay homage to an infant.

But the story above is actually not about the Magi. It is a fictional retelling of the work of a more modern and well-known figure, Johannes Kepler. This German astronomer not only revolutionized the sciences of his days, best known for the discovery of three laws of planetary motion, he was also interested in what is now been soundly debunked within the scientific community, astrological influences. The observations in question concern the conjunction of Jupiter and Saturn in 1603 and his eponymous supernova of 1604. It was also

around this time that Kepler came across scholarly arguments that would put the birth of Jesus not in 1 BCE per tradition but 4 BCE or earlier. Kepler also looked into astronomical circumstances at the new time of Jesus's birth, and one association amongst astrologers was the conjunction of, again, Jupiter and Saturn, known to have taken place in 7 BCE (based on calculations).¹ This part of the story will be returned to later on.

The efforts of Kepler have been used by many a modern researcher to amplify their own search for what was of such interest to the Magi now over 2000 years ago. Called the Christmas Star or the Star of Bethlehem, this object has garnered an immense amount of speculation about what astrological readings could have been of interest to the eastern 'wise men' and what the Star itself was.² But even though Kepler is seen as the light that many Star researchers follow, the founding figure of modern science believed the great luminary at Jesus's birth to be a miraculous object, not some distant planet, star, comet, or anything of the sort.

But Kepler believed in astrology and other miraculous things, so he was wrong about plenty. Perhaps the Star of Bethlehem, as described in the Gospel of Matthew, can be explained by some natural phenomenon that can be uncovered using modern astronomical erudition. However, in this age of greater scientific knowledge, there is also an increase in historical knowledge and methodology, and the latter subject cannot be overlooked, though it often is. Should we not

¹ The most helpful biography of Kepler remains Max Caspar, *Kepler*.

² For example, Ruth Freitag, *The Star of Bethlehem: A List of References* is over 20 pages of, as the title suggests, just references to articles and books with little or no summary, published in 1979; since then, the book could be nearly doubled in length.

ask “Was there a Star at all?” before we try to explain it? Ancient books tell of all sorts of things we are, at the very least, skeptical of, and the Bible has not been free of such inquisitions by skeptic and Christian alike.

The Star of Bethlehem is a subject that touches two of the chief intellectual forces in the world today: science and religion. Arguments to their compatibility have lasted for many decades, and part of that discussion has included events such as the Star. If it could be verified by science, it would show a dialogue between the paradigms of thought rather than conflict or avoidance. Exploring this debate can potentially reveal how the ideas of faith fit into the realm of empirical inquiry, not to mention shed light on history and one of the most influential figures in modern times.

This volume will focus on the efforts trying to explain what the Star was and why someone in antiquity would have thought it so special as to signify the birth of a king in Judea. This book can be considered a companion volume to Jonathan Pearce’s *The Nativity: A Critical Examination*, which considers matters of historicity for both versions of the canonical birth of Jesus stories more than can be done here. There, the Star was considered for several pages, but this book will be more thorough in showing the various theories put forward, and their difficulties (something that I had done in the past³). Moreover, the current book is a summary of a larger planned volume about the Star along with the historicity of the Nativity in the Gospels and the literary context of the story.

To approach this properly, the book is organized in the following way. First will be a summary of what the

³ Aaron Adair, “Science, Scholarship, and Bethlehem’s Starry Night”, *Sky & Telescope* 114, 6 (Dec 2007): 26-9.

second chapter of the Gospel of Matthew has to say about the Star. From that starting point some basics of the chronology will be considered so we know when in time anything of interest ought to be. After this, in Chapter 2 some of the methodological problems with any attempt to find the Star of Bethlehem will be discussed, including what sort of source Matthew is. From the particular details to the historical problems at hand, this chapter will show that the effort is rather fruitless even assuming the historicity of the account. From there, each following chapter will discuss a particular hypothesis that has been put forward over the decades of what may fit the description of Matthew's account; however, Chapter 6 will discuss a number of other, less popular hypotheses but have been published and discussed in the Star of Bethlehem literature nonetheless. In those chapters about the various Star hypotheses, the major problems each theory has will be demonstrated. The penultimate chapter will demonstrate that no natural phenomenon can fit Matthew's account, leaving it up to the miraculous (or perhaps nefarious ETs and their ships). Finally, several points will be levied against the historicity of this particular part of the story of Jesus's birth. Even if the miraculous is allowed and it does happen, the last chapter should demonstrate that this account is just not credible.

A few more notes should be made here. First, when referring to the stellar object from Matthew's story, the word 'star' is capitalized. Similarly this is done for canonical gospels, the version of the nativity from the Bible, and other titles associated with the Christian canon. As for the text of the Gospel, I tend to use to Revised Standard Version (RSV) as a guide, though in many cases I use multiple translations and consult the

original language directly. My own translation of the story can be found in the appendix of this volume. For dating, I have tried to be consistent with scholars that use the convention of Before Common Era (BCE)/Common Era (CE), which are equivalent to the better known BC/AD system.

When it comes to terminology, I will explain technical terms as they appear, and a glossary of various terms is provided at the end of the book. However, it is worth distinguishing here what is meant by ‘astronomy’ and ‘astrology’. While separate terms were not used in antiquity, they have meanings that are rather important for modern scientists and historians. Astronomy is the study of the stars and planets, including their motions. Astrology is a form of divination using the positions of the stars and planets. In the past, the astronomers and astrologers were the same people and profession, while today one can be an astrologer with minimal knowledge of celestial mechanics. A modern astronomer, on the other hand, is unlikely to even talk about astrology, let alone practice it. (If you chat with someone and they tell you they are an astronomer, one sure way to anger them is to ask what their Sun sign is.)

With these considerations in place, let us consider what has been one fascinating intersection with science and the Christian religion.

PART I: THE STORY

Chapter 1: Matthew's Account and Timelines

The oldest version of the story we have about the Star of Bethlehem comes from the author of the Gospel of Matthew, the first book in the New Testament of the Holy Bible. While carrying the name of one of the Disciples of Jesus of Nazareth, scholars have been in agreement that no one really knows who wrote this book. There is also considerable uncertainty as to where and when it was written. Many introductions to the New Testament and similar resources⁴ will mention that it may have been written in Antioch on the Orontes River, then in the Roman province of Syria (its ruins today rest in Turkey), but this cannot go beyond a reasonable speculation.

The dating is also said to be around 85 CE, which would already place the writing of the book nearly a century after Jesus's birth. However, the date is uncertain. Generally it is agreed that Matthew uses another book as its source, the Gospel of Mark, and that

⁴ i.e. John Barton and John Muddimen, *The Oxford Bible Commentary*, pp. 844-845; Raymond Brown, *The Birth of the Messiah*, pp. 45-48.

was written probably sometime after the Jewish Revolt from 66 to 73 CE. Mark (or whoever wrote the Gospel) mentions the destruction of the Second Temple in chapter 13, and that structure was destroyed by the Romans in 70 CE. This marks, then, the *terminus a quo*, or earliest date, for Mark, and hence the earliest for Matthew who used Mark. The author of Matthew also mentioned the Temple's destruction, so this independently forces a post-70 CE dating. As for the latest possible date, the *terminus ad quem*, this has more contentious scholarship surrounding it. For simplicity, I will point to the most unambiguous early citation of Matthew's Gospel, Justin Martyr, who wrote his works around 155 CE. He refers to the Gospels as the "memoirs of the apostles", but his citations show that he is talking about the texts we know of. He also explicates the Star of Bethlehem in his *Dialog with Trypho* (chapters 106 and 126), connecting it with various parts of the Hebrew Bible. This means that, broadly, Matthew was written between 70 and 155 CE.⁵

At this point, it is worth mentioning the writings of Ignatius of Antioch, a bishop in the second century. His letters, alleged to have been written while a prisoner about to be executed, have been pointed to as earlier citations of Matthew, but this is quite problematic. There is uncertainty that the letters are authentic, that Ignatius lived and wrote when he is said to have done (his martyrdom is said to be during the reign of Emperor Trajan), and if he actually cited any Gospel. None of

⁵ David Sim, "Reconstructing the Social and Religious Milieu of Matthew: Methods, Sources, and Possible Results", in *Matthew, James, and Didache*, eds. Huub van de Sandt, Jürgen Zangenberg (Atlanta: Society of Biblical Literature, 2008), pp. 15-19 notes that the conflict with formative Judaism seen most clearly in Matthew would make better sense in the second century.

these things can be decisively shown, and there is considerable scholarship contesting all of these points.⁶ The arguments concerning authorship seemed to have the greatest consensus (that they are authentic), but the rest is arguable. Other alleged limits from sources such as the early second century bishop Papias and the Christian treatise called the *Didache* are similarly problematic. Allegedly Papias said Matthew was a collection of sayings in Hebrew,⁷ but our Gospel is in Greek and far more than a list of sayings, while the *Didache* also fails to say it quotes from a Gospel, let alone Matthew. As such, there is minimal utility in looking to these sources to shrink the time that Matthew wrote.

For now, these details matter little. The fact that Matthew wrote in 75 or 175 CE will not be a significant factor with regard to the conclusions of the following chapters. At any rate, Matthew is writing generations after the birth of Jesus (assuming his chronology), and what matters are his methods and sources. After all, centuries after the fact we can write decent biographies of Benjamin Franklin because of the sources at hand, including his own writings. If Matthew had good sources and competently and critically used them, it matters not one iota when he wrote. On the other hand, if his sources were poor and his methods unsound, it wouldn't matter if he wrote at the earliest possible date.

⁶ Timothy Barnes, "The Date of Ignatius", *The Expository Times* 120, 3 (2008): 119-130 notes recent arguments for the inauthenticity of the letters, the incorrect dating of the letters, and his own argument that they fit best in the 140s or so. On the independence of Ignatius from Matthew, see William R. Schoedel, *Ignatius of Antioch*, p. 9.

⁷ Eusebius, *Ecclesiastical History* 3.39.16.

However, for the time being we will not question Matthew's account or bring up any inconsistencies it has with other sources about Jesus, something that Jonathan Pearce does in his *The Nativity* as well as Raymond Brown in his *Birth of the Messiah*. Let us for now consider what Matthew says.

The story of Jesus's birth is begun by the Evangelist in a rather boring way with a list of who-begat-whom, but then he gets into the narrative proper, starting with the unusual circumstances of Jesus's birth. Then, in the second chapter, we come to the story of the Magi. During the reign of King Herod the Great (37-5/4BCE), these unnumbered eastern sages came to Jerusalem saying that they had seen the star of the King of the Jews rising. Some translations say the Star was "in the east", including the King James Version, but more correctly the Greek refers to the rising of the Star, which would be on the eastern horizon, and this is reflected in several modern renditions into English. Now, the Magi believing the king to have been born, they wanted to worship him. Herod, who was the king and had been for some time, was justifiably troubled by this. He further inquired into the Star the Magi talked of, even calling on the religious leaders in Jerusalem to help figure out where the Messiah (apparently it was Herod that connected the king and the messiah together; these titles are not synonymous) was to be born. According to prophecy, he was to be born in Bethlehem, just a few miles south of Jerusalem. Herod then told the Magi to go find the Messiah and inform him of his whereabouts so that he, too, could worship him.

The Magi left, and then the Star appeared to them (perhaps it had disappeared?) and "went before them until it came and stood over where the child was" (Matt

2:9). Here is the story at its most spectacular and which all natural Star hypotheses must contend with. The Star “goes before” the Magi, arrives, and then stands over a particular locale. At this point the Magi are at Bethlehem, perhaps just outside the very house Jesus and his family were in (the Gospel specifies it was a house that the Magi enter), and they then enter in. They see the child with his mother, worship Jesus, and give gifts of gold, frankincense, and myrrh, all items of considerable value. From there, the Magi stay awhile and then have an angel telling them to leave town; Joseph has a similar message in a dream, and soon all the good guys in the story have escaped from Bethlehem and the oncoming wrath of Herod.

So, the sorts of details that the Star has, according to Matthew, are as follows:

- ✧ There was a star or something star-like (2:2, 9)
- ✧ It was visible to the Magi (“for we have *seen* his Star at rising” 2:2)
- ✧ Found in the east/at rising (2:2, 9)
- ✧ Told of the birth of a king/messiah (2:2, 4, 11)
- ✧ Went “before them” until... (2:9)
- ✧ Stands over “where the child was” (2:9)

But there remains one last detail that isn’t explicit, and that is the date range for the Star, at least according to Matthew. What we are told is that Jesus was born before the death of King Herod. The details of the story also imply that the Holy Family stayed in Egypt for a relatively short time, after which they returned to the Holy Land when Herod died in 5/4 BCE. (Some have argued that Herod died later, some even claiming as late as 1 CE, but I will remain with the standard chronology

as it fits the evidence best.) But this does not strongly establish by what date Jesus would need to have been born by.

At this point many scholars will turn to the Gospel of Luke which gives further details (though he says nothing of stars or magi at the Nativity). For example, in Luke 3 we are told that John the Baptist began to minister in the 15th year of Emperor Tiberius, which would probably be 28/29 CE. Jesus came to John sometime after he began to baptize by the River Jordan, and Jesus was “about thirty” (Luke 3:23) when he came to John. Taking 28/29 CE as the earliest date that Jesus came to John and assuming Jesus was 35 years old, then that would put Jesus’s birth in 8/7 BCE, the furthest we can go back given the Lukan information. But this assumes that Jesus started his ministry in the same year John began to preach, and Jesus was as old as possible given the language of Luke. We are at least as justified in thinking Jesus started his ministry in 34 CE at the age of 28 and thus born in 6 CE. Moreover, the above strained chronological speculation ignores what Luke does say about when Jesus was born, which was during the census of Quirinius when he was governor of Syria (Luke 2:2) which was in 6/7 CE. This is a problem discussed elsewhere, so it will not be argued about here; we only need to note the difficulty in establishing any early biographical information about Jesus and how we cannot simply blend our sources together indiscriminately.

Nonetheless, the context of Matthew is the best guide we have when it comes to dating the Star, and it implies Jesus was very young when his family escaped from Judea to Egypt and returned (though to Galilee) after a short time. So, from that we could leave a broad

period of time consistent with Matthew's account between 10 and early 4 BCE for the birth of Jesus. But this is merely a best guess, and it ignores other information we have. Because of this difficulty, a date earlier than 10 BCE is not necessarily out of bounds, though we should remain strict about the 4 BCE date as we are more solid about when Herod died and Jesus needing to be born before that, according to Matthew.

Chapter 2: Methodological Difficulties

Before we investigate what natural event may have been behind the Star of Bethlehem, there are some methodological problems that need to be highlighted which are not often well-considered in the literature. Part of that list of problems was discussed in the prior chapter, but they will be highlighted again here.

Firstly, we are rather unsure about the dating of Jesus's birth. None of the canonical Gospels provide an exact date, nor even a year, so we are forced to infer what we can from those books. The historical context Matthew provides in his story can only provide a *terminus a quo* with the death of Herod. The data from Luke could help establish a *terminus ad quem* based on the data when Jesus came to John, as noted in the previous chapter. However, that must ignore the information about Jesus being born during the 6/7 CE census of Judea when it came part of the province of Syria under Quirinius. In the Star of Bethlehem literature, the way this is dealt with is varied. For example, astronomer and historian David Hughes,⁸ in his book, looked at some of the attempts to reconcile the accounts of Matthew and Luke. Retired astronomer Michael Molnar,⁹ on the other hand, is not convinced by these attempts and sees it as a serious issue, though this does not seem to make him skeptical of the account

⁸ David Hughes, *The Star of Bethlehem: An Astronomer's Confirmation*.

⁹ Michael Molnar, *The Star of Bethlehem: The Legacy of the Magi*.

of the Star in Matthew. However, without either a reconciliation or a demonstration of which dating of Jesus's birth is more reliable (assuming either are reliable), then searching for an astronomical event or astrological circumstance is pointless unless something very definite can be cited as to what gave rise to the story of the Christmas Star.

The chronology is also confusing to modern researchers because Luke's version of the birth of Jesus has very little in common with Matthew's account. Most strikingly, Luke has no mention of the Magi, the slaughter of babies in Bethlehem, the flight to Egypt, or the Star. Luke is also the only Gospel author that even puts on the mantel of a historian, so the absence of the amazing story seen in the First Evangelist should be alarming. Did Luke not include the story because he thought it wasn't historical? Or did he just not hear of the tale? And if he didn't know of the story, and Luke claims to be using witnesses of the Jesus tradition (Luke 1:2), what does that mean about the reliability of Matthew's tale of astrologers following a strange luminary?¹⁰

This leads to the second methodological problem, one perhaps even more insurmountable than the prior one. The premise of all Star scholarship is that something in the skies indicated to the Magi that a king was born, specifically Jewish. But this requires delving into astrological interpretation, and that is extremely problematic. This is not because astrology is a failed

¹⁰ The debate about the relationship between Matthew and Luke is long and nuanced, but the arguments by Mark Goodacre in *The Case Against Q: Studies in Markan Priority and the Synoptic Problem* are suggestive that Luke did know and use Matthew for writing his Gospel. However, this possibility will not be considered here, and I reserve a greater discussion on this for future work.

science (which it is), but because portents can be almost infinitely interpreted to mean most anything. This has been noted in scientific studies of the subject as practiced by modern astrologers.

For example, consider the following study with several astrologers, all of whom were respected by their peers and had practiced their craft for years. They had participants answer questions about personality traits that fit what the astrologers wanted for analyzing their horoscopes, along with the date, time, and location of birth of the participants. The results were then mixed up and the astrologers had to match up the 23 participants' horoscopes with their personal details. The study found that the astrologers could not connect the horoscopes with the personality reports any better than chance, as expected of a subject that does not have any basis in reality. However, more interesting is the fact that the astrologers were in amazing contradiction among each other when it came to which horoscope went with each personality report. That is, one astrologer would say horoscope A fit with personality report B, but another would say horoscope A fit personality report C, and so on with the remaining personality reports. On average, the agreement between any two astrologers was only 1.4 times out of 23. That is basically at the level of chance.¹¹ A Dutch study of similar design found much the same results: accuracy was about at the level of chance and so was agreement between astrologers.¹² So, at least for modern astrologers, what a given horoscope will mean is

¹¹ John McGrew and Richard McFall, "A Scientific Inquiry into the Validity of Astrology", *Journal of Scientific Exploration* 4, 1 (1990): 75-83.

¹² Rob Nanninga, "The Astrotest: A Tough Match for Astrologers," *Correlation* 15, 2 (1996): 14-20.

just as determinate as a roulette wheel. This makes any interpretation of past skies highly suspect since most any condition of the sky could be said to mean almost anything.

This condition can be seen in the ancient records as well, including those from Christians. For example, Christian critics of astrology said that the astrologer simply looked for what their client wanted to hear as any horoscope had positive and negative conditions; just pick the parts you wanted.¹³ Pagans such as the historian Tacitus were of a similar opinion.¹⁴ In his critique of astrology, the famous Roman politician Cicero gives the example of a Babylonian astrologer who made predictions for the three members of the Triumvirate—Crassus, Pompey, and Caesar. The astrologer, apparently wanting a happy story for these figures as well as the Roman public that depended on their alliance, stated that all three would live long lives and be glorious. Cicero wrote after the assassination of Caesar, which itself followed the failed invasion of Parthia by Crassus and his demise as well as the civil war between Pompey and Caesar that ended with Pompey's death.¹⁵ For Cicero, it was obvious how astrologers were poor prophets and instead brownnosers.

Astrologers of the era were well aware of the problem of poor predictions. The most famous of the ancient Western astrologers, Claudius Ptolemy, wrote in the second century that the subject had many fraudsters, but even the best will make mistakes because of the

¹³ Cf. *The Recognitions of Clement* 10.11; Origen, *Philocalia* 23.21-2.

¹⁴ Tacitus, *Histories* 1.1.22.

¹⁵ Cicero, *De Divinatione* 2.99-100.

immense complexity of the task,¹⁶ an argument repeated by a fourth century Christian convert and astrologer, Julius Firmicus Maternus.¹⁷ In other words, because there are so many things that can be considered in putting together and interpreting a horoscope, experts are bound to disagree.

So, the ancient astrologers in the Western tradition already tell us that the act of interpretation is very difficult, and the evidence is that astrologers cannot agree even when they all follow the same rulebook. The work of the historian of astrology Tamsyn Barton also shows that the ancient treatises of ancient Western astrologers (such as from Marcus Manilius, Ptolemy and others) actually do not tell us how to interpret a star chart.¹⁸ But more damning is the incredible level at which astrologers will contradict each other and even themselves. One particular example should be examined because of its importance to this study: what constellation or astrological sign corresponded to or influenced what region of the world; this is known as geographical astrology. This is important for Star research because of efforts to find the constellation which would have been associated with Judea. The only specific list that mentioned Judea comes from Ptolemy who connected that region with the zodiac constellation of Aries. However, a century earlier in the writing of Manilius this same general region is under the purview of Aquarius. Later in the same century, Dorotheus of Sidon goes with Gemini.¹⁹ There are still other astrological geographies,

¹⁶ Ptolemy, *Tetrabiblos* 1.2 (6-7).

¹⁷ Firmicus, *Mathesis* 1.3.2.

¹⁸ Tamsyn Barton, *Ancient Astrology*, pp. 114-142.

¹⁹ Cf. Wilhem Gundel and Siegfried Schott, *Dekane und Dekansternebilder: Ein Beitrag zur Geschichte der Sternbilder der Kulturvölker*, p. 312.

and none agree more than once or twice with each other and none agree on any one thing.

As such, historians of astronomy and astrology, such as John North (who studied ancient and medieval horoscopes), would not try to interpret a given horoscope since how it was done was subject-dependent; there are too many ways to read a star chart.²⁰ Considering how much contradiction there is between astrologers on so much, from methods to meaning, there is no respectable position one can take on how a horoscope would have been interpreted in antiquity. But this problem is compounded even more by another enigma: which version of astrology is being used?

There is not one, singular version of this divination technique, and there are significant differences between how astrology is done in the Western tradition compared to the Chinese version. However, even the Western tradition is taking on an earlier form of the art from Assyrian and Babylonian traditions which exist in records of great antiquity. We have some horoscopes from these lands and times, but what has been seen is that it is not the same as the methods better known from Greek and Latin treatises from the second century BCE and onward. One of the defining features of the Western horoscope is the arrangement of the cardinal points. These are the locations of the eastern and western horizon (the ascendant and descendant), along with the point on the zodiac that is highest in the sky at the observer's location and the point 180 degrees opposite on the zodiac (the midheaven and anti-midheaven). The location of these points in a horoscope are very important to interpretation, and considerable calculations are

²⁰ John David North, *Horoscopes and History*, p. xi.

done to find and place them, yet they are absent from the older Babylonian versions of horoscopes. Moreover, we have no good information to help us even guess how ancient, eastern horoscopes were interpreted as most of them failed to include an interpretation, and we have no treatises on the subject.²¹ The closest is the astrological text called the *Enuma Anu Enlil*, written sometime around 1000 BCE, but how this source was applied in the first century is almost impossible to extrapolate.

This means we have almost no basis to discuss how any configuration of the stars would have been interpreted by eastern diviners. We also have to be more specific in which group one considers the magi. These pious figures were at first perhaps a tribe in modern-day Iran (the Medes), but by the rise of the Persian Empire it had become a priestly cast in the Zoroastrian religion. More discussion of the magi and their religion will take place in Chapter 8, but for now it is worth mentioning that the attitude towards astrology at the time of Jesus's birth is even more divergent than what we know from the Greeks and Babylonians. Overall, the hurdle of interpretation of astrological portents by moderns ought to be out of the purview of appropriate method.

Lastly, we should consider an issue that is poorly addressed in the literature by those searching for the Christmas Star: what makes us think the author is trying to tell us history and why ought we trust him or his sources (if he had any concerning the Nativity)? Besides having the correct reconstruction of the text (something which we are in reasonably good standing when it comes to Matt 2; see the translation provided in this book), we need to consider the literary purpose and

²¹ Francesca Rochberg, *Babylonian Horoscopes*. Transactions of the American Philosophical Society vol. 88, pt. 1.

context of a text. Are we dealing with a diary of events similar to Julius Caesar's *Gallie Wars*, a biography based on eyewitnesses such as Arrian's history of Alexander the Great (the *Anabasis*), or do we have folklore and romances as in the stories of King Arthur? When it comes to the Gospels, this is not an easy question and has created mounds of collected paper and ink, all with both helpful and unhelpful observations, insights, and points of method in determining what we are dealing with.²² But perhaps key is the fact that the story of the amazing portent at birth indicating greatness of the infant is part and parcel of stories told of heroes, kings, and gods in antiquity. This includes stellar features as well. Famous stars leading heroes to their destination include that of Timoleon on his way to Sicily²³ and Aeneas on his way to Italy.²⁴ We also have the parallels between Jesus's birth and Moses's, especially in non-canonical sources concerning the latter (which include a star at his birth interpreted by 'wise men' of sorts).²⁵ Put simply, how does one categorize the story? You cannot skip this step and do proper analysis; else you will simply misunderstand what you are reading.

But even assuming, upon reflection, that Matthew's account is a sober one recorded by Joe Friday trying to get "just the facts", we need to figure out what his sources were, what quality they were, and how critically Matthew treated them. Unfortunately, unlike other

²² On the genre of at least some of the Gospels, the most useful argument that has a theoretical foundation comes from biblical scholar Michael Vines in *The Problem of Markan Genre: The Gospel of Mark and the Jewish Novel*.

²³ Plutarch, *Timoleon* 8; Diodorus, *Bibliotheca Historica* 16.66.3.

²⁴ Virgil, *Aeneid* 2.687-711. See Chapter 8 for more discussion of this tale.

²⁵ John Bowman, *Samaritan Documents: Relating to their History, Religion, and Life*, pp. 287-288.

historians and biographers of the time, including the lower-quality ones, Matthew does not tell us what his sources were, nor does he do anything like a critical examination of them. Often it has been a premise of form criticism, the study of the conveyance of stories and sayings found in the Gospels, that the stories about Jesus came through some sort of oral tradition from his early followers and then through a chain of transmission (be it via plebs or pastors) to the authors of the Gospels. Unfortunately, we have great difficulty figuring out what may have come from history and what may have been good storytelling. This problem is compounded immensely for the Nativity accounts because at this time Jesus would have been little-known, having not started his ministry and gathered a following yet. As such, none of the Disciples of Jesus would have been witness to his birth. This leaves Jesus's family, at best, to tell the tale. However, this is further limited by the apparent disappearance of Jesus's father, Joseph, from the Jesus story after the Nativity. Tradition has it that Joseph was dead by the time Jesus began preaching, leaving only Mary as witness. But even she disappears at the beginning of the Christian movement, last mentioned in the Book of Acts and otherwise existing in later legend (legends that include her own miraculous birth). Any oral tradition then about Jesus's birth (if it existed) would not be guaranteed by a living witness when Matthew wrote, and we have no evidence to show the story could be traced back to any of those witnesses. But most importantly, Matthew does not tell us what his sources were and why he thought they were useful.

Compare that to a biographer from around the same time who is usually considered a less than trustworthy writer, Suetonius. He wrote biographies of the Caesars,

starting with Gaius Julius and ending with Domitian, and for his earlier works he had access to the official Roman archives. Consider his biography of Gaius Octavius, later known as Caesar Augustus. He tells us various stories about the birth of Octavius, many of which include omens that are reminiscent of portents at Jesus's birth. However, Suetonius distinguishes these tales from that which he finds in various documents that he names. Suetonius also considers the plausibility of the folktales that surround the emperor. The same thing can be seen in Suetonius's biography of Emperor Nero, often considered overly credulous. It is from him that we learn the story of Nero playing his lyre while Rome burned, confusing gossip with what really happened (Nero wasn't even in Rome according to the more reliable Tacitus).²⁶

Matthew, on the other hand, does not discuss his sources, does not separate witnesses, and provides not a hint of skepticism or critical engagement with the stories. The only source that he does cite is the Old Testament when talking of prophecies about Jesus, but if this is indicative of his efforts then Matthew isn't even a careful historian. As is well-known, he has either misinterpreted or misquoted parts of the Hebrew Bible or has even made them up (the prophecy that the Messiah had to be from Nazareth is not found, as is, in the actual Hebrew or the ancient Greek translation commonly used around this time; the prophecy of the birth in Bethlehem is also a confabulation between verses of different Hebrew authors). Since Matthew's historiographical methods are vastly inferior to that of an author considered suspect, and his access to reliable information is

²⁶ Tacitus, *Annals* 15.39.

even more remote, we have to, *a fortiori*, consider Matthew even more suspect for historical facts, and this assumes Matthew is even trying to tell us facts.

But these problems of literary context and historical method are not well-argued in *Star of Bethlehem* literature. For example, David Hughes, back in his 1979 book, spoke of the story by Matthew as having “the ring of truth”,²⁷ but this can only be the affect fallacy. Little else is said by proponents to show that the tale likely has a kernel of truth. What this means is that most who approach the subject looking for the Star as a natural event have assumed a lot about the text and its historicity. The lack of awareness of what problems the narrative has before analysis will not likely be fixed by ignoring them, and Star researchers should be both aware of and deal with them.

Still, let us bypass these issues for now, though they are significant, and instead give these investigations into the Star their best shot. If someone could show a good correspondence between Matthew’s account and some particular stellar configuration or series of events, then that could affect the probability of Matthew telling us something true. But even if a solid connection between events in the sky and Matthew’s tale can be uncovered, we may still need to be skeptical of the historicity of the story because of the uncertainties about Matthew’s sources. Other possibilities are back calculations by someone later on who used a few equations to figure out what sorts of interesting sights were in the sky around

²⁷ Hughes, *The Star of Bethlehem: An Astronomer’s Confirmation*, p. 198. In private conversation, Hughes has expressed more skepticism of the story.

when Jesus was born,²⁸ or a similar event at a later date inspired the story.²⁹ These possibilities also need to be considered before reaching the conclusion that we have helpful data about the birth of Jesus rather than theologizing of natal astronomical events well after the fact.

But before coming to these decisions we need to see what natural phenomena are alleged to fit the tale. That is the subject of the following chapters, to which I hope you will now turn.

²⁸ Cf. Lynn Thorndike, *History of Magic and Experimental Science*, Vol. 1, pp. 471-472.

²⁹ Cf. Rod M. Jenkins, "The Star of Bethlehem as the Comet of AD 66", *Journal of the British Astronomical Association* 114, 6 (June 1999): 336-343.

PART II: THE HYPOTHESES

Chapter 3: Comets and Meteors

Perhaps the oldest argument put forward for a non-miraculous version of the Star of Bethlehem comes from a German Protestant theologian, Christian Gottlieb Kühnöl, in 1807. Coming from a wave of scholars in reaction to higher criticism of the Bible, he and others were starting to reimagine what the Bible said about miracles, and Kühnöl in particular focused on what could explain the Star. Not only being one of the first in this area, the author is also one of the published few that argued the Star could be explained as a couple of meteors.³⁰ More recently, the late Sir Patrick Moore, the voice of popular astronomy in Great Britain, in one book³¹ and in BBC television conversations about the Star with David Hughes and Mark Kidger, favored the meteor explanation for the Star back in 2001.³²

What advantages does this hypothesis hold? The flash of a meteor, which is basically a rock flying through the atmosphere that came from outer space, can travel

³⁰ Christian Gottlieb Kühnöl, *Evangelium Matthaei*, p. 23.

³¹ Patrick Moore, *The Star of Bethlehem*.

³² Martin Mobberley, *It Came from Outer Space Wearing an RAF Blazer!: A Fan's Biography of Sir Patrick Moore*, p. 535.

in most any direction, so it could have first been seen overhead in Jerusalem by the Magi, and that streak of light could have traveled south towards Bethlehem. So perhaps this can fit the description of the Star “going before” the Magi. Moreover, any year for the birth of Jesus can be consistent with a meteor hypothesis since they happen most all the time, and showers of them regularly happen throughout the year.

However, this is pretty much all the criteria a meteor could potentially fit, while it significantly fails to fit other details. For one, it is hard to say how the object first seen by the Magi in their eastern skies was the same object they saw perhaps weeks or months later when traveling to Bethlehem. Moreover, because meteors are so short-lived, it is hard to talk about how it could have risen in the east at all. Considering how many read the phrase about the Star rising as being connected to sunrise, a meteor then in the east would have been invisible, drowned out by the light of dawn. Also because of their fleeting perceptibility, it is hard to see how a meteor can last long enough to “go before” the Magi during the full trip to Bethlehem from Jerusalem, which on foot would probably take on the order of two or three hours. While a series of fortuitous meteors all pointing towards the south could do this, that scenario would not fit with the singular Star mentioned by Matthew. The astronomical difficulties do not end there. We cannot see how a meteor can be said to arrive and stand over a particular location, a point that Moore fails to overcome. Such a moving flash of light could not fit this bill.

The last detail to point out is that there is simply no reason to think that a meteor would have been particularly auspicious and pointed to the birth of anyone. Meteors, as noted above, are extremely common. They

are seen throughout the sky and can be seen on most any given night so long as there is good visibility. In a shower, meteors can burn up in the sky at a rate of more than one per minute. If meteors were so auspicious, one would think that magi would be going all over the world looking for kings nearly constantly. However, we have no record to support this absurd situation even partially. The only potential solution is if there were other things in the sky that had the needed astrological import. That discussion will come later in Chapter 5. In the meantime, it is safe to leave aside this proposal as there are too many astronomical and symbolic problems.

However, there is a more prominent transient object that we know gets the attention of people: the comet. These dirty snowballs produce a tail of gas and dust when they approach the Sun; the heat causes the comet to evaporate at the surface, and the tail of the comet can be millions of kilometers long, much farther than the distance from the Earth to the Moon. A comet, such as the famous one named after Edmund Halley, will usually be visible to the naked eye for days or weeks, perhaps even months. Because of their prominence, we are fortunate that many ancient cultures, from China to Mesoamerica, record their entry into the sky and what they meant to them. This is already significantly better than the meteor hypothesis because we can know if there was such an object in the sky, where it was, and we can investigate what they were thought to mean since we have these ancient eye-witness records of comets providing the time they were seen, where in the sky they were seen, and often providing descriptions of the comet itself (and interpretations of comets are numerous).

As for the astronomical circumstances that could be consistent with Matthew's account, it is relatively easy

for a comet to be said to rise in the east. Depending on its ephemeris (the path of the object in the sky), a comet could appear on the horizon at a time that could have greater astrological significance. Considering that many interpret the phrase “at the rising/in the east” (*en te anatole*, ἐν τῇ ἀνατολῇ) from Matt 2:2 to mean the Star rose at sunrise, a comet can also fit this bill. Records of comets from the past often do not have enough details to make such an ephemeris impossible, so a comet can be consistent with most any reading of Matt 2:2.

Moving to verse 9, a comet can be seen towards the south, so it could be in the direction of Bethlehem from Jerusalem when the Magi headed out. As for the Star standing over where Jesus was, the two references which have comets standing over places that proponents point to are from the Roman historian Cassius Dio³³ and the Jewish historian Flavius Josephus.³⁴ So far, it seems we can have a match between the astronomical aspects of a comet and Matthew’s account.

Lastly, we need to consider if a comet was seen in the timeframe of Jesus’s birth according to Matthew’s chronology. At this point we are fortunate because of the excellent records of comets from China. These Far Eastern reports not only tell us when and where a comet was seen, but it even gives details of how the comet looked. As we have it, in 5 BCE there was a comet with a notable tail which the Chinese called a *hui* comet or ‘broom star’; another tailless comet was seen in 4 BCE but after Passover and the death of King Herod. Before the comet of 5 BCE is Halley’s Comet in 12 BCE, but this

³³ Cassius Dio, *Roman History* 54.29.8.

³⁴ Josephus, *Wars of the Jews* 6.289f.

is likely to be too early to fit.³⁵ Nonetheless, we have a reasonably good candidate for a comet that fits the timeline.

While this is also superior to the meteor hypothesis, the comet still has significant problems fitting the description of Matthew's account. For example, while a comet would potentially be seen in the direction of Bethlehem from Jerusalem as argued by the material scientist Colin Humphreys,³⁶ the text says that the Star moved in that direction—the Star “went before” the Magi. The movement of the comet would only be able to perceptively travel from east to west, not from north to south. Moreover, a comet cannot arrive and stop as the Star is said to have done, let alone hang over a particular hamlet. The examples mentioned above of comets standing over places do not actually fit when one looks at the original text rather than an English translation. In the original Greek they use significantly different words than does Matthew and none have a comet initially moving and then stopping. While the Evangelist uses the term *histemi* (ἵστημι) to mean stand, *erchomai* (ἔρχομαι) for arriving, and *epano* (ἐπάνω) to mean over, Dio uses *aiotheis* (αἰωρηθεῖς) that means ‘to hang’ rather than stand and *huper* (ὑπέρ) to mean over. Similarly, Josephus uses the preposition *huper* instead of *epano*. The difference between *huper* and *epano* will be discussed in Chapter 7, but in the meantime it can be said that these historians mean something significantly different than what Matthew describes. But even

³⁵ Ho Peng Yoke. “Ancient and Medieval Observations of Comets and Novae in Chinese Sources”, *Vistas in Astronomy*, vol. 5 (1962): 127-255, esp. p. 137.

³⁶ Colin Humphreys, “The Star of Bethlehem—a Comet in 5 BC—and the Date of the Birth of the Christ”, *Quarterly Journal of the Royal Astronomical Society*, 32 (1991): 389-407.

excusing the vocabulary differences, neither of these authors has a comet standing over a particular house which Matthew implies, which again will be specified in Chapter 7. This means there is no precedent in the literature to say a comet can stand over a location, let alone travel south and stop—another claim to which no parallel is provided.

But perhaps the most obvious problem with the use of a comet is that it is one of the few astrological signs that are almost universally agreed upon: they are a terrible sign! The late astronomer Carl Sagan and his wife and TV producer Ann Druyan in their book *Comet* describe how from China to the Mediterranean to Mexico comets were so often seen as signs of evil. Dozens of examples can be given, but let us see a few in the regions that are of interest to this study. The writer Diodorus Siculus described a comet which appeared to have three tails that was seen as a sign of the end of the powers of the peoples of the southern peninsula of Greece, the Lacedaemonians, better known as the Spartans.³⁷ A nephew of the Roman philosopher Seneca the Younger wrote the following: “The heavens appeared on fire, flaming torches traversed in all directions the depths of space; a comet, that fearful star with overthrows the power on the Earth, showed its horrid hair.”³⁸ Seneca would seem to agree with his nephew in his own writings about comets.³⁹ The astronomer and astrologer Ptolemy states that comets are a sign of wars, hot weather, and disturbed conditions,⁴⁰ all of which is consistent with the

³⁷ Diodorus Siculus, *Bibliotheca Historica* 15.50.

³⁸ Carl Sagan and Ann Druyan, *Comet*, p. 25.

³⁹ Seneca, *Naturales Quaestiones* 7, especially sections 1.5; 17.3; 28:2-3.

⁴⁰ Ptolemy, *Tetrabiblos* 2.9 (90f).

astrologer Manilius's opinion of the matter.⁴¹ Even Humphrey's own examples of comets that stood over things in the historical record were seen as both a sign of doom for the Jews during the revolt⁴² and a portent at the death of Augustus's friend Agrippa.⁴³ On top of these, ancient Western sources could add dozens of cases with the same explications.

Now, all of these are examples among Greeks and Romans. But the situation is the same in the Middle East. Cuneiform texts concerned with celestial signs, including letters by astrologers, give connections between a comet and the death of a king.⁴⁴ The death of a king is hardly the birth of one, so there is no hope of twisting this omen into a natal premonition, let alone a favorable one. In Zoroastrian literature, there is an object known as *Mush Parik*, and this was supposed to be a comet occasionally released from the Sun. It was a terrible sign and foretold of abysmal events.⁴⁵ Later, after the Arab conquests, the records of the Persians and Greeks would be read and pondered by Muslims and other Arab-speaking scholars. Their records indicate that comets were also ominous and ruined a horoscope.⁴⁶ From the whole of antiquity and into the medieval period, I have found in Middle Eastern records no

⁴¹ Manilius, *Astronomica* 1.874-926.

⁴² Josephus, *Wars of the Jews* 6.314-315.

⁴³ Dio, *Roman History* 54.29.7-30.1.

⁴⁴ Hermann Hunger, F. Richard Stephenson, C. B. F. Walker, et al., *Halley's Comet in History*, p. 18; Gary W. Kronk, *Cometography: A Catalog of Comets*, p. 1; Hermann Hunger, *Astrological Reports* §§ 339, 456.

⁴⁵ *Greater Bundahishn* 5.4-5; 5 A.5-7; *Shkand gumanig wizar* 4.46. See *Encyclopedia Iranica*, vol. 2, p. 867.

⁴⁶ Edward S. Kennedy, "Comets in Islamic Astronomy and Astrology", *Journal of Near Eastern Studies* 16, 1 (1957): 44-51.

exception to the rule that comets were signs of evil to come.

This problem is well-known to everyone that argues in favor of a comet as the Star of Bethlehem, so there is a heavy reliance on the very few exceptions in the literature from antiquity. The most prominent counter concerns the comets seen at the birth and ascension of King Mithridates of Pontus, a ruler in modern-day Turkey who had been a significant opponent to the Romans in Asia Minor in the first/second century BCE. The timing of these comets was at one time considered a fiction, but they seem to be confirmed in Chinese annals and coins. Unlike other despots that would avoid the use of such an ominous symbol, Mithridates had the comet on his coinage (a trait he would share with Vlad the Impaler), though only on the smallest of coin denominations.⁴⁷ However, it seems that the way the symbol was used was done *post factum*, that is, after Mithridates seized power, and the comet required reinterpretation. The coins do not simply show the comet but couch it in other astrological symbols, such as the constellation Pegasus, so it seems that the ill signs were being mitigated, as suggested by classicist John Ramsey. The comet is interpreted as a great flame rather than a comet straight-out and thus conforming to Persian prophecies.⁴⁸

The same situation happened with the Octavius's (later known as Caesar Augustus) comet; it was transformed into a symbol of power by mythologizing it into Julius Caesar's soul rising to heaven. This was also

⁴⁷ John Ramsey, "Mithridates, the Banner of Ch'ih-yu, and the Comet Coin", *Harvard Studies in Classical Philology* 99 (1999): 197-253.

⁴⁸ Ramsey, *op. cit.*, pp. 228-229.

possible because when the comet was first seen the tail was not noticeable, thus appearing as a star rather than a comet. It would be Augustus's enemies that would call the new object a comet and a sign of the coming civil war, while his allies more often referred to it as a star (*sidus*).⁴⁹ Thus, the identification and interpretation was politicized, and when viewed as a comet by Octavian's detractors the comet was given its evil classification, thus adding another example of negative interpretations of comets.

In these cases, we see that the comets were not viewed as positive omens in their own right but were construed by those in a position of power to influence how these objects would be seen after the fact as propaganda. That is completely different than what is needed in the case of the Star of Bethlehem: the Magi see the comet and know it is an auspicious sign for another nation's king. They do not know before seeing a comet that some newborn is their political master, as was the case of Mithridates, so they had no reason to try and reimagine how the comet was in fact an excellent sign and not really a nefarious comet. The interpretation by the Magi is not *post factum* in favor of a major political figure, so the examples of Mithridates and Augustus do not fit what is needed to conform to Matthew's story. But even if we ignore how the symbol was being drained of its harsh portents after the fact by a particular political figure for his own benefit, this is still an exception (two at best) in a list of many dozens that are distinctly negative. We still have to say that comets were seen as ominous signs in well over 95% of the cases. Without showing how the comet of 5 BCE could be made out to

⁴⁹ John T. Ramsey and A. Lewis Licht, *The Comet of 44 B.C. and Caesar's Funeral Games*, pp. 135-153.

be a great omen rather than an ill one via some other astrological circumstance, then we have every reason to discount the comet hypothesis based on its symbolism alone. Combined with its failure to fit the description of Matthew's Star, the comet hypothesis is a failed one.

However, there exists one potential trick up the sleeve of the adherent to the comet hypothesis: the testimony of Origen, perhaps the most scholastic Christian in the first few centuries of the Common Era. In his point-for-point response to the attack on his faith by the pagan philosopher Celsus, Origen talked about the Star of Bethlehem and comets.⁵⁰ Origen says how there are plenty of examples where a comet was seen and it wasn't that bad, so having the Star of Bethlehem being like a comet was not a poor heavenly sign. Many take this to mean that Origen thought the Star was a comet.

However, Origen only says the Star is like a comet in how it has significant meaning and how it is newly seen in the sky, unlike planets or other stars. Moreover, Origen does much to contradict the view that it was a comet that made people think a king was born. He said that he knew of no prophecy about a comet being seen at a certain time and heralding the birth of a king of any region. Considering how bookish he was, that suggests that there was no such connection in the ancient world—no one was expecting a comet to prophesy a king. Furthermore, Origen in other works shows the Star was a miraculous object that came down to Jesus just like the dove that perched on him at his baptism.⁵¹ Origen did not think the Star of Bethlehem was a comet, but instead he brought up the tailed star as a point about

⁵⁰ Origen, *Contra Celsum* 1.58-59.

⁵¹ Origen, *Homilia in Numeros* 18.4.

momentous astral symbols at the start of major events. Origen leaves the comet hypothesis for the Star with nothing as he sides with a supernatural understanding of the Star, and the significant problems with the hypothesis, mentioned here, make it untenable.

Chapter 4: Novae and Supernova

This chapter focuses on what is one of the newer yet popular hypotheses concerning the Star of Bethlehem, and this newness is because for so long the phenomena concerned were not well-understood. Here we want to look at novae and supernovae. These are different objects, but they are effectively the same for naked-eye observers: a bright light which was called a new star due to its sudden appearance, hence the name of 'nova'. The mechanisms for novae and supernovae are different, and there are different versions of each type, but they are both basically a stellar explosion. In the case of a nova, an object known as a white dwarf accumulates hydrogen on its surface from a near-by star; when there is enough gas there, nuclear fusion can take place and make a burst of light. For supernovae, there are two basic types, ingeniously named Type Ia and Type II. Type Ia are similar to novae in that they involve gas being accumulated by white dwarves, but when the mass of the white dwarf is large enough it collapse and releases a huge amount of energy. In the case of Type II, a giant star, far larger than the Sun, has exhausted its nuclear fuel, collapses due to gravity, and then the in-falling gas slams into the core of the star, bouncing and producing some of the largest explosions in the universe.

If bright and close enough, novae and supernovae will appear in the sky to the naked-eye observer as a bright point of light, not unlike a normal star. What will give them away is their recent appearance and how they slowly dim away. A nova may lose its brilliance in a matter of days, and a supernova may still be visible for

months; a year after seeing a supernova, there is unlikely anything to be seen without a telescope. Novae can be recurrent should they still have a continuous source of hydrogen gas, but supernovae are one-time events of brilliant power. A supernova can have the luminosity of an entire galaxy of stars. Unfortunately for us in the 21st century, there has not been a visible supernova in our galaxy in a long time, the last one being witnessed by Johannes Kepler in 1604. He was also fortunate to see one in 1572, while we have had a dry spell of such explosions for centuries. The supernova in 1987 from the Large Magellanic Cloud was for a time visible, but it was outside our galaxy and not nearly as brilliant as Kepler's star. Many think one is long overdue, and a red giant star like Betelgeuse in the constellation Orion could go at any time.

The question then is if such an object could be what was seen by the Magi 2000 years ago. A brilliant light in the sky captures the imagination, not to mention looks well on a Christmas card, so perhaps it makes sense that this is one of the rather popular conceptions of what the Star could have been. The most admired incarnation may be from the award-winning short story by Arthur C. Clarke,⁵² one of the top science fiction authors of the 20th century, especially known for writing *2001: A Space Odyssey*. In his story, a Jesuit astrophysicist comes to a distant planet to discover the remains of a lost and prosperous civilization that was destroyed when the sun of that world had gone nova. After some calculations, he determines that this star that exploded would have its extreme luminescence reach Earth at the time of Jesus's birth. This drove the Jesuit astronomer into a crisis of

⁵² Arthur C. Clarke, "The Star", *Infinity Science Fiction* 1, 1 (1955): 120-127.

faith, asking how his good God could destroy a civilization for the sake of making the Star of Bethlehem. The story won the Hugo award and would later be adapted into an episode of the 1980s reboot of *The Twilight Zone*.

Besides fictional and popular conceptions, the idea of a supernova as the Star was put forward by professional astronomers in the peer-review literature in 1977⁵³ and was also supported recently by the astronomer Mark Kidger.⁵⁴ The theoretical physicist Frank Tipler also has his own version of the supernova hypothesis.⁵⁵ What features do they see that can explain the Christmas Star? Firstly, the newness of the nova would be spectacular to those familiar with the sky; a new object could well fit what Matthew meant by “his Star” to distinguish it from all the other stars and planets in the sky. Like a star, it can rise in the east, and again it can fit the meaning of “at the rising” from Matt 2:2 just as well as a comet. Also like a comet, the nova could be in the direction of Bethlehem from Jerusalem, which is then seen to be before the Magi. On the other hand, Tipler believes the supernova was at the zenith so it could stand over Bethlehem, fulfilling a different aspect of the description of the Star. In fact, Tipler never even attempts to explain the movement of the Star, leaving his hypothesis weakened even before critical analysis.

As for what the nova would have meant to the magi, unlike the case for comets, we have effectively nothing to work with in knowing how novae were interpreted

⁵³ David H. Clark, John H. Parkinson, and F. Richard Stephenson, “An Astronomical Re-Appraisal of the Star of Bethlehem—A Nova in 5 BC”, *Quarterly Journal of the Royal Astronomical Society* 18 (1977): 443-449.

⁵⁴ Kidger, *The Star of Bethlehem: An Astronomer's View*

⁵⁵ Frank Tipler, “The Star of Bethlehem: a Type Ia/Ic Supernova in the Andromeda Galaxy?” *The Observatory* 125 (2005): 168-174.

astrologically. The divinatory treatises of Ptolemy and others say nothing about them, and records of novae in ancient Greece, Rome, and the Middle East are very sparse. One of the few records of a nova by someone from this time and region comes from Hipparchus of Rhodes in the second century BCE,⁵⁶ but we have no interpretation of the object besides how it inspired Hipparchus to make a careful map of all the stars to see how much the sky changed over time, countering Aristotle's notion that the heavens were unchanging.

However, if Chinese records are any indication, a nova may have been indistinguishable from a comet. In the catalogue of Chinese comet reports, three categories are noted: *hui* or "broom star" comets, *po* or "shining" comets, and 'guests stars'. The latter are considered novae, while the first in the list are certainly comets as they have tails. The *po* comet, on the other hand, is ambiguous; it could be a comet that has its tail pointing directly towards or away from the Earth so it looks like a singular object like a nova. Some *po* comets today are thought to be novae. So it may be the case that novae would have been thought to be a comet in the West; even if distinguished by European or Babylonian astronomers from comets, it may have been seen as similar in meaning to a comet because of their sudden appearance. At best we cannot know what the ancient magi would have thought of a nova, and at worst it would have been seen in the same way as a comet, hence a negative rather than positive omen (see Chapter 3).

Before considering how well novae actually do fit the details of Matthew's account, we need to see if any such object was observed in the timeframe of when Jesus may

⁵⁶ Pliny, *Natural History* 2.96.

have been born. The Chinese records as put together by Ho Peng Yoke show no 'guest stars' anywhere near the timeframe of 10-4 BCE, but there is a *po* comet recorded in 4 BCE. This *po* comet comes too late in the year to have happened before Herod's death (no later than early 4 BCE), so it won't fit the timeframe. The comet in 5 BCE is said to be a *hui* comet and with a tail, so it should not even be a potential nova/supernova candidate. That is, unless one believes Kidger and the researchers he follows. He believes that the object in 5 BCE and the other in 4 BCE are the same object; there was a mistake in the date of the record and in fact it was a nova in 5 BCE.

This should be automatically a suspect argument for several reasons. The first point to consider is the distance the objects are from each other in the sky. Treating the stars as lying on a sphere, astronomers talk about the angular distance two objects are from each other; like lines of longitude, there are 360 degrees to go around in a circle, and difference in longitude indicate angular distance. A good rule of thumb is that the Moon is about half of a degree across as seen from the surface of the Earth; your thumb when held out at arm's length will cover the moon and is about 2 degrees wide, making the phrase 'rule of thumb' very apt in this circumstance. The 4 BCE object was 20 degrees (40 moon diameters) away from the location of the 5 BCE object, which is simply too amazingly large of a mistake in placement. The 5 BCE object was away from the plane of the galaxy (that is, what is commonly seen in the sky and called the Milky Way) in which the vast majority of the stars are located, making it less likely a nova; with far fewer stars, especially massive ones, there are fewer opportunities to have an exploding star. Lastly, the 5 BCE object is said

to have had a tail, so it cannot be a nova which do not have tails. If the astronomers making this argument were correct, it means that the ancient eastern astronomers got the time, location, and identification of the object wrong. If so, then why trust them for any of this data? But this is only worth considering if the astronomers actually had a good reason to think these things.

However, the only premise that this strange line of inquiry stood upon was that the Chinese did not record the 4 BCE object, while the Koreans did, though with a nonsensical date (a sort of February 31st); since the Chinese comet records are far superior in quantity and quality, it seems likely that the Korean record was a mistaken duplicate of the 5 BCE comet/nova. All of this was soundly refuted back in 1979 by Chinese historian Christopher Cullen who showed that the reasons given were completely wrong; the error was only in a Korean copy of the original 4 BCE Chinese record which we also have but without the date slip-up. The Chinese did record the 4 BCE comet (which medieval Korean scholars miscopied), and it was mentioned on the very page of the list from Ho Peng Yoke's compilation of Chinese comet records Kidger and others used to make their argument (the same paragraph, no less).⁵⁷ Kidger and those he follows have produced nothing to contradict Cullen, and they cannot because their argument is based on the premise that there is no Chinese record of the comet in 4 BCE, which is false. What this eventually means is that there was no observed nova for the time of Jesus's birth.

The hypothesis of Tipler tallies even less with the historical record. His supernova in the Andromeda

⁵⁷ Christopher Cullen, "Can We Find the Star of Bethlehem in Far Eastern Records?" *Quarterly Journal of the Royal Astronomical Society* 20 (1979): 153-159.

galaxy has not even the 5 BCE comet to connect it, and it is unlikely it was even visible. Based on current physics models of supernovae, Tipler calculated that the maximum theoretical brightness of such an explosion could be observable at the naked-eye level (though he underestimated the distance of Andromeda to us), but the brightness is even less than the apparent brightness of the galaxy itself, and that assumes no dust or gas is blocking the light of the object (a very unsafe assumption); it would not register for someone without a telescope. We should also realize that the Andromeda galaxy somehow went unobserved until it was sighted in the Arab-speaking world in the 10th century,⁵⁸ hardly then a place anyone was looking or finding any supernovae. Considering no supernova has ever been seen in Andromeda without the telescope, and only one has been witnessed with such a tool (in 1885), and while such explosions are supposed to happen on the order of once per century, then the complete lack of records of sighting of the galaxy and any explosions in it proves no one was noticing or could notice anything happening in that spot in the sky. We can be reasonably certain no supernova could have been nor was seen between 10 and 4 BCE.

Ignoring the problems of interpretation, and if such an object was actually seen at the time of Jesus, could the other criteria of a nova or supernova really fit what Matthew says? Consider again the point of the Star “going before” the Magi. Unlike a comet, a nova cannot even move relative to the stars in the sky, and all stars move from east to west, not north to south as Matthew implies when he says the Star “went before them”, and the Magi are traveling south to Bethlehem from

⁵⁸ George Robert Kepple and Glen W. Sanner, *The Night Sky Observer's Guide*, Vol. 1, p. 18.

Jerusalem. Again, the Star ‘goes’, not ‘is’ before the Magi, so motion is implicit and must travel in the same direction as the path of travel to Bethlehem. The notion of the Star standing over where baby Jesus was is also not easily explained. On Tipler’s hypothesis of the nova at the zenith, it would no more be over Bethlehem than over Jerusalem without extremely accurate measuring equipment unheard of in the first century. Kidger, on the other hand, just says the Star was high in the sky towards the south.⁵⁹ But this fails to explain the implicit motion of the Star, the fact that Matthew says it stops, and how it can stand over anything if, on Kidger’s reading, it’s not over the town at all, let alone a particular house, when the Magi are there and the Star is still south of them.

While a supernova at the birth of Jesus would have been spectacular, it simply doesn’t fit what Matthew describes. What is equally important is that no such event can be shown to have happened that fits the timeline, and there is no evidence to show it would be associated with the birth of a king, Jewish or otherwise.

⁵⁹ Kidger, *The Star of Bethlehem: An Astronomer’s View*, p. 35.

Chapter 5: The Planets and their Positions

Perhaps the most popular of the hypotheses for the Star of Bethlehem have dealt with the planets, their motions against the background stars, and their meaning to the astrologically minded. In the literature there are copious musings about the planets Jupiter and Saturn, the constellations Pisces and Aries, and individual stars such as Regulus in Leo. Many attempted interpretations of these objects have created a mass of confused and conflicting thoughts on the subject, but a few common and basic ideas will come to the surface in this analysis. The motions of the planets have also persistently been correlated with the phrases used by the author Matthew, especially how the Star “went before” and “stood over”.

There are numerous versions of this attempt to explain the Star that I will split these into two larger groups: a conjunction of two planets (i.e. Jupiter-Saturn, Jupiter-Venus), and the horoscope in its entirety (that is, considering all the planets and the location of the cardinal points). There are fewer scholars in the last camp, but it also includes one of the most extensive attempts to connect the Star, Matthew’s account, and history. There is significant overlap between the hypotheses as well, so it is worth reading both sections to understand the advantages and disadvantages of a general hypothesis that uses the planets to elucidate the Star of Bethlehem.

Part A: Conjunctions

First, a point of definition is needed since this is not usually understood correctly. In the system of stellar coordinates, there are the equivalents of latitude and longitude called declination and right ascension. Declination, like latitude, measures from what is called the celestial equator, which is simply the Earth's equator projected out into space. Right ascension measures along the celestial equator similar to longitude. Using this system, a conjunction means that two planets orbiting the Sun (or a star and a planet) seen from Earth have the same right ascension in the sky; basically, they are on the same line of longitude. It does not mean that the two objects are necessarily close to each other, but when it comes to the planets this is more often the case. Because all the planets are about in the same plane as the Earth about the Sun, they will have minimal latitude differences and so at a conjunction they can be close. Sometimes planets can be so close that one can cover the other, though this is very rare; more common is an object such as the Moon blocking a star or planet which is called an occultation. These points are highlighted here so that way a thesis about the Star is not dismissed simply because the planets are not close enough; closeness is not a criterion of conjunctions, though it may be for some arguments about the Star. The closeness of the conjunction should only matter if a proposed hypothesis must have the objects so close that they are indistinguishable and appear as a single object (as some have argued).

When it comes to the conjunction hypothesis, there are two major contenders in the literature and in

planetarium shows. The first was the one connected to Johannes Kepler, that of the conjunction of Jupiter and Saturn in 7 BCE (actually, there were three in that year), and it has the longest and most interesting history. The second hypothesis concerns Jupiter and Venus, along with the star Regulus; this idea came into notice first in the 1960s and then greater mainstream prominence in the 1980s. The latter hypothesis will be considered first, and more attention will be given to the Jupiter-Saturn conjunction(s) since it has had the greater consideration (including a partial endorsement by the Benedict XVI, Pope emeritus)⁶⁰ and diversity of thought.

The conjunction of Venus and Jupiter was first brought to light by Robert Sinnott,⁶¹ a writer for the popular astronomy magazine *Sky & Telescope*, but the first person to consider it in depth was a meteorologist and follower of the evangelical Hebert Armstrong, Ernest Martin.⁶² The events in question take place between 3 and 2 BCE, which is outside the allowed time period for Jesus's birth; because of this, Martin's writing on the subject is quantitatively more on arguing for a different chronology in which King Herod dies (not before the Passover in 4 BCE but in 1 BCE). So Matthew's story is historical, so long as you change history! The arguments of Martin and others will not be considered here as they have not been convincing to historians.

Nonetheless, let us suppose the dating of Herod's death is not well-known. Ignoring the chronological issue, what was it that was so important about this

⁶⁰ Joseph Ratzinger, *Jesus of Nazareth: The Infancy Narratives*, pp. 97-101.

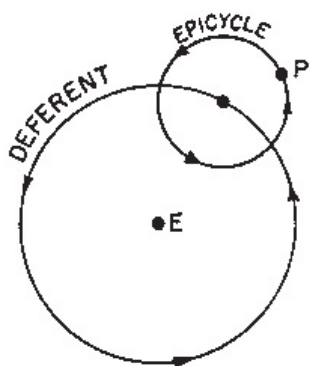
⁶¹ Robert Sinnott, "Thoughts on the Star of Bethlehem", *Sky & Telescope* 36 (1968): 384-386.

⁶² Ernest Martin, *The Star that Astonished the World*.

conjunction? What is interesting is that it combined two symbols in the sky with kingship: Jupiter, the king of the gods, and the star Regulus, a name derivative from the Latin for king (*rex, regis*) found in the constellation Leo, which Martin argues was associated with the Jews. After a conjunction with Venus, Jupiter hangs about Regulus for several months, appearing to go backward and forward around the star, perhaps highlighting what is to come. Then, in 2 BCE, Venus and Jupiter had such a close conjunction that they could have appeared to be a single object. With some effort, evidence from the Bible is used to show that Leo is the constellation associated with the Jews, making conjunctions there important for that people or their religion. The first conjunction is also visible on the eastern horizon before sunrise, thus in line with the Star seen “at rising”. As for the business of the Star “going before” the Magi, Martin either says it was symbolic or that Jupiter alone was seen southward, similar to what has been said of comets and novae in the previous chapters. As for the Star standing, this is in reference to Jupiter reaching the apparent stationary point relative to the other background stars in the sky exactly on December 25, 2 BCE. Merry Christmas!

Because the point of a planet being referred to as stationary or in retrograde will come up several times more, let me explain what is meant here. To observers on Earth, the planets of the solar system are star-like objects that appear to move against the background of the other stars; the name ‘planet’ comes from the Greek for wanderer as they appeared to not just move but to move in strange ways. Usually the planets will travel westward relative to the background constellations, but occasionally they slow down, stop, travel backward to the east (what is called retrograde motion) for a bit and

make a loop, stop again, and finally travel westward again. This odd motion we understand as due to an optical illusion because the planets go about the Sun at different rates and the Earth passes them from time to time. It is a similar observation when you overtake another car; at first the slower car is ahead and appears to move forward, but then as you begin to pass it seems to go backward for a bit. In ancient times, this was a significant mystery and one of the popular theories of the time was used by Ptolemy; this was an earth-centered (geocentric) system (actually, Ptolemy set the Earth a little bit off from the center of motion of the planets), and the planets went about on circles, but not so straightforwardly. In fact, the planets stayed on rotating circles whose centers went about in a circle, creating what is called an epicycle—a cycle on top of another cycle. As such, sometimes the planet's overall motion was backward compared to the motion of the larger cycle.



The basic geocentric model of Ptolemy. The Earth (E) is at the center of the deferent circle, while the Planet (P) rotates about on the epicycle. The center of the epicycle travels along the deferent circle. The combined motions of the planet on the epicycle and the movement of the epicycle give the apparent backward motion of the planet in the sky to earth-based observers every once in a while. This model of retrograde motion, with other complications such as the equant, lasted for centuries, though modified by Arab astronomers and finally abandoned during the Scientific Revolution.

This sort of system was apparently used in the ancient computer known as the Antikythera mechanism which used the eponymous epicyclic gearing to model planetary motion. While strange-sounding to moderns, the model was very successful in its day in predicting the future location of the planets, and even the solar system of Copernicus with the Sun in the center (heliocentric) was not superior.

This whole business is important to various Star researchers because they will talk about the retrograde motion of the planets as related to the “going before” of the Star, and the stationary point as related to it standing still. While Martin in his Jupiter-Venus hypothesis does not talk about retrograde motion in this way, others do, and more discussion will concern these.

With this preliminary background out of the way, let us consider how well the thesis actually fits the description of Matthew’s Star. On the point of the Star “rising”, there is nothing apparently at fault with the hypothesis. However, the text specifies a single Star, not two as Martin has it with the close conjunction of Jupiter with Venus. Perhaps it was Jupiter that was the important star, as others theorize, so let us accept this and simply move on to more important aspects. When it comes to the motion of the Star, Martin’s explanation fails in the same way as it does for novae and comets. Jupiter does not travel north to south, nor does any other planet; moreover, in Martin’s hypothesis, Jupiter is at a stationary point and not moving at all with respect to the stars. As for the condition of stopping, a stationary point is not something that can be seen happening in an instant, but instead a planet will appear to be stopped for days. Matthew talks about the Star going before and then stopping, a sequence of events taking place in the

couple hour walk to the small town where Jesus was; this specifies that the stopping happened in the short time it takes to travel from Jerusalem to Bethlehem, probably even less. Moreover, the text of Matthew uses a verb form (discussed in Chapter 7) that specifies an instantaneous stopping action, not one that could be spread out for days.

We also have no good explanation as to how the Star is “over where the young child was” given this hypothesis. In Martin’s reading, Jupiter happened to be over Bethlehem as seen from Jerusalem just before dawn, and that, argued Martin, is what Matthew meant. However, the text says that the Star came to stop over Bethlehem *after* it “went before” the Magi. For Martin, the events are happening at the same time—going before and standing over. These difficulties will be further explored more fully in Chapter 7, but the key word that upsets Martin’s hypothesis is ‘until’: the Star went before the Magi “until” it stood. Even without considerations of the Greek, we can see in English that the events of the Star moving and stopping are not simultaneous, and the stopping action took place after the movement of the Star.

But what about the possible symbolic importance of the events in the sky? Unfortunately, Martin gives no ancient evidence that a Jupiter-Venus conjunction would specify the birth of anyone; instead, cuneiform literature says that such a conjunction was a sign of war and hostilities toward the king, not the birth of one.⁶³ Nor is there evidence that Jupiter retrograding about Regulus had any positive meaning for eastern astrologers. Martin only has speculations. Had he actually looked into

⁶³ Hermann Hunger, *Astrological Reports* §§ 212, 448.

eastern records, he would have found that this very retrograde motion about Regulus by Jupiter was a sign of someone overthrowing the king, a terrible omen, rather than the birth of one.⁶⁴ If anything, the Magi should have been worrying about a civil war in their own land, not a baby in Judea.

As for the constellation Leo being the sign of the Jews, Martin looks to two verses from the Bible, Genesis 49:9 (“You are a lion’s cub, O Judah; you return from the prey, my son. Like a lion he crouches and lies down, like a lioness—who dares to rouse him?”), and Revelation 5:5 (“Then one of the elders said to me, ‘Do not weep! See, the Lion of the tribe of Judah, the Root of David, has triumphed. He is able to open the scroll and its seven seals.’”). Unfortunately, this has three obvious problems. One is that neither of these verses is about the stars or constellations, so the connection between Judea being like a lion (and what nation *doesn’t* want to have such a symbol for their own?) is not the same as Leo representing the nation. The second is that there is no reason why eastern Magi would be using the Bible for their astrological geography. Instead of going to an actual magian text, selective exegesis of the Bible is used to fill in the gap, but that is pointless unless it can be shown the Magi would have been doing the same thing. No evidence exists, leaving a connection between Leo and the Jews by the Persians as speculation, not fact. A third point worth mentioning is that we could have done this same exegesis for most any constellation we wanted to. Revelation 5:6 says Jesus is the Lamb, so perhaps the constellation of choice should be Aries the Ram? To see how suspect Martin’s exegesis is, one can turn to the

⁶⁴ Hunger, *Astrological Reports* § 279.

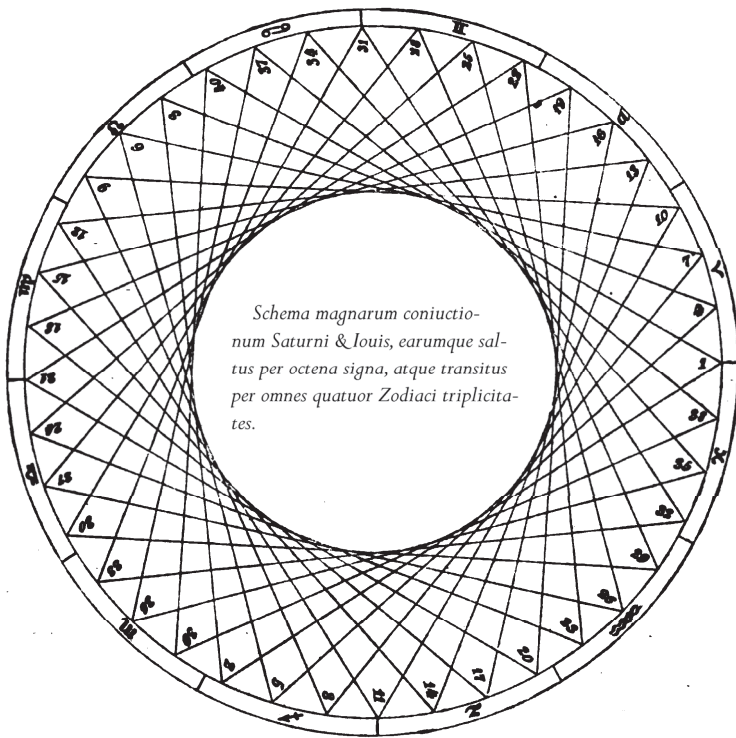
Talmud for a debate about astrology and Judaism. According to Rabbi Johanan, there is no *mazal* (astrological fortune, meaning, in the context, no planet or constellation) for Israel, but only for other nations that do believe in such things.⁶⁵ One cannot say that some particular star sign represents the Jews or the Jewish religion when we have such testimony denying any such connections. When European rabbis in the Middle Ages took up astrology, they also chose different constellations than Leo, such as Aquarius.⁶⁶ This will be discussed more below. As such, even if the Bible were used by the Magi to determine a constellation representative of Israel, there is nothing that shows Leo was the preferred interpretation.

Martin's entire case is constructed on a series of baseless speculations, many of which can be shown to be false. It doesn't help that the events are also outside of the plausible timeframe for Jesus's birth and do not fit Matthew's description of events. The problems with Martin's description have not even been fully explored, so how far off he is from actually explaining the Star will be shown even more starkly in Chapter 7.

Assuming the Jupiter-Venus conjunction does not fit the bill for astrological auspiciousness, it is now time to consider the longest-running conjunction connection to Jesus. While many associate it with Kepler, the idea actually goes back centuries before his time and to a different land. The thesis actually rests on the idea that the periodic conjunctions of Jupiter and Saturn, about 20 years apart, have an important effect on world affairs. Those effects are strongest when the conjunctions are

⁶⁵ Babylonian Talmud *Shabbat* 156a.

⁶⁶ Bernard R. Goldstein and David E. Pingree, *Levi ben Gerson's Prognostication for the Conjunction of 1345*, pp. 35, 37.



A zodiac map of the successive conjunctions of Jupiter and Saturn, starting at the first degree of Aries. Each conjunction is numbered up to 40, after which the conjunctions repeat. This image is reproduced from Johannes Kepler's *Mysterium Cosmographicum* (1596), and it was this astrological diagram that led to Kepler's first (though erroneous) epiphany into the structure of the solar system. The text in English reads "A schema of the great conjunctions of Saturn and Jupiter, and their leaps through eight signs, and crossing through all four quartiles of the Zodiac."

first seen in a new set of three constellations called a trine. Each trine is associated with one of the ancient four elements of earth, wind, fire, and water (and unfortunately not with heart), and the Jupiter-Saturn conjunctions tend to bounce around in a single trine for decades. When a conjunction happens for the first time in the Fiery Trine (Aries, Leo, and Sagittarius) this is the most powerful, and this happens only about every 960 years (or 800 when considering the effects of precession). Because such a cycle is nearly a thousand years, this system of astrology (the great conjunctions) became particularly interesting to Zoroastrians, the religion of the magi, who had an affinity for millennialism.

Does this mean then we are in luck? Can a great conjunction actually be said to have interested the Magi? Unfortunately, there are things we know about this hypothesis, and part of that is when this practice began. While it did originate in Persian territories, it did not happen until the Sassanid period in Persia which began in the third century CE. Then, kings such as Shapur I would send out scholars to gather up the scientific knowledge of the Greeks, Romans, and Indians; afterward, they used their astronomical and astrological works to create the great conjunction system.⁶⁷ At best, it comes several centuries too late. We also don't see this system applied to individuals until the rise of Islam.

The first time the great conjunction was connected to Jesus appears to be with the Jewish Persian astrologer Masha'allah in the 8th century, whom we know about concerning this subject from the 9th century Christian

⁶⁷ David Pingree, "Astronomy and Astrology in India and Iran", *Isis* 54, 2 (1963): 229-246; "Historical Horoscopes", *Journal of the American Oriental Society* 82, 4 (Oct-Dec 1962): 487-502.

astrologer Ibn Hibinta.⁶⁸ However, when Masha'allah did use this method, his timeframe was far different than what any modern researcher believes fits the lifetime of Jesus, and this is because he followed the method correctly and later scholars researching Jesus and his birth do not. As mentioned above, the most important conjunctions of Jupiter and Saturn take place in the Fiery Trine for the first time, and this happened in Leo in 26 BCE. According to our source, Masha'allah said Jesus was born in 13 BCE, though he also gives the more traditional date of 2 BCE.⁶⁹ Later on, a Muslim astrologer famous in medieval Europe, Abu Ma'shar, would fit a great conjunction closer to the time of Jesus's birth, the now-famous one in 7 BCE. This idea would later travel to Europe, and it would be the ultimate source for Kepler's speculations on the subject.⁷⁰

So while the idea of a great conjunction has considerable antiquity in its association with Jesus, it is not ancient enough by a significant margin and it doesn't fit the astrological theory correctly. To put it another way, the Magi would not have been interested in such conjunctions because no one believed they had such importance until centuries later. In the same way, scientists didn't know about the importance of the Higgs boson until it was theorized in the 20th century; no one would expect Isaac Newton to incorporate such a hypothesis into his work, so we would not expect astrologers to incorporate theories before they were invented. Add in the fact that the hypothesis does not naturally lead to the 7 BCE conjunctions, and the

⁶⁸ Kennedy, Pingree, *The Astrological History of Masha'allah*.

⁶⁹ Kennedy, Pingree, *op. cit.* p. 72.

⁷⁰ Adair, "The Star of Christ in the Light of Astronomy", *Zygon* 47, 1 (March 2012): 7-29, esp. pp. 9-12.

chances of the Magi caring about the astrological events then are very small. What we know about the Zoroastrian religion before then is also damaging, but that will be discussed in Chapter 8.

So the actual astrological theory that would give weight to a conjunction of Jupiter and Saturn comes too late, but most Star researchers are not aware of the theory of great conjunctions anyway, and their arguments are independent from this. What is mostly stated is that Jupiter is the planet of kings, while Saturn is the planet of the Jews. Also, it is alleged that the constellation of Pisces (the two fish) was similarly associated with the Jews. On this line of reasoning, the Jupiter-Saturn conjunction stated there was a new, royal event concerning a Jewish king, supposedly his birth.

Several attempted lines of evidence are brought in to support the stellar connections, but the arguments that prove to be particularly faulty concern the constellation of Pisces. The only source provided for the link between this sign and Judaism, Judea, or the Jewish people is a late medieval rabbi, Issac Abrabanel of Spain, with some suggesting that he inspired Kepler.⁷¹ Abrabanel writes towards the end of the 15th century, so he is literally 1500 years too late to have any likely knowledge of archaic astrological beliefs.⁷² Rather, the rabbi shows no signs of holding onto an ancient tradition as he needs to argue for his association between Pisces and Judaism and his chronology to have things match up with the

⁷¹ Hughes, *The Star of Bethlehem: An Astronomer's Confirmation*, pp. 68, 96, 184-186; Roy Rosenberg, "The 'Star of the Messiah' Reconsidered", *Biblica* 53, 1 (1972): 105-109. Hughes and others have spelled the rabbi's name 'Abarbanel' among other variants.

⁷² Molnar, *The Star of Bethlehem: The Legacy of the Magi*, pp. 27-8 give a few arguments not to consider Abrabanel's hypotheses as fitting to classical times.

birth of Moses. His position is further undermined by other rabbis from around the same time and earlier in the High Middle Ages who all contradicted each other on this very point. Some thought the constellation of the Jews should be Taurus, others Aquarius, and some the entire Earthly Trine (Taurus, Virgo, Capricorn). This allowed another famous Jewish scholar of the early modern period, Azariah dei Rossi, to argue that using astrology is a futile pursuit (and with it, predicting the coming of the Messiah).⁷³ So why trust Rabbi Abrabanel any more on Pisces than Levi ben Gerson on Aquarius as the constellation of the Jews?⁷⁴ In fact, why assume they know anything about what non-Jews a millennium ago thought on occult matters? Importantly, and perhaps unknown to most Star researchers, Abrabanel and these other rabbis were using the great conjunction hypothesis themselves, so all their speculations come after its introduction into Europe in the 12th century or so. It has nothing to do with ancient beliefs but medieval concerns. There is nothing worth calling evidence that favors Pisces as the constellation representative of the Jews to eastern soothsayers.

But what about Saturn? Can any evidence be provided that this planet was linked with Judaism? Here two major sources are considered, and both are ancient. First is the prophet Amos from the Old Testament, one of the less famous Hebrew prophets but perhaps the oldest. Second is the Roman historian Tacitus from the second century CE. Let's consider them in order.

⁷³ Azariah ben Moses dei Rossi, Joanna Weinberg, *The Light of the Eyes*, pp. 548-550.

⁷⁴ Bernard R. Goldstein and David E. Pingree, *Levi ben Gerson's Prognostication for the Conjunction of 1345*, pp. 31, 37.

The verse in question for the Old Testament is Amos 5:26: "You shall take up Sakkuth your king, and Kaiwan [Saturn] your star-god, your images, which you made for yourselves." The verse is cited by Peter in Acts 7:43, so it was not obscure in the first centuries of Christianity. However, the context of the passage is not about Saturn representing the Jews or the planet as worshipped in Judaism. It was about the Jews failing to worship correctly; worshipping the star-god Kaiwan was sinful, not kosher practice. In fact, God specifies in the following verse how this will be a sort of apostasy and leading to exile, a punishment. Earlier in the same chapter (vv. 8-9), God is said to control the stars and the coming of day and night; he is the Lord of Hosts (v. 27, meaning he rules the sky). The point is that Kaiwan/Saturn is *not* God, who is far beyond any of the stars in the sky, and God has cursed his people to worship an idol instead. This means to use Amos 5:26 as support for Saturn being the Jews' planet is to assume precisely the reverse of what the text reads; the exact opposite conclusion ought to be reached. If one wonders why the star-god Kaiwan/Saturn was specified rather than any other planet, there may be a simple reason: in Hebrew, the word for 'star' is *kockab* (כוכב), so the author may well just be using alliteration between *kockab* and Kaiwan and nothing more.

As for Tacitus, his talk of Saturn and the Jews is related to various speculations about why they had their Sabbath on the day of Saturn (Saturday).⁷⁵ In actuality, the entire line of reasoning that Tacitus reports (and which he does not actually believe) is anachronistic. It was not until after the calendar reforms of Julius Caesar

⁷⁵ Tacitus, *Histories* 5.4. Cf. Dio, *Roman History* 37.16.2,4; 49.22.4; 66.7.2; Frontinus, *Stratagems* 2.1.17.

and perhaps not until his adopted son Octavian (Caesar Augustus) came to power that the weekdays were connected to the planets; the seven-day week was itself a product of Julius Caesar's calendar.⁷⁶ In the East, there were also no seven planet-named days of the week, making the speculation even more anachronistic and culturally irrelevant.⁷⁷ Even ignoring this, Tacitus does not say anything about the planet of the god Saturn representing the Jews. If anything, people were saying that Jupiter was the god of the Jews, not Saturn.⁷⁸ The entire line of Roman guesswork is very much out of place and has nothing to do with what anyone in the Middle East would have thought. Tacitus provides no evidence at all for any connection between the Jews and Saturn, at least for Eastern magi.

With these particular sources found to be wholly lacking in their utility to support the premise, a more general point ought to be observed. One should consider why anyone in the East would associate one of the planets with the Jews. There are only seven planets in the ancient system (Sun, Moon, Mercury, Venus, Mars, Jupiter, Saturn), and there are lots of potential regions, kingdoms, or civilizations that could have something represent them; the same can be said for the twelve zodiac signs. But why would we think that Babylonian or Persian astrologers would have cared so much about the small area between traditionally more powerful regions (Egypt, Phoenicia, and Syria/Assyria) and assign it a

⁷⁶ Barton, *Ancient Astrology*, p. 52; Roger Beck, *A Brief History of Astrology*, p. 73; Eviatar Zerubavel, *The Seven Day Circle: The History and Meaning of the Week*, pp. 15f.

⁷⁷ Zerubavel, *The Seven Day Circle*, p. 14.

⁷⁸ Augustine, *Harmony of the Gospels* 1.30. Augustine also denies that the Jews worship Saturn multiple times. Cf. *Contra Faustum* 18.2,5; 20.13; *Harmony of the Gospels* 1.42.

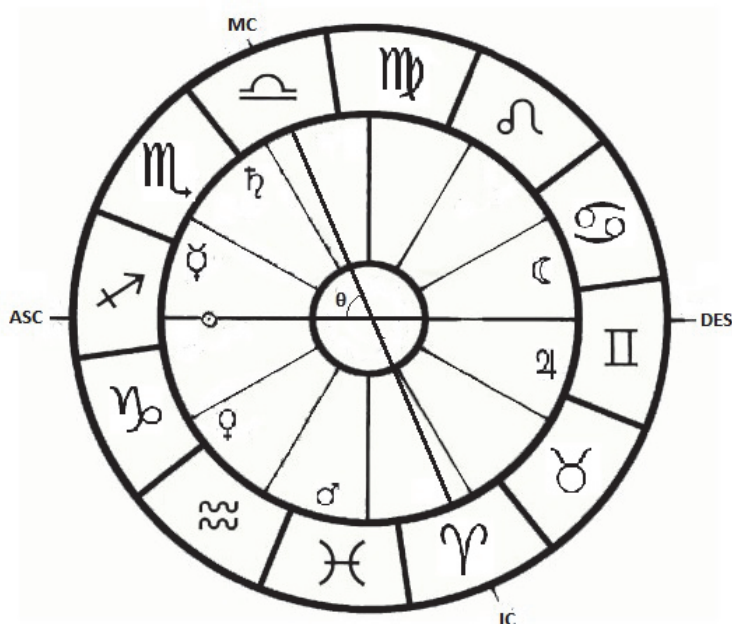
planet? The importance of Judaism seems to be much more in the eyes of the modern Star researcher, and that has become anachronistic when placing the same importance of this religion among Eastern sages. So perhaps it's not that telling that no Eastern record has been put forward to show that *any* star, constellation, or planet was specifically assigned to the Jews or Palestine.

This problem then impacts upon the issues in the next section about horoscopes.

Part B: Horoscopes

So far, Star of Bethlehem hypotheses have looked at a singular object or conjunction. Instead, the horoscope that astrologers have used for millennia consider the location of all the planets as well as the zodiac constellations (or ‘signs’). The oldest horoscopes come from the fifth century BCE in Babylonia. These horoscopes come in ancient cuneiform writing and date up to the first century BCE.⁷⁹ However, almost all our knowledge about using horoscopes comes from Greek sources, a system that came to exist after the conquests of Alexander the Great and greater intellectual contact between East and West. It would seem that the methods of Western horoscopic astrology began in the second century BCE in Egypt and then spread from there. The oldest known Greek treatise on the subject is that of Nechepso and Petosiris (pseudonymously using the names of ancient Egyptian figures) which exists only in fragments, while the oldest extant book on how to do horoscopy comes from Marcus Manilius from around 14 CE, though the work is unfinished. Later, we have other astrologers that detail their ideas about astrology and horoscopes, notably Claudius Ptolemy of Alexandria, Egypt, his near-contemporary Vettius Valens in Antioch, Syria, and the Christian convert Firmicus Maternus.

⁷⁹ Rochberg, *Babylonian Horoscopes*.



Example horoscope with cardinal points and positions of planets

Our Western sources provide many ways to look at the relative position of the planets in the sky, and as discussed in Chapter 2 it makes figuring out anything of what would have been meaningful to an ancient astrologer nearly impossible. However, the most thorough attempt at this in recent years has been by the retired astronomer Michael Molnar.⁸⁰ He uses many of the sources mentioned above and others in his reconstruction of what Jesus's horoscope would have been, along with his ideas of what in particular was "his Star

⁸⁰ Molnar, *The Star of Bethlehem: The Legacy of the Magi*.

in the east” (Matt 2:2). His discussion of ancient astrological practices is probably the best introduction on the subject with its technical details communicated in an understandable way; on the other hand, his treatment conceals the difficulty of interpreting a horoscope, especially how ambiguous and contradictory our textual sources are in figuring out how charts were used by astrologers.

However, Molnar is not the first to create horoscopes for Jesus. The earliest extant one that we have comes from the 15th century by a French cardinal, Pierre d’Ailly, and many others came to exist before the 19th century.⁸¹ It has also been attempted by one Bible scholar in the early 20th century whose date for Jesus’s birth was almost the same as Molnar’s,⁸² though the latter seems to have worked without knowing this. There are also a plethora of modern Jesus horoscopes crafted by professional astrologers.⁸³ As you can guess, these horoscopes do not match each other in time but are all somehow interpreted favorably. Hitherto I will focus on Molnar’s version since his work is the most popular of the lot, the most thorough, and the best as an exemplar of the difficulty of the approach.

The date for the Star of Bethlehem that Molnar comes up with is April 17, 6 BCE, focusing on the

⁸¹ Ornella Pompeo Faracovi, *Gli oroscopi di Cristo*.

⁸² Heinrich Voigt, *Die Geschichte Jesu und die Astrologie: Eine religionsgeschichtliche und chronologische Untersuchung zu der Erzählung von den Weisen aus dem Morgenlande*.

⁸³ Max Tschudin, “Das Horoskop von Jesus-Christus—ein Versuch”, *Astrologie Heute* 52 (Dec/Jan 1994/5): 8-11; J.-C. Weiss, “2000 Jahre Jesus Christ”, *Astrologie Heute* 52 (Dec/Jan 1994/5): 12-16; Patrice Guinard, “L’étoile de Bethléhem: Un scénario organize par des astrologues”, *C.U.R.A.* (2002) at <http://cura.free.fr/16christ.html>; Dieter Koch, *Der Stern von Bethlehem* (see the appendices).

morning sky. He comes to this date based on not just a close conjunction of Jupiter and the Moon, but the Moon passed in front of Jupiter as seen from Earth, blocking it, which is called an occultation. This event happened in the early dawn rays of the Sun, which would connect with the dawn rising of the Star as implied in Matt 2:2 (Molnar particularly believes the text uses a technical term for a heliacal rising). Moreover, the occultation happened in the constellation of Aries, which Molnar argues was the constellation associated with Palestine. The other planets are also alleged to be in a formation that was particularly auspicious when considering Western texts on astrology.

However, the first major roadblock to considering the Jupiter-Moon occultation as propitious comes from Eastern records, the ones that ought to be more telling of what people in that region believed. Instead of such an occultation acting as a sign of the birth of a king, it was considered the opposite—the death of a king!

When the Moon occults Jupiter (*Sagmígar*), that year a king will die (or) an eclipse of the Moon and Sun will take place. A great king will die. When Jupiter enters the midst of the Moon there will be want in Ahurrû. The king of Elam will be slain with the sword: in Subarti... (?) will revolt. When Jupiter enters the midst of the Moon, the market of the land will be low. When Jupiter goes out from behind the Moon, there will be hostility in the land.⁸⁴

This comes from the *Enuma Anu Enlil*, the most important of the ancient Babylonian astrological texts

⁸⁴ S. J. Tester, *A History of Western Astrology*, p. 13. R. Campbell Thompson, *Reports of the Magicians and Astrologers of Nineveh and Babylon in the British Museum* (London 1900) II, 192, p. lxvii.

and was likely still used by horoscopic astrologers in the East.⁸⁵ We can see that the beliefs encapsulated in this approximately 3000 year old text were on the minds of the East's intelligentsia. The panic that such an occultation caused was so great that we have letters between scribes in the 7th century BCE warning of it in the future and what should be done to protect the monarch.⁸⁶ The trouble this sign wrought was so great that even having Jupiter close to the Moon could cause stress about the potential death of the king.⁸⁷

Molnar defends his premise that occultations were auspicious to the Magi, and thus consistent with Matthew's story of an all-important stellar event, from these records by arguing that the Western practices he was reliant upon had replaced the more archaic ones used before in Mesopotamia.⁸⁸ Unfortunately, the only evidence that he provides says nothing of the sort, only that natal (that is, horoscopic) astrology was done by some in Babylonia.⁸⁹ In fact, as mentioned in Chapter 2, we can actually see from the cuneiform horoscopes right up to the year 69 BCE (well into the time when Hellenistic methods had come into existence) that they are significantly different, missing features that no Western horoscope would ever leave out. Molnar is simply stuck with the assumption that the astrology he examines has anything to do with that done by first century Zoroastrians. Not only does it seem unlikely that the Magi were

⁸⁵ Rochberg, *Babylonian Horoscopes*, pp. ix-16.

⁸⁶ Hermann Hunger, Simo Parpola, "Bedeckungen des Planeten Jupiter durch den Mond", *Archiv für Orientforschung*, 29/30 (1983/4): 46-49; Hunger, *Astrological Reports* § 100.

⁸⁷ Ulla Koch-Westenholz, "The Astrological Commentary *Šumma Sin ina tamurtišu* Table 1" in Rika Gyselen, ed., *La Sciences des Cieux: Sages, Mages, Astrologues*, p. 154.

⁸⁸ Molnar, *The Star of Bethlehem: The Legacy of the Magi*, p. 39.

⁸⁹ Strabo, *Geography* 16.1.6.

practicing astrology in the same way as did the Greeks and Romans, given the current evidence, but in Chapter 8 I will prove that the Magi did not use such methods at all.

But even if we do stick to ancient Greek and Latin texts on the subject, there are plenty of problems for Molnar's thesis. That Aries has influence over Palestine exists in only one astrological geography, that of Ptolemy in around 150 CE, but he also uses that same constellation for what is modern-day France, Belgium, Germany, and Britain.⁹⁰ As pointed out in Chapter 2, all the other astrological geographies contradict Ptolemy, and not even a majority agrees on which of any constellation has influence over any particular region. In fact, most astrological geographies do not specify Palestine at all, making an already contradictory record ambiguous. Molnar also cites the astrologers Valens and Manilius for support, but the former is ambiguous at best and the latter is misread and in fact contradicts the premise Molnar defends;⁹¹ there is no support to be found here that Aries was the constellation of the Jews and thus not evidence for what the Star of Bethlehem may have been.

Molnar also considers another line of evidence for Aries acting as the sign of the Jews, a set of coins from

⁹⁰ Ptolemy, *Tetrabiblos* 2.3 (62, 73-74).

⁹¹ Valens, *Anthology* 1.2; Manilius, *Astronomica* 4.620-7, 797-8. The Loeb English translation of Manilius includes a map of the world with its astrological geography that shows that Aries was not the influencing sign of Israel. As for Valens, his mention of Coelē Syria may refer not to southern Syria but northern as it was named in the late second century CE after the reforms of Emperor Septimus Severus. But even assuming otherwise, Coelē Syria is not the Holy Land (it better fits modern-day Lebanon), and Phoenicia may more accurately fit Israel to Valen's geography.

the city of Antioch on the Orontes,⁹² the third largest city in the Mediterranean World in the first century and capitol of Roman Syria. The coins in question come from the first half of the first century CE and on one side include Aries. Molnar conjectures that these coins were minted upon the annexation of Judea into the Province of Syria in 6/7 CE, and the coins use Aries to symbolize the recently acquired region. In Molnar's theory, the constellation was chosen because it was the well-known astrological sign of the Jews and their land, and thus Aries was the constellation the Magi would have been looking to, and so this is where to search for the Christmas Star.

However, this entire line of speculation is simply in contradiction of what the oldest coin says; the coin declares it is "of the metropolis of Antioch", mentioning nothing about Israel. Using a Jewish symbol would be very odd. It would be like the US state coins including the Statue of Liberty labeled 'Michigan' or Euro coins with an Italian minting and label celebrating France's win in the 1998 World Cup. Moreover, the earliest coins were civic coins, for city purposes, not the legate class of coins that were ordered by the provincial governor, so there is no reason to think they have anything to do with celebrating the annexation of Judea. In fact, Molnar's source for dating the earliest coin puts it the year *before* the events that caused Israel to become part of Syria.⁹³ Assuming Molnar's source is wrong, there is still the difficulty that these sorts of coins were minted in Antioch with Aries on the reverse for centuries after the annexa-

⁹² Molnar, *The Star of Bethlehem: The Legacy of the Magi*, pp. 52-53.

⁹³ George MacDonald, "The Pseudo-Autonomous Coinage of Antioch", *The Numismatic Chronicle* 4 (1904): 110-111.

tion and even after Palestine was in a different province; they also include the city goddess, Tyche, further indicating that the coins were for and about Antioch.⁹⁴ Molnar even provides examples of coins from Antioch with the same astrological symbol during times when Judea was a client kingdom and not part of any Roman province.⁹⁵ There is simply nothing here but an erroneous use of speculation to generate data. Far more likely, Aries was representative of the city because of the horoscope of the founding of Antioch; according to city tradition, the town was founded in late April (22 Artemisius in the old Macedonian calendar) in 300 BCE around dawn,⁹⁶ and the Sun and ascendant of the horoscope would have been in Aries.

There are other arguments Molnar has for Aries being Israel's star sign,⁹⁷ but they are also speculative. For example, he claims that common knowledge of Emperor Nero's horoscope was what caused people to believe he would rule in the East and conquer Jerusalem, but this is almost impossible since emperors rarely published their horoscopes for fear they be used against them. Moreover, the theory Molnar uses is dependent still on Ptolemy's astrological geography, thus Molnar's "evidence" isn't even independent confirmation of what sign related to Israel. Molnar also relies on details from

⁹⁴ Examples include, from the British Museum Catalog (BMC), 4, 21, 446, 527, 535.1, 537, as well as Prieur 167.

⁹⁵ Andrew Burnett, Michael Amandry, P. P. Ripollés Alegre, and Marguerite Spoerri Butcher, *Roman Provincial Coinage* 4286-4287. On the status of Israel at this time with King Agrippa II and a Roman procurator for tax purposes, see Josephus, *Jewish Antiquities* 20.104, 137-138.

⁹⁶ Malalas, *Chronographia* 8.12; Sacha Stern, *Calendars in Antiquity: Empires, States, and Societies*, p. 243 n. 27.

⁹⁷ Molnar, *The Star of Bethlehem: The Legacy of the Magi*, pp. 104-116.

Suetonius on Nero, perhaps the worst source for this emperor—including the point that Molnar needed confirmation on.⁹⁸ Molnar also thinks that the astrologer Firmicus in the fourth century had Jesus's actual horoscope but didn't reveal this knowledge even after becoming a Christian. Unfortunately, the details of the horoscope mentioned by Firmicus don't match Molnar's horoscope for Jesus, while it best matches that of Caesar Augustus.⁹⁹ Also, considering that early Christians could not even agree on the year Jesus was born, or the day, it is baffling that there was documentation in the fourth century that would have given the correct time of birth to the hour!

This is sufficient to show that there is no evidence, besides Ptolemy's contradictory statements, to support Aries as the constellation of Judea. The same can be said for using ancient records to support any sign as that of Israel in the sky, and as stated in the previous section of this chapter it is also strange to think astrologers could care enough to give this one region such prominence in their astrological geographies; the importance of Judea is an artifact of the importance it has to modern researchers.

So far we haven't even gotten much into interpreting Molnar's horoscope for Jesus and how it had all sorts of auspicious characteristics for the person born under those stars. Sure, the occultation of Jupiter by the Moon

⁹⁸ K. R. Bradley, *Suetonius' Life of Nero: An Historical Commentary*, p. 247; Andrew Wallace-Hadrill, *Suetonius: The Scholar and his Caesars*, pp. 63-64; Barry Baldwin, *Suetonius*, pp. 174-180. B. H. Warmington, *Suetonius: Nero*, pp. 76-78.

⁹⁹ Firmicus, *Mathesis* 4.3.9 says that the moon is full and waxing, but Molnar's Jesus horoscope has a New, waning moon. Similarly Jupiter was not in its sign of exaltation, nor is it in trine aspect with the Sun.

was seen as terrible in the East, but what about the rest? Unfortunately, you can see just how easy it is to re-interpret the same horoscope into something terrible. For example, the Moon is approaching the Sun and is close enough for a conjunction by ancient astrologers, which is said to bring on sickness and insanity;¹⁰⁰ in addition, the Moon moves away from Saturn, bringing on elephantiasis.¹⁰¹ Considering Mercury, because it and Saturn were in or near feminine signs (Taurus and Pisces), the male is predicted to become a eunuch or prostitute, which is exacerbated by the Moon moving away from Venus towards Mercury.¹⁰² One can go on, using a variety of sources and potential correlations from those works to interpret a given horoscope any way that is desired, which is entirely consistent with the results from studies of modern astrologers as mentioned in Chapter 2. That makes the entire method worthless.

But what about using a horoscope in general? When considered more carefully, the process of using the horoscope to predict the birth of anyone is completely backward. A horoscope is drawn up based on the location and time of a birth, and from the chart you predict things about that child's personality and future. In other words, horoscopes are made *after* a birth, not before, and they do not predict births. So even assuming that Western astrology was practiced by the magi (it wasn't), and the Greek and Latin records were not contradicting each other (they do), and our ability to interpret them was possible and unambiguous (it is certainly not!), it wouldn't matter anyway because the entire method is being done wrong right from the

¹⁰⁰ Firmicus, *Mathesis* 4.5.1.

¹⁰¹ Firmicus, *Mathesis* 4.9.6.

¹⁰² Firmicus, *Mathesis* 3.9.1; 4.13.1-2.

starting gate. Making something more and more wrong doesn't suddenly make it right!

At this point, it may be worthwhile switching from considering how a horoscope could have been interpreted to how the theory could fit Matthew's description. There is also trouble at the start because Molnar's occultation of Jupiter was happening so close to the Sun it would have been impossible to see. While the astrologers could have calculated the event, the Gospel does state that the Magi had seen the Star. But let us instead consider the aspects of horoscopic theories rather than Molnar's particular version.

If, in fact, we can get astrology to fit the details of Matt 2:9, then it may encourage us to keep the planetary interpretation hypothesis alive. Again, the biggest proponent for this in detail is Michael Molnar who sees Matthew using actual, technical jargon from astronomers and astrologers. For example, Molnar says the word for "going before" in Matthew is the same or related to the one used for retrograde motion, *proegeseis* (προηγῆσεις); similarly, "stood" is related to the word for the stationary point of a planet.¹⁰³ On Molnar's reading, Jupiter is the Star which is first in retrograde motion and then reaches its stationary point while the Magi are on the move to Bethlehem.

Unfortunately for him, his arguments about terminology show that, with all due respect, he doesn't know what he is talking about. Already in the published literature has a biblical textual critic, J. Neville Birdsall, shown that Molnar is completely wrong on all substan-

¹⁰³ Molnar, *The Star of Bethlehem: The Legacy of the Magi*, p. 90.

tive points when it comes to “went before” and “stood”.¹⁰⁴ In fact, it seems Molnar cannot tell the difference between a noun and a verb in Greek, let alone details of conjugation and declension. Molnar says that *proegeseis* means “to go before”, but the term is not a verb but a noun (and he provides the plural rather than the singular, *proegesis*), and even so it’s not the word used by Matthew or even closely related to it (see below). There is simply no correlation between the terms used by Matthew about the Star’s motion and the terms used by astrologers.

However, Molnar is not the only one to argue that the terms are about planetary motion, and one other person in particular is even worse in his understanding of classical languages. A former Jesuit astronomer at the Vatican Observatory, the Hungarian priest Gustav Teres argues similarly to Molnar, though the two seem to have concluded independently.¹⁰⁵ But his reasoning shows how uninformed he is. For example, the term used by Matthew for the Star “going before” the Magi, *proago*, is claimed to be a different verb, *proegeomai*, as Molnar had done, minus his noun-verb confusion. The verbs are not related, and most of their similarities come from their prefix (*pro-*) and the rules for how verbs are conjugated. The root verbs are *ago* (ἄγω) and *hegeomai* (ἡγέομαι), which are obviously different, as noted by Birdsall. Teres also claims that the verb *proegeomai* is in the aorist imperfect tense. If you are a classicist, you are probably doing a double-take with that statement. Everyone else should as well because there is no such thing; Teres has

¹⁰⁴ Owen Gingerich, Michael Hoskin, David Hughes, and J. Neville Birdsall, “Review Symposium: The Star of Bethlehem”, *Journal for the History of Astronomy* 33, 4, (2002): 386-394.

¹⁰⁵ Gustav Teres, *The Bible and Astronomy*, p. 20.

conflated two different verb tenses. In Latin, the equivalent would be the perfect imperfect—what I call the round square—tense. If the Jesuits are teaching their students Greek, not all of them are remembering their lessons. Teres also tries to look at a Syriac translation of Matthew (which is translated from the Greek) because it is then in the same language family as Hebrew and hence closer to Matthew's original Gospel. Not only is there no solid evidence that Matthew was originally in Hebrew (his use of Mark in Greek belies the hypothesis), but the method is absurd; it's like translating the current book into Hebrew to know what it was like in the hypothetical original Arabic. This is nothing but a ramshackle intellectual edifice built upon foundations of quicksand.

So, the linguistic argument isn't working out. But even contextually there is nothing to work with. In the case of retrograde motion, which is alleged to conform to Matthew's statement that the Star "went before" the Magi, the planets move against the background sky from east to west. However, the Star is said to "go before them", the Magi, who are traveling from north to south. To use terminology from physics, Matthew has established his reference frame, and it is orthogonal to the one concerning the motion of planets. As for standing still, this is something so slow and imperceptible that it takes days to notice the motion again. The astronomer and astrologer Ptolemy says just as much: "stations cannot be fixed at an exact moment, since the local motion of the planet for several days both before and after the actual station is too small to be observable."¹⁰⁶ There is no way the Magi could have specified the Star had stood

¹⁰⁶ Ptolemy, *Almagest* 9.2.

still in the timeframe of their walk to Bethlehem from Jerusalem, a matter of perhaps a couple of hours. This point can't be emphasized enough, since it effectively pulls the rug from under the feet of any physical explanation. One of the few claims of Matthew with regard to the Star is clearly incoherent with respect to the laws of nature. But again, even ignoring this, the reference frame is that of the Magi, and even if a planet was at its station it would still travel with the rest of the stars in the sky from east to west, never stopping relative to the ground.

While in Chapter 7 the details of the motion of the Star will be specified, it should be clear from the above considerations that normal planetary motion has no connection with the description of Matthew. It is only by ignoring the context that it even seems plausible, but even a cursory look shows the significant observational impossibilities. I haven't yet considered what is meant by "stood over" where Jesus was in this chapter, leaving that discussion for Chapter 7.

Let me lay out, then, exactly where this leaves us. The astrological theory of the Star of Bethlehem comes up against two major problems: there is no way to actually show some particular combination of planetary positions was in fact considered auspicious to the magi, and the details of planetary motion do not fit Matthew's context. This should be sufficient for a death sentence to be administered to these sorts of speculations.

Chapter 6: What (Other) Stars do Spangle Heaven with Such Error? Last Ditch Efforts

Back in the 1950s, journalist and Bible enthusiast Werner Keller observed “[a]nything that has ever moved across the canopy of heaven, as well as much that has only existed in men’s imaginations, has been dubbed the Star of Bethlehem.”¹⁰⁷ And so, there are many other hypotheses that have been floated by various proponents that have neither found traction within the public sphere nor within the intellectual literature. However, both for the sake of completeness as well as to show that the other avenues proposed will not work better than the main ones discussed in previous chapters, I will talk about some of the odd ideas, including the outright bizarre yet non-miraculous.

One odd idea is that the Magi discovered that one of the stars in the sky was a variable star.¹⁰⁸ A number of the stars in the night sky change in brightness for a variety of reasons, most commonly because there is actually a star pair; two stars orbit about their center of gravity and can periodically block the light of the other, making the combination seem dimmer. One such star is Algol, situated in the constellation of Perseus where the head of Medusa is held. I have personally observed this dimming, but the dimming and return to full brightness

¹⁰⁷ Werner Keller, *The Bible as History*, p. 325.

¹⁰⁸ Costantino Sigismondi, “Mira Ceti and the Star of Bethlehem”, *Quodlibet: Online Journal of Christian Theology and Philosophy* 4, 1 (Winter 2002).

takes several hours and may not be noticed if one isn't aware of its variations. There are three major problems with such a theory: there is no reason to think a variable star would translate into "a king is born"; no variable stars are recorded by ancient astronomers (the star catalog of Ptolemy, inherited from Hipparchus, shows no sign of such knowledge and are not unambiguously known until the Scientific Revolution); and such a star cannot move in the way Matthew describes. The variable star hypothesis is basically just as flawed as the nova hypothesis.

Another suggestion is the discovery of a planet, namely Uranus.¹⁰⁹ But this has the same flaws as the variable star hypothesis: no reason to think it connects to the birth of a king, Jewish or otherwise; it doesn't move like Matthew's Star; and there is no record of the planet being discovered until the advent of the telescope, such as the one used by William Herschel in 1781. Uranus can enter the lowest reaches of what is naked-eye observable, but it is still almost impossible to find unless you know where to look, something that didn't happen until it was discovered.¹¹⁰ To give it the needed astrological power (and this is also the case for the variable star thesis), it is connected to the Jupiter-Saturn conjunctions in Pisces where Uranus was residing at that time. But this assumes the astrological importance of that conjunction, something found to be wanting, as shown in Chapter 5, Part A.

¹⁰⁹ George Banos, "What the star of Bethlehem Uranus", *Astronomy Quarterly* 3, 12 (1980): 165-168.

¹¹⁰ If you know where to look, you can see very dim objects. I have seen Venus during the day, but I had to know almost exactly where to look.

Instead of planets, asteroids have been alleged to have been discovered. Little more needs to be said: there is no evidence that any were observed then; there is no reason to think it means “king of the Jews was born”; and the motion doesn’t fit what Matthew describes.

Two more diffuse light sources have also been considered: the aurora borealis (or Northern Lights) and the zodiacal lights.¹¹¹ The former is due to high-energy particles from the Sun interacting with the magnetosphere of the Earth. Though most prominent in the arctic, they can be seen much farther south, even to states such as Texas. The zodiacal lights are due to sunlight bouncing off dust in the solar system that is concentrated in the plane of all the other planets going about the Sun;¹¹² it can become a diffuse pillar of light at dawn and dusk. The former light source doesn’t fit well since, if anything, it will be seen in the north, so it won’t rise in the east or even be seen in the south towards Bethlehem. As for the zodiacal lights, the oldest observations come from Muslims, and otherwise it was unknown before.¹¹³ Like the aurora borealis, the lights cannot “go before” anyone, nor arrive at a place, stop and then stand over a locale. The main proponent of the zodiacal light hypothesis, the late Austrian astronomer

¹¹¹ Konradin Ferrari d’Occhieppo, “The Star of Bethlehem”, *Quarterly Journal of the Royal Astronomical Society* 19 (1978): 517-520; “Neue Argumente zu Anfang und Stillstand des Sterns in der Magierperikope Matthäus 2, 1-12”, *Sitzungsberichte der Österreichische Akademie der Wissenschaften, Abteilung II. Mathematische, Physikalische und Technische Wissenschaften*, 206 (1997): 317-344.

¹¹² Interestingly, the study of the motion of this dust earned a Ph.D. in astrophysics for Brian May, the lead guitarist for the band Queen.

¹¹³ Donald W. Olson, “Who First Saw the Zodiacal Light?” *Sky & Telescope* 77, 2 (Feb 1989): 146-148.

Koradin Ferrari d'Occhieppo, used the planet Jupiter for some things, such as the stopping, and the lights for others, namely the "going before" aspect of the Star. But this all ignores the singular "Star" in Matthew, which also doesn't fit a diffuse light cloud on the horizon.

An interesting phenomenon that remains uncertain in origin has not been properly published in relation to the Star, but astronomer David Hughes knows of some who have considered explaining the Star with ball lightning.¹¹⁴ These objects are bright and spectacular, manifest themselves in our terrestrial plane, and can move about. They are also said to explode after a time. So, here is something star-like, able to move about, and it could also stand in place (potentially). However, these objects are short-lived on the order of seconds to minutes, so it couldn't "go before" the Magi for the entirety of their trip to Bethlehem. Also, it couldn't be the same object that rose up in the eastern sky, and we cannot say how it could have made anyone think "king". Furthermore, such an explosion would probably have made it into Matthew's record, if not for theological reasons but for the fact that it would be awesome, and the spectacular is something Matthew does not shy away from (zombies anyone? Matt 27:52-3). So ball lightning is probably a no-go.

Now, so far these ideas have been coming from either biblical scholars or scientists, but some theories are more on the fringe yet still try to explain the Star. One concerns the precession of the equinoxes.¹¹⁵ This is an effect of the wobble of the earth much like a spinning

¹¹⁴ David Hughes, *The Star of Bethlehem: An Astronomer's Confirmation*, pp. 170, 2f.

¹¹⁵ Garry T. Stasiuk, "The Star of Bethlehem Reconsidered: A Mythological Approach." *Planetarian* 10, 1 (1980): 16-17.

top; the axis of the planet draws out a circle while remaining about 23 degrees tilted related to straight up from its orbit about the sun. This cycle lasts about 26,000 years and is credited to have been first discovered by Hipparchus of Rhodes in the second century BCE. What is noticeable to earth-based observers over a considerable period of time is that the stars slowly drift together in the same direction when compared to where the Sun is at a given day of the year, and we will eventually have a different North Star. This has the effect of changing where the Sun will be among the background stars on the important first day of spring. This drifting of where the Sun is on the first day of spring (the vernal equinox) has an effect on astrological predictions; for most modern astrologers, the signs (such as Aries, Pisces, Leo, etc.) no longer line up with the actual constellations you see in the sky because they base their zodiac on where the Sun is at the vernal equinox rather than the actual stars. This means that the Sun sign you have in the newspaper (such as Sagittarius for those born between Nov 22 and Dec 21) does not indicate where in the sky the Sun is for those dates (the Sun is more likely a Scorpio). Many skeptics have pointed this out though not knowing that astrologers have been aware of this for millennia.

But how can this possibly explain the Star? Well, the theory has to rely on a bit of questionable scholarship from the 1960s, a book called *Hamlet's Mill* which postulated a much, much earlier discovery of the precession of the equinoxes, and this knowledge pervades myths from all over the world. Since the hypothesis requires a hyper-diffusion model of culture with one super-culture from the past that left no archaeological record, and its methodology is very loose

with the facts, it has no standing among scholars. Nonetheless, the idea is that Jupiter and Saturn had their conjunction where the Sun would be on the first day of spring, which is extremely important to astrologers (even though the ancient astrologers never said that), and then Jupiter is at the same important location when it becomes stationary. So, we have to rely on the stationary point idea to explain the “stood” aspect of the Star, which in Chapter 5, Part B was shown to not work (see also Chapter 7), but the thesis is also problematic because the conjunctions do not happen on the vernal equinox, just somewhat close. In fact, there was a better such conjunction in 54 CE; why no Magi then? As for how the Star “goes before them”, no actual explanation is forthcoming.

And last, but certainly not least, when it comes to strange ideas about the Star is an alien UFO. Yes, this has been seriously proposed. First it was genuinely argued independently by Paul Thomas (a.k.a. Paul Misraki), Rev. Barry Downing, and R. L. Dione in the 1960s,¹¹⁶ it later received support by UFO contactee George Van Tassel,¹¹⁷ and it has appeared in an episode of *The History Channel's* (now on *H2*) continuing scholastic embarrassment, *Ancient Aliens* (Season 3, Episode 8). Because of its popularity, the show may make the UFO hypothesis the trendiest one for the Star, at least amongst a certain crowd.

However, not all in the alternative scholarship world buy into UFOs. For example, Douglas Elwell, who has a master's degree from Wheaton College in biblical studies,

¹¹⁶ Paul Thomas, *Flying Saucers through the Ages*, pp. 45, 138. Barry Downing, *The Bible and Flying Saucers*, p. 134; R. L. Dione, *God Drives a Flying Saucer*, pp. 44-45, 101.

¹¹⁷ George Van Tassel, *When Stars Look Down*, p. 140.

argues that the Star was the reappearance of Planet X, which itself was a product of a collision with the proto-Earth and an interstellar object in the distant past.¹¹⁸ In effect, it is a meshing of the ideas of Immanuel Velikovsky's *Worlds in Collision* with New Testament fantasy. Also related is a supposed connection between the Christmas Star and Nibiru,¹¹⁹ the imaginary planet of pseudo-scholar Zecharia Sitchin and his misreading of Sumerian texts. Besides the unlikely solar system history, the thesis is still not unlike the nova or Uranus hypotheses, and it thus can no more explain the Star than those failed efforts.

Still other alternative writers are more traditional, in a sense, when it comes to ideas about the Star. The New Age *Urantia Book*, for example, provides all sorts of biographical information about the life of Jesus, including his Nativity. The massive tomb states that the story of the Star is a legend but is based on the conjunction of Jupiter and Saturn.¹²⁰ In an interview on the radio program *Coast to Coast AM*, novelist and ufologist L.A. Marzulli, who otherwise believes the Bible tells of inter-dimensional beings, also argued for a conjunction as the Star of Bethlehem rather than a flying craft, but he went for a conjunction between Jupiter and

¹¹⁸ Douglas A. Elwell, *Planet X, the Sign of the Son of Man, and the End of the Age*.

¹¹⁹ Barbara Hand Clow, *Chiron: Rainbow Bridge between the Inner and Outer Planets*, p. 21; Al McDowell, *Uncommon Knowledge: New Science on Gravity, Light, the Origin of Life, and the Mind of Man*, p. 185.

¹²⁰ *Urantia Book*, paper 122.8.6-7. About this book, including its take on the Christmas Star, see Martin Gardiner, *Urantia: The Great Cult Mystery*, esp. pp. 206-209.

the star Regulus in 3 BCE.¹²¹ So, you can't please everyone it seems.

But, strange as this idea is, at the very least the UFO can match up with what Matthew says. A ship can appear like a star when far away and assuming it is reflecting or sending out light. If the ship wanted to, it could be seen in the eastern skies. And a ship can travel from north to south, stop in place, and hover over a location, even a very specific one—at least according to Spielberg. If aliens were doing this, perhaps then they could also have had chats with the Magi (or the aliens used “thought transference” as Van Tassel believed) so the Magi could know what they were up to, why it's important to follow, and so on. There is the issue though of how the Magi would have seen the UFO over a house and still not notice it is a ship, so the fit to Matthew's Gospel is still not perfect. Of course, the real sticking point is that we don't know if advanced civilizations even exist in our galaxy besides our own, and the immense distances between stars make travel prohibitive without the laws of physics being curiously out-maneuvered by unknown technologies. The issue of motivation for the aliens is also something not to be put aside. But the only way to properly discount the UFO hypothesis is either to prefer the miraculous version of the story or deny the historicity of the event altogether. So, if you buy into the alien hypothesis, you will need to consider Chapter 8.

¹²¹ A summary of the show with Marzulli can be found at <http://www.coasttocoastam.com/show/2010/12/22>.

PART III: FATAL PROBLEMS

Chapter 7: Failure of all Natural Hypotheses

While the individual chapters about particular ideas regarding the Star can show the significant weaknesses of those hypotheses, this chapter can, on its own, show why none of them will ever be able to work. It is simply a problem of the details of Matthew's account; and many of these issues various Star researchers try to reinterpret while doing harm to the context and meaning of the words. Part of the problem is that most recent authors on the subject are unable to read the story in its original language, Koine (common) Greek. While many do talk about particular words or phrases from the manuscripts, the understanding is not broad, and they are still primarily reliant on translations. As such, I will need to talk about the words and phrases carefully to make the point of what the text does indeed say.

The phrase that perhaps receives the most attention concerns the Star "at the rising" (*en te anatole*), which has been variously interpreted to mean rising at sunrise, rising at sunset, or a very particular rising with the sun called a *heliacal* rising—the first visible rising on the eastern horizon before being blotted out by the sun. The

first person to argue that Matthew was actually using a sophisticated phrase concerning the heliacal rising was the biblical scholar Heinrich Voigt in 1911.¹²² However, this was contested by one of the foremost experts on ancient astrology of the time, Franz Boll, who stated that the argument was “naïve”.¹²³ Modern translators of ancient Greek astrology texts also read it the same way, that it is not a special phrase from astronomers or astrologers in antiquity.¹²⁴ But this matters little in the scheme of this chapter, because there are plenty of natural phenomena that can conform to this phrase in any interpretation found in translations or Star of Bethlehem literature.

It is when considering verse 9 of chapter 2 of Matthew’s Gospel that things get hairy. The word that describes how the Star “went before” the Magi is the verb *proago*, which means to lead forward. But in the context of Matthew, it is even more specific because the verb takes a direction object—that which the verb is acting on—and that direct object is clearly the Magi. As such, the Star was leading the Magi, bringing them forth to their destination; the Star is doing more than standing in a certain direction or even moving about, but it is actually leading the Magi on. This can be similarly seen in Matt 21:9 with the crowds that “went before” Jesus when he entered into Jerusalem; it was like a parade procession. This point is expressed in some translations of the Bible better than in others, such as the Holman Christian Standard, the New Living Translation, The

¹²² Heinrich Voigt, *Die Geschichte Jesu und die Astrologie*.

¹²³ Franz Boll, “Der Stern der Weisen”, *Zeischrift für die neutestamentlich Wissenschaft*, 18, Jahrg., Heft 1/2 (1917): 40-8.

¹²⁴ Courtney Roberts, *The Star of the Magi: The Mystery that Heralded the Coming of Christ*, pp. 120-1.

Message, as well as translations by particular scholars such as Hugh Schonfield and Robert Miller.¹²⁵

But even ignoring the particulars of the meaning of the word, the context has the Star “going before” the Magi who are traveling from north to south, while all objects in the sky travel instead from east to west. Perhaps one can imagine a scenario with the road to Bethlehem turning in just the right way as the Magi walk at just the right speed with an object in the sky in just the right position so it remains over Bethlehem over the time journey. But even so, that does not fit the motion specified by the verb *proago*.

And indeed it does move, and then it stops. This is fixed by Matthew saying that the Star was “going before” the Magi “until” something happened. That “until” (using the Greek word *heos*) means there is a termination of the previous circumstance. If in many readings of Matt 2:9 by scientists that the Star is simply to the south and so in the direction of Bethlehem, then Matthew is saying that situation stopped when the Star “stood over where the child was”. This makes no sense on the natural phenomenon hypothesis. This “until” was noted by David Strauss in 1835 when he challenged many sorts of rationalizations of miracles from the Bible in his day.¹²⁶ The only person who has tried to deal with this is the Jesuit astronomer Gustav Teres who was mentioned in Chapter 5, Part B, and who was found to be linguistically incompetent. Here it is no different.¹²⁷ He instead wants to translate *heos* to mean ‘while’, which is possible depending on context. Here it doesn’t make sense. If we

¹²⁵Hugh J. Schonfield, *The Original New Testament*, p. 58; Robert J. Miller, *The Complete Gospels: Annotated Scholars Version*, p. 61.

¹²⁶ David Strauss, *Das Leben Jesu kritisch bearbeitet*, pp. 220-53.

¹²⁷ Teres, *The Bible and Astronomy*, p. 21.

use his translation, then the text says the Star was going while it stood; it goes while it is not going. Not even a spiritualized reading by the Gnostics is going to fix this contradiction, and I am not even considering issues of grammar here, just basic logic.

Another word of interest concerns how the Star “arrived” or “came”. Some translations like the New International Version do not include this because in English it seems to only add verbiage, but for our purposes it is worth noting. The operative word here is a very common verb, *erchomai*, which means ‘to come/go’, depending on context. As the verb is here conjugated (as a participle), it means that in an instant (the aorist tense) it came to its destination. Again, we have a verb used that describes motion. That is in utter contradiction to any hypothesis that the Star was simply seen in the south. Both *proago* and *erchomai* both clearly indicate that the Star was moving during the Jerusalem-to-Bethlehem journey. Moreover, the Star arrived at a location, one that is specified, over which it stood still.

This is now leading up to the standing of the Star, which already the word *heos* (‘until’) indicated was coming. The term here is *histemi*, another very common verb that means ‘to stand/make stand’. The verb is again conjugated like *erchomai* to indicate that the Star came to a standstill in an instant using the aorist tense. That would disallow the stationary point of a planet which for an ancient observer was unperceivable in how long it took to stop and go again. But more importantly, there is nothing in the sky that comes close to traveling south for a couple of hours and then coming to a halt.

But perhaps it is the last part of the description of the Star that dooms any theory of an astronomical or astrological phenomenon. The phrase used by Matthew

says the Star “stood over where the child was”. The word ‘over’ here is using the preposition *epano*, but there is another aspect that is invisible when translated into English. The term *epano* has a particular meaning when it is followed by a noun, participle, or adjective in a particular declination called the genitive. Prepositions in Greek usually only take certain declinations as their partners in meaning, though some take multiple possibilities which allow for different meanings. Here *epano* when followed by the genitive takes a meaning close to what we get with one of *epano*’s root words, *epi*. This preposition is known even in English in words such as epidermis (top layer of skin) and the epicycles discussed earlier, and it has the meaning of being on top of something. And that is the sort of meaning Matthew’s text implies; the Star is not simply above Jesus, it is hovering right over the house he is in. For comparison, the same construction is used by Matthew in 5:14 concerning a city *on* a hill, in 23:18, 20 about gifts placed *on* alters, in 23:22 about God sitting *on* a throne, in 21:7 when Jesus rides *on* a colt and ass which walked *on* a robe-covered road, in 27:37 about the sign *on* the cross *above* the crucified Jesus, and in 28:2 about the angel that sits *upon* the stone door that had closed the empty tomb. In all these cases, Matthew uses *epano* with the genitive to mean something that is on top of or slightly above an object. This meaning is consistent in other places used in the Bible, including Luke 4:39, 10:19, 11:44, and Revelation 6:8, and the same can be said for other Greek literature.

Compare this to another possible preposition Matthew could have used, *hyper*. As discussed in Chapter 3, this preposition was used to describe comets that hung over cities. Josephus had done this in reference to a star

over (*huper*) Jerusalem, but this can mean the object was up in the heavens. The same is done by astronomers like Ptolemy that speak of astronomical or astrological events “above the earth” (*huper gen*, ὑπὲρ γῆν).¹²⁸ Not true for *epano* with the genitive which implies close proximity. The term is distinct, and there is no astronomical or astrological work that uses this construction when it comes to objects in the sky relative to objects on the ground. If you want to check this, you can use the database program *Thesaurus Linguae Graecae* and examine all the astronomical and astrological works in Greek in the catalog; you will find that *epano* itself is very rarely used at all, and even more rarely followed by the genitive. More often the word is used with respect to the text of their volume (i.e. “as was shown *above*”), and Ptolemy used it with respect to stars relative to other stars in the heavens, and those stars were still close to each other on the celestial sphere of the night sky.¹²⁹ For example, Ptolemy spoke of the star 18(e) Ursa Major as “above the right knee” of the bear,¹³⁰ about 3.5° away. So in one of the very few places where an astronomer uses *epano* with respect to the stars, it is in relation to another nearby star in the night sky.

This demonstrates beyond reasonable doubt that Matthew is describing the Star as in close proximity to where Jesus was, which is said, in verse 11, to be a house. In another infancy gospel attributed to Jesus’s brother James, the situation is also made stark as the Star enters into a cave where Jesus is and stands over his head.¹³¹ Similarly, this is how it has been understood

¹²⁸ Ptolemy, *Tetrabiblos* 2.5 (76).

¹²⁹ Toomer, *Almagest*, pp. 15-6.

¹³⁰ Toomer, *Almagest*, p. 342.

¹³¹ *Protoevangelium of James* 21.

by generations of readers, from theologians to astronomers. Let me give just two examples, both of which show that it was not due to lack of knowledge that led to the miraculous reading. First, consider St. Augustine of Hippo. Before he was a Christian he belonged to a religion that was heavily involved in astrology, and Augustine also read about astronomy and astrology and conversed with astronomers on their craft.¹³² So when he said that, unlike the stars in astrology, the Star of Bethlehem left the course of the rest of the sky and came down to a newborn, and astronomy and astrology have nothing to explain,¹³³ we should take seriously that he understood the science involved as well as the meaning of Matthew's account.

But also consider the person so many scientists refer to when they make their own journey into Star research, Johannes Kepler. His work on connecting Jesus's birth with the great conjunction in 7 BCE certainly involved astrology, but Kepler, who in his schooling learned Greek and Latin, also specified the miraculous in the tale. "This star was not one of the ordinary comets or new stars, but by a special miracle moved in the lower regions of the air."¹³⁴ Kepler was no deist, and when he read of a miracle in his holy book, he believed it was a miracle, and he believed the Star to have traveled down low to the ground for the Magi to see and follow.

¹³² Augustine, *Confessions* 4.1-3; 5.3. See Leo Ferrari, "Astronomy and Augustine's Break with the Manichees", *Revue des Études Augustiniennes* 19 (1973): 263-276.

¹³³ Augustine, *Reply to Faustus* 2.6f.

¹³⁴ Christian Frisch, *Joannis Kepleri Astronomi Opera Omnia* IV, p. 346. Translation of the following Latin: *Stella haec non fuit e numero communium cometarum aut novorum siderum, sed accessit illi privatim miraculum motus in inferiori regionis aeris.*

Countless others from antiquity, medieval times, and the early modern period can be cited, including Thomas Aquinas, Martin Luther, and John Calvin. To find these three agreeing on anything ought to indicate what the text is actually saying. I document this interpretation through the centuries and by modern scholars in a full article on the subject which includes the first attempts to understand the Star as something natural beginning in the early 19th century.¹³⁵ Today, one is hard-pressed to find any recognized Bible scholar publishing in peer-reviewed literature taking the view of the Star as something non-miraculous, the last example I can find coming from the mid-20th century. Among those that can read the text critically, there is no question about its presentation.

What this demonstrates overall is that nothing fits with what Matthew actually talks about, something few Star researchers are able to even access because classical languages are being taught less and less, especially to scientists. But Matthew talks about a Star that travels south towards a particular destination, leading on eastern sages, until it comes to its destination, stops and hangs over a particular hovel in the small town of Bethlehem. No object in the sky can do such a thing, not by a long shot. The only two natural phenomena that we could even talk about fitting such motions are ball lightning (but see chapter 6) and a UFO. However, we really are left with the impression by the author that something miraculous has happened.

At this point it ought to be clear: we have a miracle as described. So, if we accept that this is what is actually

¹³⁵ Aaron Adair, "The Star of Christ in the Light of Astronomy", *Zygon* 47, 1 (March 2012): 7-29.

recorded by Matthew, can we say if the events happened or not? This idea is discussed in the next chapter.

However, considering the consistency of interpretation of the miraculous nature of the Star for centuries and by modern biblical scholars, it raises the question as to why anyone thought it could be something naturally explained. I explore this in my article mentioned earlier, but here is a summary of the history of such explanations.

Up until about 1800 everyone who said something about the nature of the Star agreed on its supernatural qualities, and no exception exists until the 19th century. Why the change? For one, the premises of the Enlightenment had meant for many that miracles simply do not occur. Either they are impossible, as argued by the philosopher Spinoza, or reports of them are almost never believable, as argued by David Hume. English deists and other skeptics had an influence on many in continental Europe, and the most important example of this influence is a German scholar of the Bible, Hermann Samuel Reimarus. In his writings that were published posthumously in the 1770s, he took those beliefs about the miraculous to their logical conclusion and made a huge swath of the Bible, Old Testament and New, to be credulous clap-trap. He even suggested that the Disciples were conmen of sorts. It was he who first seems to have argued in print that the Star of Bethlehem was a fiction.¹³⁶ The published fragments of his work caused outrage in Hamburg, the religiously conservative town that saw to their first publication, and it sent shockwaves through the rest of academia.

¹³⁶ Gerhard Alexander, *Apologie; oder Schutzschrift für die vernünftigen Verehrer Gottes*, Vol. 2, p. 536.

In reaction to the wave of supernatural skepticism among those who still considered the Bible, especially the Gospels, to be trustworthy accounts of what really happened, an unusual compromise of ideas took place. The Bible was true, but so were the laws of nature as described by Newton and other scientists. To reconcile, the miracles of Scripture were not actual aberrations of God's perfect laws of the universe, but events which the 'noble savages' saw and misunderstood. The prime example of one of these scholars from the early 19th century was Heinrich Paulus. He famously argued that the miracle of Jesus walking on water was nothing so exciting; it was dark and foggy, so the Disciples on their ship thought they were out at sea rather than at the shore, so when Jesus was walking by on the beach they thought their master was doing the impossible. Now, imagine this sort of thinking for every miracle in the Bible, and you have the Rationalist movement among Protestants. And the logic is simple: miracles are impossible, but miracles misunderstood are possible, and if it's possible and we know in our hearts that the stories are true then what possibly happened probably happened.

Now, this movement did not last for a long time, though echoes of it still exist—if you want to see this sort of thinking in action today, look at the Ancient Alien Theory (AAT) crowd.¹³⁷ It fell out of the mainstream scholarly world by the middle of the 19th century, but it seems that this sort of thought process is going on when it comes to the Star today for many scientists. Because the idea is ultimately a Christian apologetic, a defense of the faith, the naturalistic Star exists for that purpose.

¹³⁷ Cf. Robert M. Price, *Night of the Living Savior*, pp. 44-45.

Some people, such as Robert Newman, are explicit that they use the naturalistic theory for apologetic ends, even when they believe in the supernatural.¹³⁸ Besides the various scientists that publish on the subject, the theory is touted around in other Christian apologetic outlets. In the late 19th century, Francis William Upham used Star research to fight atheism,¹³⁹ and those who attack the “New Atheists” do much the same.¹⁴⁰ Fox News anchor Bill O’Reilly has also recently used the Star as historical proof for the stories in the Gospels by arguing for a comet,¹⁴¹ though his argument is even more confused on the details than in other comet literature. One of the most popular productions out there defending the physically-explicable Star comes from a team of evangelicals. Lawyer and preacher Fredrick Larson has a video presentation and website defending the thesis of Ernest Martin about the Star, which was produced by the same person who produced Mel Gibson’s *The Passion of the Christ*. Larson himself was also a student of the late Francis Schaeffer, one of the most influential evangelical apologists of the 20th century. Perhaps you can see why his presentation on the Star has been broadcast multiple times on the Christian Broadcasting Network (CBN) and featured on the Trinity Broadcasting

¹³⁸ Robert Newman, “The Star of Bethlehem: A Natural-Supernatural Hybrid?” *Interdisciplinary Biblical Research Institute* 2001, pp. 1–16.

¹³⁹ Francis William Upham, *Star of our Lord: or, Christ Jesus, King of all worlds, both of time or space. With thoughts on inspiration, and the astronomic doubt as to Christianity*, p. 306.

¹⁴⁰ Michael Poole, *The ‘New’ Atheism: 10 Arguments that Don’t Hold Water*, p. 43. Rev. Downing in his sermons and in our email exchanges also used UFO explanations for biblical miracles to combat liberal theologians and atheists.

¹⁴¹ Bill O’Reilly and Martin Dugard, *Killing Jesus: A History*, chapter 1, n. 6.

Network (TBN). So we see a strong contingent of Christian apologists using the Star explained by science to defend the faith.

Now, the motivations of the scientists who promote these ideas may not be so evangelical, but it is certainly so for some. This is definitely the case for Frank Tipler whose book on the subject is all about proving Christianity through physics.¹⁴² The evangelical Francis Collins, former head of the Human Genome Project, current director of the National Institute of Health in the United States, and a major voice that tries to find a place for Christian belief in science, cites work on the Star of Bethlehem as proof of reconciliation between science and faith, not to mention support for the historicity of the stories of the Bible.¹⁴³ On the other hand, the motivations of other scientists are probably more along the lines of their own personal beliefs and motivated reasons to support them. However, all of these explorations into the Star as a non-miraculous object stand out because the same procedure does not exist for most other miracles in the Bible, such as the Virgin Birth or the Resurrection. These Star hypotheses are a hold-over from early 19th century Rationalism, but incompletely. But why is there no effort to make the resurrection of Jesus a natural phenomenon?

What this Star research ultimately attempts to do is prove that something from the Gospels actually happened, some guarantee to modern Christians that something about the story of Jesus can be demonstrably shown to be true with the tools we use to prove galaxies are 'island universes' millions of lightyears away and

¹⁴² Frank Tipler, *The Physics of Christianity*.

¹⁴³ Karl Giberson and Francis Collins, *The Language of Science and Faith: Straight Answers to Genuine Questions*, p. 108.

planets exist around other stars. And if one thing can be proven true in the Gospels, it makes all the other stories more plausibly true, including the miraculous parts, and it provides salvation for the theological message of the Bible.

Consider why this research only rarely happens outside of Christian circles and without the positive results. It is telling that no skeptics or atheists have use for the research on the Star of Bethlehem. One of the few nonbelievers to study Star research was science fiction writer and prolific author Issac Asimov. He looked at the various hypotheses about the Star as interesting what-if scenarios, but he ultimately concluded the story was a fiction.¹⁴⁴ The skeptical “Bad Astronomer” Phil Plait also figures the tale is likely a fiction.¹⁴⁵ Historians of ancient astronomy also ignore the attempts to find the Star; as the influential historian of the exact sciences Otto Neugebauer said of tales like the Star of Bethlehem, “I think it better simply to discard such sources for the reconstruction of historical data.”¹⁴⁶ Overall, the investigations into the Star are only used to make the biblical story more plausible and more believable rather than being “explained away” like the psychic abilities of Uri Geller—the difference being that the skeptics think there is nothing that happened to be explained. So it is understandable why the story in Matthew is breached from its context and made into something that could potentially find coherence with the sciences.

¹⁴⁴ Issac Asimov, *The Planet that Wasn't*, pp. 183-194.

¹⁴⁵ Phil Plait, “Starry, Starry Night,” in Adriane Smith, ed., *There's Probably No God: The Atheist's Guide to Christmas*, pp. 59-67.

¹⁴⁶ Otto Neugebauer, *A History of Ancient Mathematical Astronomy*, p. 608.

One anecdote to consider is the email conversation I have had with David Hughes, whose book on the Star of Bethlehem is still probably one of the best starting points on the subject (though I disagree with his conclusions). He basically says that it is worrisome if Matthew's account of the birth of Jesus is not historically true, even though the Evangelist seems like such an honest guy; then it throws much else into doubt. What do we believe and disbelieve, then; and are we prone to special pleading and other methodological biases in so doing; what criteria can differentiate truth from falsity? These are not easy questions to answer, but they are the questions of the historian, and we may have to leave behind a lot of stories as fiction when we do history. Obviously for someone that considers the Bible authoritative, theologically or otherwise, it is troublesome to believe that some, many, or perhaps all the stories are not factual, and that makes for some difficult introspection to understand what one should believe. This is highlighted for scientists who study the world as acting without miracles, yet they are reading about what all their studies indicate is impossible. Such anxieties explain the motivation for making sense of tales for those who already believe on faith yet must square with their reasoning and understanding of natural law.¹⁴⁷

In a strange way, this situation is best attested in the writings of those that propose a UFO as the Star. It is especially helpful to have the view articulated by Barry Downing who has a doctorate in theology as well as a science background. In his book, *The Bible and Flying*

¹⁴⁷ Hughes has expressed more skepticism recently towards the story of the Star and Nativity in general, but he seems uncertain where to stand. He also likes to give talks on the subject, a perennial favorite among holiday audiences.

Saucers, he looks at the 'demythologizing' program of theologians such as Rudolph Bultmann and John A. T. Robinson. In this paradigm, the worldview of the writers of the Gospels is not that of modern cultures, especially when it comes to symbolic narrative compared to literal histories of events. Unlike ancient people, moderns do not believe in supernatural powers and principalities, and the antique believer was not scientific and critical. Bultmann, in particular, wanted to no longer believe in the mythological and instead wanted to get to the core of the meaning of the tales. There were still core historical facts, such as the crucifixion of Jesus, but faith was more about existential reflections upon the tales of Jesus rather than recounting the history of Jesus. Bultmann himself was following in a tradition of other 19th century theologians, such as Friedrich Schleiermacher, Ludwig Feuerbach, and the young David Strauss, though Bultmann also combined his approach with existential philosophy. Faith is not about believing that a series of impossible events happened, but it is about accepting the meaning and purpose of the narrative.

Downing, however, says this leaves the stories of the Bible without power. In the preface to the 1973 edition of his widely-sold paperback as well as in his sermons and our email conversations, he notes how the aliens, as emissaries of God, provided humanity with morality and spiritual purpose. He needs the Bible to be true for there to be spiritual and moral potency, not to mention the promise of eternal life in heaven, and it is the belief that we cannot trust the miraculous tales in the Bible that undercuts this. UFOs save the phenomenon of the text, then, and with it our souls. But why is believing in miracles not sufficient? As another UFO proponent puts

it, then any ghost story is just as well justified¹⁴⁸ (not to mention other, non-Christian religions). Science (or more science fiction) is what makes the stories become realistic to a modern mind and not mere myth; there is no inconsistency in believing in light bulbs as well as supreme beings.

This can probably account for many of the efforts, past and present, to explain the Star consistently with physics and why Star research has a popular following, even all the way back in the 1930s with the earliest planetariums and their Christmas shows. There are people who want science, the best way to know things about the world, to support their beliefs, especially when science is perceived to undercut so many other religious dogmas.

Perhaps, then, it is understandable why there are these attempts to read Matthew's account about something astral, something science has a lot to talk about, as a natural light rather than a supernatural one. But as shown, one cannot analyze Matthew's account seriously and critically and still believe that there is something physically explicable behind it. Yet, even if we suppose against all else that the tale is about a real object, scientifically-explicable or otherwise, did it really happen? The final chapter will explore just how plausible the story actually is.

¹⁴⁸ Dione, *God Drives a Flying Saucer*, p. 79.

Chapter 8: Historical Issues

Consider now that the Star of Bethlehem, as told to us by the Gospel of Matthew, is a miraculous object, moving about with the aid of forces not explained by physics. Can we believe it happened? Should we? Some adherents to the truth claims of Christianity will complain that discounting a story because it is miraculous is a bias; yet, if we are to use that term, it is an appropriate bias. We tend not to believe in the miracle claims of other religions, since otherwise we would end up converting. Followers of the Abrahamic faiths do not believe in the miracle statues of India or the healing and resurrecting powers of Asclepius. Modern Christians don't believe a lot of the stories told from the Middle Ages, such as St. Christopher being transformed from a dog-head into a full human, and they don't believe in the ability of Rev. Jim Jones to walk on water and raise the dead as he was claimed to do. Ask yourself, how much evidence would you need to believe Jim Jones raised the dead? Just two congregation members' testimonies? You are probably not going to believe in him based simply on that, so why accept similarly miraculous claims from even more dubious reports? Perhaps you don't care for this history, but even for Jesus there are things said of him that you probably don't believe. Most Christians don't accept the stories of Jesus as a boy in the non-canonical infancy gospels, such as the *Infancy Gospel of Thomas* that had Jesus forming living birds out of clay and resurrecting his friends among other miracles. You probably don't believe the story in *The Gospel of Pseudo-Matthew* that has Jesus taming dragons while still an

infant at his mother's breast. So if not believing in a miracle claim without evidence is a "bias", it is one shared by everyone except the most credulous. This is because we have realized that not everything we read or hear is true, and most of the amazing claims are probably false. In reality, we are all skeptical of such claims except those we already believe, but that is not a bias but a hypocrisy.

But even if we ignore what happens to the prior probability of a story that contains miraculous elements, could the story still be true? Let's even suppose for the sake of argument that some natural phenomenon can account for the Star or Matthew's account is so confused to render it useless in determining if he meant something miraculous (in which case, why believe him at all?), so we don't actually disqualify the story for having too heavy a divine hand in the action. And let's even suppose that Matthew is actually trying to tell us what really happened rather than inventing myths. Given such conditions, are there reasons to reject the story as it is?

As discussed briefly in Chapter 2, even if Matthew is actually trying to write a history or biography, we are still unsure about his sources and methods. If he is relying on people at the end of the first century (or later perhaps), and if they don't have a connection to the events, then it doesn't matter what Matthew's intentions were. As was also mentioned, the number of living witnesses for Matthew to contact by the time he wrote was zero; Joseph and Mary would have been dead due to age (Mary must have been at least 12 when Jesus was born c. 6 BCE, so she would have to be at least over 100 by the time Matthew wrote, a near impossibility in antiquity with an average lifespan not better than 50 for someone who reached adulthood; Joseph was even older

and traditionally understood to be dead before Jesus started his ministry). The brothers and sisters of Jesus were also likely dead at this point due simply to age (and James is said to have been martyred before the end of the Jewish War). Considering also how Matthew relies heavily on Mark who was not an eyewitness even traditionally, let alone according to critical scholars, it doesn't look like Matthew had access to such witnesses.

Combine that with how Matthew doesn't do anything to show he was treating his sources with a hint of critical thought. After all, he says nothing of the plausibility of his story, and since it is significantly different to Luke's (contradictions or no) he ought to be saying why he trusts his sources over others. Again, that assumes Matthew is even trying to write history, but even if he was, he proves to be poor on method. Since the author of this Gospel is using even worse methodology than one of the most notorious biographers from the Roman Era, Suetonius, and Matthew's access to reliable sources is even less likely than was the case for Suetonius, we have to trust Matthew much less. So, at best, if Matthew is trying to be a biographer, his story of Jesus's birth has to be treated with significantly more skepticism than with most any other biography from antiquity—and we have biographies of people who never even existed, like Romulus.

However, let's leave even that aside and consider if the story is plausible, even if we have no qualms with the miraculous. If we bend over backwards to accommodate the claims to this degree, can we still believe the tale? There are several features that I will highlight which show, even if miracles are possible, that the story goes against what we know about the past.

Let us first consider the Magi themselves. According to Matthew, they came looking for the King of the Jews because they saw a Star in the east/as it rose and came to worship. First, this is strange because the Magi are not Jews but Zoroastrians from Persia. The magi were in fact the caste of priests in this religion that is older than Judaism, and they had their own king and their own belief in a savior figure much like the Jewish messiah. So we have to wonder why they would have given two hoots about the birth of a Jewish king which they may not have even connected to a savior figure—it is Herod in the story that connects the King of the Jews to the Messiah. Historical circumstance already makes the tale suspect, because now we need magi that want to become Jews because of a Star.

But what about Zoroastrian interest in the stars and astrology? Here we find another significant problem: before the Sassanid period in Persia (3rd-7th century CE), the Zoroastrians did not practice astrology.¹⁴⁹ The scriptures of the magi, such as the Avesta literature, show only the most basic of stellar interests, and it says nothing of the planets. When in the third century the astrological treatises of the Greeks and Indians were brought to Persia for translation into their language,¹⁵⁰ we see a visceral reaction to astrology by the priesthood. They considered all the planets, including the Sun and

¹⁴⁹ Gerard Mussies, "Some Astrological Presuppositions of Matthew 2: Oriental, Classical and Rabbinical Parallels", in Peter Willem van der Horst, ed., *Aspects of Religious Contact and Conflict in the Ancient World*, pp. 25-44, esp. pp. 28-37; Albert de Jong, *Traditions of the Magi: Zoroastrianism in Greek and Latin Literature*, pp. 397-398; R. C. Zaehner, *Zurvan: A Zoroastrian Dilemma*, pp. 147-165, 369, 377f, 400f, 410-411; *The Dawn and Twilight of Zoroastrianism*, p. 238.

¹⁵⁰ Bayard Dodge, *The Fihrist of al-Nadim: A Tenth-Century Survey of Muslim Culture*, vol. 2, pp. 573-575.

Moon when eclipsed, as corrupted by Ahriman, the Zoroastrian version of Satan. This was because of the irregular motion and changes of the planets, unlike the never-changing stars. Change in the heavens was a sign that those objects were corrupted, possessed by evil, and no good was to come of them. This was also, in part, why the Zoroastrians thought comets were evil (as seen in Chapter 3)—their appearance was irregular. The magi then were like many fundamentalist pastors with respect to astrology: either indifferent or hostile. This discussion lasted for some time, though later the magi came to accept and practice astrology, but only after its re-introduction into the Middle East. No direct evidence of Zoroastrian interest in astrology exists before then.¹⁵¹ So, if there was some astrological portent in the first century, the magi wouldn't have noticed it; if there was a moving Star, they would have called it evil, not something to lead them to worship.

So why are there magi in Matthew's story if they had no astrological interests? There was actually a misconception about the founder of the religion of the magi. Known as Zarathustra in the Persian language, in Greek and Latin he was called Zoroaster. The latter has the Greek word for 'star' in it, so we can see why some would associate the practice with the man with *aster* in his name. There was also a blending of the differences between the magi and Babylonians, the latter who invented astrology and transmitted it to the Greeks. This is not unlike modern Westerners not knowing the difference between Arabs and Persians. In ancient Western sources, Zoroaster is called a Babylonian and

¹⁵¹ Ehsan Yarshater, *Encyclopaedia Iranica* Vol. 2, pp. 258-259; Mussies, "Some Astrological Presuppositions of Matthew 2", p. 30 n. 31.

the founder of the magi. Because of such confusions, a vast body of literature on astrology was pseudonymously penned by Zoroaster.¹⁵² As such, if Matthew had a source, that person seems to be someone influenced by Greco-Roman misconceptions and not relating history to us.

But suppose, nonetheless, that the Star was even more miraculous than Matthew describes. In other infancy stories, including ones about the Magi, all sorts of crazy things were happening in the East that told the Magi of the newborn king. In the recently translated *Revelation of the Magi* from the late second or third century, the Star is baby Jesus himself, who comes to spread the news of his Incarnation—talk about an early self-promoter!¹⁵³ So let's suppose even more miracles, which are killing the probability of this story even if miracles happen. If all sorts of miraculous machinations were happening in Persia to get the magi interested and cause at least a few of them to travel to Palestine, then one would think that the great king they sought would have had great reverence in their land and his followers respected. Even if they did not see Jesus as the ultimate savior they could have still revered him in the way Muslims do today. Reality, on the other hand, couldn't be further removed from this. In inscriptions from the third century, we have the leader of the magi conducting persecutions of Jews *and Christians*!¹⁵⁴ We find nothing

¹⁵² Roger Beck, "Thus Spake Not Zarathustra: Zoroastrian Pseudopigrapha of the Greco-Roman World" in Mary Boyce (ed.), *A History of Zoroastrianism*, Vol. 3, pp. 491-565.

¹⁵³ Brent Christopher Landau, "The Sages and the Star-Child: An Introduction to the *Revelation of the Magi*, an Ancient Christian Apocryphon", Th.D. Thesis, 2008.

¹⁵⁴ Solomon Nigosian, *The Zoroastrian Faith: Tradition and Modern Research*, p. 34; Mary Boyce, *Textual Sources for the Study of*

that shows any deference for the Christian faith or the religion it flowered from, but we see figures harassing those that follow the man the Magi were supposed to have worshipped. So it seems when it comes to Magi honoring Jesus, no one told the actual magi.

The above considerations are already forcing the probability of this story down to zero with great speed, but let us suppose even against what has been noted that some group of magi were interested in some Star, that they did come and praise Jesus at his birth, and so on. Let's just say that our record of Persia isn't good enough to have on hand the actual report of the Magi's journey and things went awry in the time after their visit so that the importance of Jesus wasn't recognized (could it be *Satan!*?). But what about the consequences of such a journey to Israel, had it happened? Would this have been of note and reported and exist in our extant records of the past?

In actuality, it should and would have been noteworthy. Sure, a small group of eastern sages traveling about may not seem like much, but consider that the Magi are not just your local pastor or assistant rabbi. These people were part of a religious caste that not only had significant authority on matters of the divine, but they also were part of the selection of the king of Persia. For example, Strabo, a writer from just before the birth of Jesus, wrote that the council of magi would pick the king from familial claimants.¹⁵⁵ Further in the past, the father of history, Herodotus, said the magi had led an overthrow of the government with their own chosen king

Zoroastrianism, pp. 112-113; Philippe Gignoux, "L'Inscription de Kartir à Sar Mashad", *Journal Asiatique* 256 (1968): 387-418.

¹⁵⁵ Strabo, *Geography* 11.9.3.

back in the sixth century;¹⁵⁶ even though they failed, the magi were still given patronage by Darius, the king who had defeated them.¹⁵⁷ Other authors from the past considered the magi a little less than kings.¹⁵⁸ Perhaps, then, it is understandable why the Christian writer Tertullian in around 200 CE could say that the magi were “almost kings.”¹⁵⁹

So now imagine that you have people coming from the group that declares kings in Persia all the way to a land that is under the thumb of Rome, the most powerful rival to the Persians, and declare a toddler is the true ruler of Israel. Those Magi are in fact not only overriding the authority of King Herod, an infamously paranoid tyrant, they are overriding the authority of Caesar Augustus. This is not an idle incident but a declaration of war by the most powerful nation Rome knew. It would be similar if, during the Cold War era, the Soviet Union sent delegates to set up a new governor of Puerto Rico; this would be Cuba’s communist takeover squared in the international community. (Perhaps it is no wonder why Herod as well as the people of Jerusalem, according to Matt 2:3, were all filled with dread by the announcements of the Magi.)

In fact, an incident like this happened during the reign of Nero concerning the territory of Armenia, though not with the boldness of the Magi as portrayed in the Gospel. Like Judea, Armenia was a sort of buffer region

¹⁵⁶ Herodotus, *Histories* 3.61-74.

¹⁵⁷ M. A. Dandamaev, *Persien unter den erster Achämenidian* (6. Jh. v. Chr.), p. 238.

¹⁵⁸ Strabo, *Geography* 12.2.3, 3.32; Caesar, *De Bello Alexandrino* 66. See also *Cambridge History of Iran*, vol. 3, part 1, pp. 85-86 (on Zoroastrianism patronage from Vologeses I); part 2, pp. 689-691 (on the supreme council), 867-868 (on recognition of Zoroastrianism in 1st century BCE), 904-905 (on taxes for fire temples).

¹⁵⁹ Tertullian, *Contra Marcion* 3.13.

between Rome and its rival, so who ruled this land was a matter of border security. When the Persians appointed a new ruler to this region undesirable to the Romans, Emperor Nero sent in the troops; this led to a long war and was noted by several historians.¹⁶⁰ Decades earlier, there was a significant diplomatic showdown over the same region and as to who was to appoint its ruler; this did not lead to a war, in part thanks to the negotiating skills of the Roman governor of Syria, but again it made it into several histories from antiquity, including the Jewish historian Josephus.¹⁶¹

So if something like this had happened in c. 6 BCE concerning Israel, the response by Rome should have been sending, at the very least, strong condemnations, if not soldiers, and all of Jerusalem should have been afraid (Matt 2:3). Instead, silence, as if nothing happened. And it is not a silence due to a lack of sources for this time as we pretended in the case of eastern records. It is impossible to imagine that Josephus, who had eyewitness records to the life of Herod the Great, would have not mentioned the major diplomatic nightmare the coming of the Magi would have brought. Moreover, the historical reports from the past that are lost to us, such as the writings of another Jewish historian in the first century, Justus of Tiberius, would almost certainly have been preserved by Christians, at least in part, if they contained evidence of the events mentioned in the infancy stories of Jesus. Similarly, some of the writing of Nicholaus of Damascus, a friend of Herod and Josephus's major source for the king's life and final years,

¹⁶⁰ Tacitus, *Annals* 13; Dio, *Roman History* 62; Suetonius, *Nero* 57.

¹⁶¹ Josephus, *Antiquities of the Jews* 18.96-105; Tacitus, *Annals* 2.58; Suetonius, *Caligula* 41.3; *Vitellius* 2.4; Dio, *Roman History* 59.27.3-4.

should have been preserved by the Christians if he talked about this event with the Persian Magi. These unpreserved sources, or even quotes from them, further add to the absence of evidence for this amazing event from taking place, even though it would have been of international importance. One must also wonder how the Gospel of Luke, which styles itself as a historical work, provides no mention of this incredible story. The fact that no source independent of Matthew mentions this story is inexplicable if it really happened, but this silence is exactly what is expected if it never happened. This is a powerful argument from silence because we *expect* to hear something,¹⁶² yet it is not there.

Let's now combine this with everything else that has been established. Western records show no sign of a diplomatic showdown or war expected due to such a trip by the Magi, eastern records show no love for Jesus and instead persecution of his followers, the magi were not interested in astrology or a Jewish king or savior, and our only record for the tale comes from a figure of unknown provenance with unknown sources using terrible historical methodology and relating a story that is filled with the physically impossible. And this is all just considering several verses in Matthew; I haven't even included the story of the Virgin Birth, the contradictions with Luke's story and chronology, and the generally amazing tales that Matthew relates, especially at the death and resurrection of Jesus. We don't even need to consider these objections to see that the

¹⁶² Cf. J. P. Moreland and William Lane Craig, *Foundations for a Christian Worldview*, pp. 156-157; Richard Carrier, *Proving History: Bayes's Theorem and the Quest for the Historical Jesus*, pp. 117-119.

historical record is decisively against Matthew's account of the Star being factually correct.

On the other hand, we can see what would have more likely inspired the story, and it has far less to do with events in 6 BCE. There has been a considerable amount of scholarship directed at this question, though not all possibilities are as likely as others, and they may all be to some extent incomplete. However, with the complex nature of any skillfully written story, there are likely several influences, and we need to try and see what were the prospective items that the original audience for the Gospel would have been able to connect to and appreciated. We also do not expect an author to simply copy another story and make small changes, but rather the author communicates their message by what they borrow *and* by what they change.¹⁶³ As such, it is necessary to consider several potential links to uncover what the Evangelist was doing.

One idea taken from the historical record concerns another journey of magi, this time though to Emperor Nero in Rome. This event took place in the same year as Halley's comet (66 CE), an eastern potentate comes to receive kingship from Nero, and the new king and his parade of magi return to their homeland via another route (cf. Matt 2:12). Entertained as a possibility by the famous Catholic biblical scholar Raymond Brown and others before him, the most recent treatment is presented by independent scholar Rod M. Jenkins.¹⁶⁴ However, this hypothesis seems to be too weak to explain

¹⁶³ Thomas L. Brodie, *The Birthing of the New Testament: The Intertextual Development of the New Testament Writings*, pp. 43-49.

¹⁶⁴ Raymond Brown, *Birth of the Messiah*, p. 611; Rod M. Jenkins, "The Star of Bethlehem as the Comet of AD 66", *Journal of the British Astronomical Association* 114, 6 (June 1999): 336-343.

the story. For instance, it does not clarify why the Star was seen “at the rising”, let alone any of the miraculous movements of the Star. When it comes to the historical record, none of the ancient sources connect the comet to the arrival of the eastern king (Tiridates) to the court of Nero, the comet had disappeared months before Tiridates’s arrival in Rome, and the comet had nothing to do with the ascension of the new king or any prophecies. The only interesting connection to Matthew’s story, that Tiridates and his magi left for home by another route, is trivial; besides, there is already an internal narrative logic to why the Magi left by another way: to avoid Herod, a reason having nothing in common with Tiridates’s return home. It may have been difficult then for the readers of Matthew to see this minor connection as it would require memory of the coming and goings of kings long dead, not to mention that a comet happened to be seen in that year. With the additional symbolic problems of emulating a story with infamous Nero as the bestower of power along with the evil sign of the comet, there are other possibilities to consider for how the story of the Star of Bethlehem was crafted by the Evangelist or his sources.

Returning focus to the Star itself, one hypothesis put forward first by the early critical New Testament scholar David Strauss in 1835 has us pay attention to one of the most important national myths written for Rome, the *Aeneid* by the poet Virgil during the reign of Caesar Augustus.¹⁶⁵ Early on in the tale, the last surviving Trojans in their war with the Greeks pray for a sign to shepherd them to a new homeland. Suddenly then a blazing shooting star, gliding over the highest housetops,

¹⁶⁵ David Straus, *Das Leben Jesu kritisch bearbeitet*, p. 248.

becomes their guide.¹⁶⁶ The star in question has properties like a comet, which is probably because Virgil is incorporating the 'star of Caesar' into his legendary flattery of the Romans and the new emperor; the comet seen at the games of Julius Caesar after his death was interpreted as the soul of the dictator rising up to heaven. More importantly, however, the star seems to be Venus as the morning star and following in an older tradition of Venus as a star guiding her son (Aeneas) to Italy.¹⁶⁷ Considering that the Star of Bethlehem was also supposed to have been a morning star (see the translation in this book for details), then this makes for some dense links between the famous star of Aeneas in a story well-known in the Roman Empire and the story of the Christmas Star in Matthew: a morning star, one that miraculously guides its followers while in the atmosphere, and it leads to a new king(dom) in the West. The myth of Aeneas being led by the morning star was old in the days of the Evangelist, and Virgil's narrative was widely disseminated throughout the empire, so the parallels here would have been recognizable to the intended audience. There is also the notable reversal between the stories: while in Virgil the star guides Aeneas and leads to him coming to power in Italy, for Matthew the Star leads the Magi for the purposes of worshipping another. The reversal is then filled with meaning: the Messiah is the power to follow, even over the authority of other kings and Rome, that the infant Jesus is more important than the founding warrior Aeneas. Worship of the Son of God is paramount compared to earthly sovereigns. While Virgil's story, or

¹⁶⁶ Virgil, *Aeneid* 2.687-711.

¹⁶⁷ Servius, *In Virgilio Aeneidos* 1.382; Michael Paschalis, *Virgil's Aeneid: Semantic Relations and Proper Names*, pp. 94-95.

other tales of guiding stars in classical sources, cannot explain the entirety of the creation of the Star of Bethlehem, it is one of several good possibilities for what inspired the tale better known to modern Christians.

Of those other possibilities, there is a particular hypothesis widely considered by scholars that relies on the source we know Matthew was dependent upon: the Hebrew Bible and its numerous prophetic passages. One of the well-known prophecies in antiquity from the Old Testament is Numbers 24:17, which speaks of a star corresponding to the rise of a king for Israel. "A star shall rise out of Jacob, a scepter out of Israel." This was popularly used by various Jews and sects concerning the messiah¹⁶⁸ and is alluded to in an earlier Christian text, Hebrews 7:14. We already see how Matthew cites Old Testament verses to support his story and how his account "fulfilled prophecy", so another example of Hebrew Scripture providing the foundation of one detail of the story is hardly unlikely, and certainly more likely than believing the tale actually happened. Along with this verse, Isaiah 60 talks about eastern kings coming to the rising light of Israel, bringing gold and frankincense, two of the gifts of the Magi. The word that means 'rising' in Matthew 2, *anatole*, is from the same root of the word referring to the rising star in Num 24:17 and the rising light of Israel mentioned in Isaiah 60 in the Greek

¹⁶⁸ Cf. David Instone-Brewer, "Balaam-Laban as the Key to the Old Testament Quotations of Matthew 2", in Daniel M. Gurtner and John Nolland, *Built Upon the Rock: Studies in the Gospel of Matthew*, pp. 207-227; Martin Dibelius, *Botschaft und Geschichte: Gesammelte Aufsätze*, I, p. 42 n. 68; Mussies, "Some Astrological Presuppositions of Matthew 2", pp. 25-27; David Sim, "The Magi: Gentiles or Jews?" *Hervormde Teologiese Studies* 55, 4 (1999): 990-996. See also Josephus, *Jewish Wars* 3.399-404; 6.310-315 who applied the prophecy to the Roman general and later emperor Vespasian.

translation, *anatello*. Such a linguistic link helps persuade us that this is what probably brought the tale together, along with other details. In other words, the story is very explicable as a literary creation, including particular points (what gifts are brought) and the choice of words.

On top of these particular verses, we can see Matthew applying the mold of Moses to that of the Jesus story, especially the embellished stories of Moses's birth that were in existence in the first century. Josephus provides such a version in which a scribe who did fortunetelling discovered that a child of the Israelites would be born and would ruin Egypt. Pharaoh became so afraid of this that he ordered the execution of all male Israelite children. In this same legend, there is also Moses's father-to-be worrying about what to do because of the orders from Pharaoh, but in his sleep God told him that all would be okay and his son would be the deliverer of the Hebrew nation.¹⁶⁹ The parallelism with the Jesus birth story in Matthew is unmistakable, a series of tropes that anyone in the ancient world would have seen. But these are not the only similarities between stories of Moses's birth and Jesus's. As noted earlier, there is also a Samaritan version of Moses's birth of uncertain date that included a star that was interpreted and related to the birth of the savior figure.¹⁷⁰ The form of these stories is in obvious conformity with the form of the story of Jesus and the Star, along with Herod acting the part of mad Pharaoh. While Matthew may not have been dependent upon any of these particular versions of Moses's birth, we see that these were the sorts of fictions invented and prevalent at this time, and

¹⁶⁹ Josephus, *Antiquities of the Jews* 2.9, §§ 2-3.

¹⁷⁰ Bowman, *Samaritan Documents*, pp. 287-288.

Matthew appeared to be weaving these themes into his narrative.

Outside the Jewish literature, the pagan writers of this era had stories with many of the same tropes: prophecies of the birth of a new leader or savior, miraculous signs at his birth, attempts on his life by an authority figure with the escape of the infant. These things were told of the heroes Perseus and Jason, the superman Hercules, the prophet Zoroaster, the emperor Augustus, the military leader Alexander the Great, the brothers Romulus and Remus, and many others. When these elements were included in Matthew's story of the Nativity, they were already cliché in the first century and fit into a larger category of an ancient Western mythical hero archetype.¹⁷¹

But beyond the use of Hebrew Scripture and similar tales to explain the details of the Nativity story, we can see how the account is truly molded in the way expected of fiction. For example, we hear about the mental states of figures such as Herod, the people of Jerusalem, and the Magi. We have conversations between Herod and the Magi (which were held secretly) as well as between Herod and the scribes. These are all things that could not have been witnessed even by the only potential sources for Matthew—Mary and Joseph. However, this exactly fits what is expected from the omniscient narrator—he knows because he is writing the story. There is also the apparent quote from Scripture about how the Messiah was to be born in Bethlehem, but the prophetic verse from the Gospel is a blending of two different verses, 2 Samuel 5:2 and Micah 5:2. This makes for a rather

¹⁷¹ Cf. Robert A. Segal, *In Quest of the Hero*; Richard Horsley, *The Liberation of Christmas: The Infancy Narratives in Social Context*, pp. 162f.

contrived quotation to get the connection the author wants. Conversely, there is no reason that the priests and scribes would have made up the quotation themselves. Other scholars have noted this oddity and the style of the “prophecy” being much like that of the Evangelist, a strong sign that this was not the product of the chief scribes but Matthew.¹⁷² The narrative also shows a fruitful relationship between the chief priests and Herod, while the historical record indicates these two groups were antagonistic; the scribes would not be at the beck and call of the king.¹⁷³ Instead, it strikes one as authorial intent guiding the story rather than historical plausibility.

The story as is also contains two inanities. There is no effort on the part of Herod to have spies or any other escort with the Magi which would have guaranteed the destruction of Jesus in the crib (or manger). It's implausible that Herod was paranoid enough to find a baby a threat, and yet he unreservedly trusted the very people that came to contest his claim to the throne. This becomes a plot hole, and one that is necessary for the story to continue. There is also the plot hole as to why the guiding Star failed to guide the Magi to Bethlehem in the first place and avoid Herod from even knowing about the birth of the Messiah. The trip to Jerusalem by the Magi and making their announcement fulfills the purpose of bringing in a villain into the story and produce drama. Obviously this version of the story with Herod is infinitely more memorable compared to the Star

¹⁷² William D. Davies & Dale C. Allison Jr., *A Critical and Exegetical Commentary on the Gospel According to Saint Matthew: Introduction and Commentary*, vol. 1, p. 242.

¹⁷³ Brown, *Birth*, p. 188.

alone guiding the Magi to Jesus, but it belies how the story is history rather than a concoction.

In fact, we could treat the story of the Star, Jesus and the Magi's escape, and the plot against Jesus's life, as a *pericope*¹⁷⁴ or singular literary unit and analyze it with the method of determining genre, chronotope—how space and time are used in a story to deliver its meaning to the reader. It is how you know that George R.R. Martin's *A Song of Fire and Ice* is in the same genre as J.R.R. Tolkien's *The Lord of the Rings* rather than a modern history of the Crusades or even a novel by Paul Auster or Jane Austin. While chronotopic analysis is usually done over an entire work, chapter 2 of Matthew can also be examined by this method; after all, the Nativity story should contain the same message as the rest of the Gospel, and it is fruitful to use chronotopic evaluation to see what sort of literature we are dealing with here.

Using the analysis by Bible and literary scholar Michael Vines concerning the Jewish novels from antiquity,¹⁷⁵ we can apply the same observations and see that they also fit Matthew's famous infancy narrative. In the Jewish novels (i.e. Greek Daniel, Greek Esther, Judith, Tobit, *Joseph and Aseneth*), Vines describes a realistic-apocalyptic chronotope (heavenly salvation but in an earthly, historical setting), which has the features of divine versus human sovereignty, God's response to conflict against his preferred people, and the use of human agents to execute His plans to re-establish His sovereignty, with all events happening not because of the human's own resources or skills but through the power

¹⁷⁴ Cf. Albrecht Dieterich, "Die Weisen aus dem Morgenlande", *Zeitschrift für die Neutestamentliche Wissenschaft* 3, 1 (1902): 1-14.

¹⁷⁵ Michael Vines, *The Problem of Markan Genre*.

of God. The expanded stories of Moses mentioned above also conform to this paradigm. Matthew 2 fits the chronotope perfectly: the setting is late-first century Judea under Herod; this is a well-known period of conflict with the land ruled by non-Jewish forces; there is conflict between Jesus as the divine King of the Jews and Herod as the human king of the Jews; God allows Jesus to be recognized as the true sovereign, and he avoids an early death and so defeats the king who dies; the Magi succeed not because of their own skills but through the agency of God using both a guiding Star as well as angelic announcements. The conflict or crisis in Jewish novels is the force in the story, and how the characters resolve it is the crux, namely if they remain loyal to God. The Magi fit this: they are instructed by Herod to return to him, yet instead they listen to a messenger from God, which protects Jesus, and hence the integrity of God's plan and sovereignty.

These are not the only literary features in common between the Jewish novels and Matt 2. The element of the weak defeating the strong is also on stage in both: Jesus, a lowly infant, is able to usurp Herod's claim to royalty, while the Magi, without any use of force, are able to defeat the powerful king's desires. We also have the particular connection to the story of Daniel where a foreign king venerates God, and with the Magi involved. Additionally, we see the defeat of Herod's army as they failed to capture the infant Christ all because of the pious actions of Joseph and the Magi; this is comparable to Judith, who all on her own defeats the army of Nebuchadnezzar using her beauty and piety.

In addition, the Jewish novels use irony that shows the superiority of the faith. In Matt 2, there is the irony of foreigners correctly worshipping God and his Son,

while the king of the Jews and the chief priests and scribes fail to do so and instead (later in the Gospel) become the chief rivals to God's Son. We have the subversion of the Jews of Palestine and their leaders, a point that is only more pronounced throughout the rest of the Gospel of Matthew. The conflict with rabbinic Judaism and the crowd that has Jesus's blood upon them when they called for his death (Matt 27:22-5) all shows the reversal inherent in the Gospel's message of the Gentile mission (Matt 28:19). Matt 2 already encapsulates the irony of the entire Gospel, of foreign piety over Jewish hypocrisy. As such, we ought to place this story (and perhaps the entire Gospel of Matthew) into the category of novel. Not history, not critical biography, but edifying, theological fiction as pseudo-historical as were the novels mentioned above. That the elements of the tale can also clearly be derived from the Old Testament overwhelmingly demonstrates the purpose and method of the story's creation, and it has nothing to do with historicity.

So, no matter what prior probability you may have reasoned initially for the truth of the tale (unless you irrationally claim 100% certainty or near it), even if you accept the miraculous and don't think it can affect the probability of the tale (itself unreasonable), the evidence is overwhelming against the historicity hypothesis. However, it is possible to make sense out of how such a story could have come about when examining the only source we know the authors of the Gospels used—the Hebrew Bible—and the literary universe the Evangelists existed in. All this decisively shows that the tale of the Star of Bethlehem is untenable as history, even if you believe in miracles or still claim it was a natural

phenomenon. The evidence is extraordinary, but extraordinarily against historical truth in Matt 2:1-12.

At this point I have not considered things such as the Slaughter of the Innocents or other historically questionable aspects of the Nativity in the Gospel of Matthew, leaving that to Jonathan Pearce in his book on the subject amongst other critical books and articles about the Nativity. However, because in this case we can be effectively certain that Matthew has concocted material about Jesus's birth, it follows that this will also lower the chances that any of his other stories about Jesus are likely to be true, especially the infancy narrative. Consequently, this excursion into just one part of the story does in fact affect the plausibility of the rest—if this was made up, then it is more likely other things were also invented.

In conclusion: No, Virginia, there was neither a Star of Bethlehem nor a journey of the Magi. It's a story of significant importance theologically, perhaps, but it has nothing more to do with history than the non-canonical tales of Jesus's infancy and childhood where he makes living clay birds and fights dragons.

APPENDIX: THE TEXT OF MATTHEW

The primary language of the world the New Testament found itself in was Greek, largely because of the expansionist conquests of Alexander the Great centuries before the Gospels were composed. The Romans, though certainly not loathsome of their own tongue, also learned this language, and it became the *lingua franca* of the Empire. Most people who needed to trade would know at least some Greek, and students were taught it in written form, as well as learning their native language, such as Latin or Hebrew. So it is no surprise that the Gospel of Matthew is written in the Greek language, and as such it could reach the widest possible audience. We are also fortunate that the chapters that open the Gospel are reasonably well-preserved in the manuscripts we have, though they are all dated to centuries after when the autograph was composed (very broadly around 100 CE). Among the manuscripts we have, there is little variance besides missing lines or misspellings, the normal and expected situation when all books had to be copied by hand.

Our primary manuscripts are the codices from the fourth century, such as Codex Vaticanus and Codex Sinaiticus. These are known as a part of the Alexandrian text-type, which are some of the oldest texts and they are considered the most reliable. The other major text-type categories are the Western and Byzantine. The latter makes up the greatest number of copies of the New Testament, but they are all younger and less revealing of what the autographs of the Gospels would have looked like. The Western text-type is older than the Byzantine type and covers a wide geographical spread of manuscripts.

However, these manuscripts tend to paraphrase and have textual corruptions. Perhaps the most noticeable difference between the Western and Alexandrian text-types is that the Book of Acts is about 10% longer in the Western version compared to the Alexandrian, and it is uncertain which is closer to the original.

For the most part, textual critics are reliant upon Alexandrian text-types because of their antiquity, their greater uniformity (better controls in copying, so more likely to reflect the earliest versions of the Gospel they had for copying), and less tendency to paraphrase, expand, or otherwise “improve” the texts. The codices mentioned above are written without diacritical marks (dashes and curves that indicate how words are pronounced; e.g. *ā*, *á*) and in all capital letters without spaces between words or punctuation marks that we take for granted. Modern critical editions of the Greek New Testament are usually more forgiving: they use upper and lower case as common English users do, they use punctuation marks, and they include diacritical marks.

The Greek text here is that agreed upon by the United Bible Societies (UBS) in their 4th edition of the Greek New Testament. For chapter 2 for the Gospel of Matthew, the USB version of the text does not differ from the older, classic version compiled by the textual critics Brooke Foss Westcott and Fenton John Anthony Hort published in 1881. There is also almost no difference between this edition of the Gospel chapter and the Greek text that was used for the King James Bible, the Textus Receptus, the only difference of note being in verse 9, using *ἔσται* instead of *ἐσθήσεται*—the active and passive forms of the verb *ἵστημι* (to stop, stand). In the context of the verse, it makes no substantial difference. Fortunately for the interpretation of this story, we have less reason to worry about textual

uncertainties and interpolations that in other parts of the Bible, and we can avoid dogmatic debates about what editions of the Greek New Testament should be given theological weight.

As for the English translation, this is my own, though it is informed by the numerous, modern English versions on the market, especially the Revised Standard Version (RSV). However, a few points in the Greek tend to not be translated with respect to the Star of Bethlehem as they make the phrasing clumsy and not terribly informative. However, I will try to include these points since they have had an effect on my analysis of the Star. There is also some uncertainty about the phrase ἐν τῇ ἀνατολῇ, which needs its own discussion, but my understanding here is sufficiently in line with other translations and research to not be unsoundly rendered as “at the rising of the sun”.¹⁷⁶ This can be compared to how scholars translate the phrase in the works of the historian and essayist Plutarch (*Moralia* 284e9) and the astronomer Ptolemy,¹⁷⁷ both of whom were broadly contemporary with the Evangelist. Other common translations include “in the East” and “at its rising”, with the latter more often found in modern translations.

¹⁷⁶ Cf. Roberts, *The Star of the Magi*, pp. 120-1. Roberts contacted two translators of ancient Greek astrological texts for their opinions on the phrase in question.

¹⁷⁷ Johan Ludvig Heiberg, *Claudii Ptolemaei opera quae exstant omnia I: Syntaxis mathematica, Part 2*, p. 595; G. J. Toomer, *Ptolemy's Almagest*, p. 638.

1 Τοῦ δὲ Ἰησοῦ γεννηθέντος ἐν Βηθλέεμ τῆς Ἰουδαίας ἐν ἡμέραις Ἡρώδου τοῦ βασιλέως, ἰδοὺ μάγοι ἀπὸ ἀνατολῶν παρεγένοντο εἰς Ἱεροσόλυμα

2 λέγοντες, Ποῦ ἐστὶν ὁ τεχθεὶς βασιλεὺς τῶν Ἰουδαίων; εἶδομεν γὰρ αὐτοῦ τὸν ἀστέρα ἐν τῇ ἀνατολῇ καὶ ἠλθομεν προσκυνῆσαι αὐτῷ.

3 ἀκούσας δὲ ὁ βασιλεὺς Ἡρώδης ἐταράχθη καὶ πᾶσα Ἱεροσόλυμα μετ' αὐτοῦ,

4 καὶ συναγαγὼν πάντας τοὺς ἀρχιερεῖς καὶ γραμματεῖς τοῦ λαοῦ ἐπυνθάνετο παρ' αὐτῶν ποῦ ὁ Χριστὸς γεννᾶται.

5 οἱ δὲ εἶπαν αὐτῷ, Ἐν Βηθλέεμ τῆς Ἰουδαίας; οὕτως γὰρ γέγραπται διὰ τοῦ προφήτου:

6 Καὶ σύ, Βηθλέεμ γῆ Ἰούδα, οὐδαμῶς ἐλαχίστη εἶ ἐν τοῖς ἡγεμόσιν Ἰούδα: ἐκ σοῦ γὰρ ἐξελεύσεται ἡγούμενος, ὅστις ποιμανεῖ τὸν λαόν μου τὸν Ἰσραὴλ.

7 Τότε Ἡρώδης λάθρα καλέσας τοὺς μάγους ἠκρίβωσεν παρ' αὐτῶν τὸν χρόνον τοῦ φαινομένου ἀστέρος,

8 καὶ πέμψας αὐτοὺς εἰς Βηθλέεμ εἶπεν, Πορευθέντες ἐξετάσατε ἀκριβῶς περὶ τοῦ παιδίου ἐπὶ δὲ εὑρητὲ ἀπαγγείλατέ μοι, ὅπως κἀγὼ ἐλθὼν προσκυνήσω αὐτῷ.

9 οἱ δὲ ἀκούσαντες τοῦ βασιλέως ἐπορεύθησαν, καὶ ἰδοὺ ὁ ἀστὴρ ὃν εἶδον ἐν τῇ ἀνατολῇ προῆγεν αὐτοὺς ἕως ἐλθὼν ἐστᾶθη ἐπάνω οὗ ᾧ τὸ παιδίον.

10 ἰδόντες δὲ τὸν ἀστέρα ἐχάρησαν χαρὰν μεγάλην σφόδρα.

11 καὶ ἐλθόντες εἰς τὴν οἰκίαν εἶδον τὸ παιδίον μετὰ Μαρίας τῆς μητρὸς αὐτοῦ, καὶ πεσόντες προσεκύνησαν αὐτῷ, καὶ ἀνοίξαντες τοὺς θησαυροὺς αὐτῶν προσήνεγκαν αὐτῷ δῶρα, χρυσὸν καὶ λίβανον καὶ σμύρναν.

12 καὶ χρηματισθέντες κατ' ὄναρ μὴ ἀνακάμψαι πρὸς Ἡρώδην, δι' ἄλλης ὁδοῦ ἀνεχώρησαν εἰς τὴν χώραν αὐτῶν.

1 After Jesus was born in Bethlehem of Judea in the days of Herod the King, behold, Magi from the East came into Jerusalem

2 saying, "Where is the one born King of the Jews, for we saw his Star at the rising of the sun, and we have come to pay homage to him.

3 Hearing this, King Herod was disturbed and all of Jerusalem with him,

4 and bringing together all the chief priests and scribes of the nation he began enquiring of them where the Messiah was to be born.

5 They told him, "In Bethlehem of Judea, as it is written in the writings of the prophets:

6 'And you, Bethlehem of the land of Judah, who by no means is least of the rulers in Judah, from you shall come a leader who will nourish my people, the Israelites.'"

7 Then Herod secretly invited the Magi and discovered from them the time of the appearance of the Star.

8 And he sent them to Bethlehem and said, "Depart and search diligently for the child, and when you find him bring me word so that I may also come revere him."

9 After hearing the king, they left for their journey, and lo! The Star which they had seen at the rising of the sun led them on until it arrived and stood over the place where the child was.

10 Upon seeing the Star, they greatly rejoiced with great joy.

11 And entering into the house they saw the child with Mary his mother, and prostrated themselves and paid him homage, and opening their treasures they gave him gifts: gold, frankincense, and myrrh.

12 And being commanded in a dream not to return to Herod, they traveled back to their country through another route.

GLOSSARY

Achaemenid Empire: the first Persian Empire. Established by Cyrus the Great in his defeat of the Babylonian Empire in the 6th century BCE, it became the largest empire in the world, stretching from Afghanistan to Egypt and into Europe. Cyrus is called the messiah in the Bible (Isaiah 45:1) and helped in rebuilding the Jewish temple (2 Chronicles 36; Ezra 1). The empire fell because of the military conquests of Alexander the Great in the 330s BCE.

Ascendant: one of the **cardinal points** of a **horoscope**. This is the location of the eastern horizon at a certain degree of an astrological sign.

Aspect (astrology): the special angular relations between **signs** and/or **planets** in a **horoscope**. The most common aspects used in antiquity are **conjunction** (0° separation), sextile (60° separation), square (90° separation), **trine** (120° separation), and opposition (180° separation). The angular separations need not be exact to count for an aspect, but the size of the margin around these angles (called orbs) is debated by astrologers.

Astrology: the divinatory technique of predicting things based on the arrangements of the stars and **planets**. One of the chief tools used for such predictions is the **horoscope**. Most practicing astrologers today are not astronomers and use tables or computers to compose a horoscope. Starting in ancient Babylonia, astrology has gone from a scientific endeavor by many competent researchers to a pseudoscience existing outside of standard astronomical research. Johannes Kepler (1571-1630) is

often considered the last great scientific astrologer. In well-formed tests of astrology, professionals show an accuracy level no better than chance and a level of agreement almost no better than chance.

Astronomy: the study of the universe beyond the Earth. This includes other **planets**, stars, galaxies, and the entire universe itself. The subject includes the use of physics, chemistry, geology, and biology. With prehistoric roots, the science of astronomy began in Babylonia, advanced under the Greeks, Romans, and medieval Arabs, and had a strong effect in igniting the Scientific Revolution. Popular areas of research include planetary formation, stellar evolution, and cosmology (origins, structure, and fate of the universe). The primary tools of the modern astronomer are telescopes that use various wavelengths of light (visible, infrared, radio, etc.) and computers, both for data analysis and for modeling. Planetary scientists also use probes that travel to solar system bodies, including the Cassini spacecraft around Saturn and the rover Curiosity on Mars.

Cardinal points: the four compass points on a **horoscope**. Those points are the **ascendant** (Asc), the descendant (Des, 180° opposite the ascendant), the **midheaven** (MC), and the *imum coeli* (IC, 180° opposite the midheaven). These points are important for the interpretation of a horoscope.

Chronotope: a tool in literary theory to describe the genre of a work. Devised by Mikhail Bakhtin in the 1930s, it looks at how time (Gk. *chronos*) and space (Gk. *topos*) are represented in a narrative. Chronotope can be established by the setting, the plot, the actions of characters, how

events change characters, the relation of the narrator's voice to the story, etc. As Bakhtin describes: "Time, as it were, thickens, takes on flesh, becomes artistically visible; likewise, space becomes charged and responsive to the movements of time, plot and history." The analysis of chronotope is done by comparison with the literary context the work finds itself in and how they establish their respective chronotopes; in the case of the Gospels, one considers the literature of that time that was influential (i.e. the Old Testament, the works of Homer, popular novels, etc.) and how they used chronotope (i.e. Homer's use of sea voyages and divine intervention, the ancient Greek novels' use of the element of chance, etc.).

Comet: from the Latin for hair (*coma*), these are balls of water ice, ammonia, carbon dioxide, as well as rock and dust and thus called "dirty snowballs" in popular literature. The nucleus of a comet will begin to boil off gas and dust when heated up on its approach to the Sun, producing the tail (or tails) of the comet, the 'hair' of the star. Often two tails can be seen, one composed of dust and the other of gas; the latter tail is most strongly pushed by the solar winds and points away from the Sun. Tails can reach lengths over 100 million kilometers long. The dust of comets remains behind and can produce meteor showers for Earth-based viewers at certain times of the year. Comets are theorized to come from a sphere of icy bodies surrounding the Sun in what is called the Oort cloud; gravitational perturbations from other objects in space occasionally send a comet falling towards the inner solar system. Because they are orbiting the Sun, most comets are periodic in their appearances, but gravitational effects by the planets of the solar system, Jupiter especially, can change the orbit and the period. Since antiquity, comets

have almost universally been interpreted as frightening omens.

Conjunction: an astrological **aspect** when two objects have the same **right ascension**. In a **horoscope**, they would appear in the same spot. Conjunctions can happen between stars and **planets**, between planets, etc.

Constellation: a collection of stars that imaginatively look like or represent something. Different cultures have different constellations, but the official International Astronomical Union (IAU) system has 88 constellations, most of which come from Greco-Roman traditions.

Cuneiform: one of the oldest forms of writing, emerging in the 4th millennium in Sumer. The script is formed using wedge shapes and was used by the Babylonians, Assyrians, and other Ancient Near Eastern cultures. It fell out of use progressively in the first millennium BCE and disappeared by the 2nd century CE. It was unlike the modern alphabet where each letter had a particular phonetic (sound) value but instead some were words, some syllables, and some consonants. After it went out of use, cuneiform was not deciphered until the 19th century. One of the oldest extant works of fiction, the *Epic of Gilgamesh*, was written in cuneiform.

Declination: a part of the (celestial) equatorial coordinate system equivalent to latitude on Earth. The zero line of declination is called the celestial equator and is the projection of Earth's equator into space. Above and below the celestial equator gives positive and negative declinations respectively, ranging between -90° and +90°.

Epicycle: literally a cycle on top of another cycle. A **planet** on an epicycle follows a circular orbit around a point which is itself moving in a circle (called a deferent) around another point. A planet doing this could look to an observer who is on or near the center of the deferent as if it slowed, reversed direction, slowed again, and returned moving the way it did before. The system of epicycles was the method primarily used to produce a geocentric model of the solar system.

Form criticism: the method of labeling literary units and determining their functions and origins. For much of 20th century biblical studies, this meant determining the oral traditions upon which the Bible authors relied on to produce their stories. Some of the most famous form critics of the New Testament include Rudolf Bultmann, Martin Debelius, and Karl Ludwig Schmidt.

Great conjunctions: the **conjunctions** of Jupiter and Saturn, which take place approximately every 20 years. Successive great conjunctions take place about 120° apart on a horoscope and thus remain in the same **trine**. When such a conjunction takes place in a different trine than the previous conjunction, it is called a greater conjunction; these take place approximately every 240 years. When a great conjunction first takes place in the fiery trine, it is called the greatest conjunction. Without considering the effects of **precession**, which moves the background stars as seen by an Earth-based observer and thus the location of the constellations, greatest conjunctions take place about 960 years apart or nearly a millennium. With the effects of precession, the period is about 800 years and will have 40 great conjunctions. Astrologers believed these conjunctions had effects on world affairs. This astrological

theory developed during the **Sassanid period** in the Middle East (3rd-7th centuries CE).

Horoscope (Western): a map of the **zodiac signs**, the **planets**, and the **cardinal points**. The relative positions of these various bodies are what lead to the interpretation of the horoscope. The earliest horoscopes known come from the 5th century BCE in Babylonia, but the Western form described above did not come to be until the 2nd century BCE in Hellenistic Egypt. The oldest extant treatise on horoscopic astrology comes from the early 1st century CE.

Magus/Magi: priest(s) in the **Zoroastrian** religion. The name seems to derive from the name of the religious caste of the Medes in what is now western Iran. Through the Greek *magoi* and Latin *magi*, the title would become the modern word 'magic.'

Meteor: the visible streak of light in the sky caused by a falling rock from space. Meteors can be seen at any time of the night in any direction, but during meteor showers they will appear to come from a certain part of the sky. Meteor flashes can last from a second to over a minute, though usually they are sudden appearances. If any part of the meteor reaches the ground, it becomes known as a meteorite.

Midheaven (*medium coeli*, MC): one of the **cardinal points** of a **horoscope**. It is the location of the highest part of the **zodiac** seen at the time and location of the horoscope's object of consideration. The angle between the midheaven and **ascendant** (the horizon) is usually less than 90°, but often in ancient horoscopes the angle is simply treated as a right angle.

Nova: a stellar explosion that happens when gas from another star is accreted on to the surface of a white dwarf star. Gravitational pressure on the gas causes fusion, and there is a fast explosion on the surface of the white dwarf. To an Earth-based observer, it appears as if a star has just appeared (hence the Latin word for 'new', *nova*, used for the object). In a short while, usually days, the nova will dim and become invisible to the naked eye. Some novae are recurrent and explode periodically.

Parthian Empire: also called the Arsacid Empire, it developed out of uprisings against the dynastic empires created by Alexander's generals, especially that of Seleucus. Established by Arsaces I of Parthia in 247 BCE, it would become the chief rival power to Rome in the East. It finally fell in the early 3rd century CE and replaced by another Persian empire, that of the **Sassanids**.

Pericope: a set of clauses or verses that can form a literary unit or thought. For much of the 20th century, Bible scholars employing **form criticism** used pericopes (or pericopae) to determine different traditions and sources for the Gospels.

Planet (classical world): from the Greek for wanderer, one of seven objects that are star-like in appearance but move against the background **constellations**. In antiquity, the seven objects were (in the common order given as their relative distance from the Earth): Moon, Mercury, Venus, Sun, Mars, Jupiter, and Saturn.

Precession of the equinoxes: the slow movement of the location of the Sun in the **zodiac** on the first day of

spring (the vernal equinox). The effect is due to the “wobble” of the Earth, similar to how a top wobbles or precesses, which is caused by the gravitational forces between the Earth and the Moon and Sun. Precession is cyclical with the north pole of the Earth drawing out a circle in the sky over a period of about 26,000 years. The discovery of the effect is credited to Hipparchus of Rhodes in the 2nd century BCE, and the oldest extant discussion of precession comes from Claudius Ptolemy in the 2nd century CE.

Right ascension: a part of the (celestial) equatorial coordinate system equivalent to longitude on Earth. It measures from the vernal equinox towards the east. Instead of using degrees, it is measured by using 24 hours for a complete circle instead of 360°.

Sassanid Empire: the last Persian empire before the rise of Islam. Founded by Ardashir I in the 3rd century CE, the **Zoroastrian** religion received significant state support while occasionally suppressing other faiths. Christianity would be recognized in 409/410 by King Yazdegerd I, though the relation between the empire and the Christians would swing back and forth between acceptance and suppression. The Sassanid Empire would continue to rival Rome and the Byzantine Empire for eastern territories until its fall during the Arab conquests of the 7th century.

Sign (astrology): similar to a **constellation**, one of 12 regions of the **zodiac** 30° across. The names of the signs come from the zodiac constellations. Signs do not necessarily line up with the visible constellations because of the precession of the equinoxes and because constellation are not of equal size. In what is called a tropical

zodiac, the defining point of the sign Aries is the location of the vernal equinox (where the Sun is on the first day of spring), and from there all the other signs get their locations. Signs have various attributes, such as gender and element.

Supernova: the most colossal form of stellar explosion. The two major types are Type Ia and Type II. The former happens when a white dwarf star accretes enough material from another star to reach the Chandrasekher limit (~1.4 solar masses) and collapse, releasing a huge amount of energy. Type II supernovae happen when a massive star runs out of fuel for fusion and collapses, the infalling gas bounces off the core of the star, and then explodes outward. Supernova explosions can be more luminous than an entire galaxy of stars. Seen from the Earth, it would appear as a particularly bright star, which over time would dim and disappear. The last naked-eye supernova in the Milky Way was seen in 1604 (Kepler's nova).

Trine/Trigon/Triplicity: three **signs** that are 120° apart in a **horoscope** and are assigned one of the four classical elements. The fiery trine, for example, consists of Aries, Leo, and Sagittarius. A trine **aspect** refers to two **planets** that are about 120° apart or in two signs in trine aspect (i.e. Jupiter in Aries, Mars in Leo). A trine aspect is usually considered positive by astrologers.

White dwarf: the remnant core of a star after all the gas that could be fused is exhausted and the compact remnant remains. Consisting largely of carbon, oxygen, and heavier elements, white dwarves have an extreme density of about 100,000 times that of lead. The gravitational force keeps the star so extremely dense, while it is

the quantum mechanical effect of electron degeneracy pressure that prevents it from collapsing. If a white dwarf receives more mass, such as gas from another star, it can become so massive it overcomes the degeneracy pressure and collapse into a denser object, such as a neutron star or black hole and in the process creates a **supernova**. White dwarves are very dim, and none are seen by naked-eye observers on Earth. An example of a white dwarf is the companion star to the brightest star in the sky, Sirius; this dwarf is known as Sirius B.

Zodiac: the ring of **constellations** or **signs** in which the classical **planets** move through. Traditionally there are 12 constellations, most of them animals (hence the zoo in zodiac). The Sun will move through the entire zodiac once a year. The zodiac was identified first in the 7th century BCE by the Babylonians.

Zoroastrianism: a religion traditionally founded by the prophet Zarathustra (to the Greeks and Romans 'Zoroaster') in the 2nd millennium BCE, in which there are two chief gods, one good (Ahura Mazda, later Ohrmazd) and one evil (Angra Mainyu, later Ahriman). Participants in the religion take part in beliefs of a cosmic battle between these gods, including a final confrontation and the destruction of Ahriman. In each of the last millennia, a savior figure called the Saoshyant would appear, the last one bringing about the final revelation and victory for Ahura Mazda. It was a major religion in Persia, and its priests were the **magi**. The Avesta is the primary collection of sacred literature, consisting of liturgical works (*Yasna*), hymns to particular gods (*Yasht*), catalogues of evil spirits and the maladies (*Vendidad*), and more. The belief in an evil anti-god, a savior figure, and a final judgment of the

living and dead all likely influenced Judaism during the **Achaemenid** Period. There are perhaps as many as one or two million followers of this faith today. In modern times, the best known Zoroastrian is probably the late Farrokh Bulsara, better known as Freddie Mercury.

BIBLIOGRAPHY

- Adair, Aaron. "Science, Scholarship, and Bethlehem's Starry Night", *Sky & Telescope* 114, 6 (Dec 2007): 26-29
- "The Star of Christ in the Light of Astronomy", *Zygon: Journal of Science & Religion* 47, 1 (March 2012): 7-29.
- Alexander, Gerhard. *Apologie; oder Schutzschrift für die vernünftigen Verehrer Gottes*. Frank am Main: Insel Verlag, 1972.
- Asimov, Issac. *The Planet that Wasn't*. Garden City, NY: Doubleday, 1976.
- Baldwin, Barry. *Suetonius*. Amsterdam: A. M. Hakkert, 1983.
- Banos, George. "What the star of Bethlehem Uranus", *Astronomy Quarterly* 3, 12 (1980): 165-168.
- Barnes, Timothy. "The Date of Ignatius", *The Expository Times* 120, 3 (2008): 119-130.
- Barton, Tamsyn. *Ancient Astrology*. London: Routledge, 1994.
- Barton, John. Muddimen, John. *The Oxford Bible Commentary*. Oxford: Oxford University Press, 2001.
- Beck, Roger. "Thus Spake Not Zarathustra: Zoroastrian Pseudopigrapha of the Greco-Roman World" in *A History of Zoroastrianism*, Mary Boyce, ed. (Leiden: Brill, 1975-1991) Vol. 3, pp. 491-565
- *A Brief History of Astrology*. Malden, MA: Blackwell Publishing, 2007.
- Boll, Franz. "Der Stern der Weisen", *Zeischrift für die neutestamentlich Wissenschaft*, 18, Jahrg., Heft 1/2 (1917): 40-48.
- Bowman, John. *Samaritan Documents: Relating to their History, Religion, and Life*. Pittsburgh: Pickwick Press, 1977.
- Boyce, Mary (ed.). *A History of Zoroastrianism*. Leiden: Brill, 1975-1991.
- Bradley, K. R. *Suetonius' Life of Nero: An Historical Commentary*. Bruxelles: Latomus, 1978.
- Brodie, Thomas L. *The Birthing of the New Testament: The Intertextual Development of the New Testament Writings*. Sheffield: Sheffield University Press, 2004.

- Brown, Raymond E. *The Birth of the Messiah: A Commentary on the Infancy Narratives in the Gospels of Matthew and Luke*. New York: Random House, 1993.
- Burnett, Andrew. Amandry, Michael. Alegre, P. P. Ripollés. Butcher, Marguerite Spoerri. *Roman Provincial Coinage*. London: British Museum Press, 1998.
- The Cambridge History of Iran*. Cambridge: Cambridge University Press, 1968-1991.
- Carrier, Richard. *Proving History: Bayes's Theorem and the Quest for the Historical Jesus*. Amherst, NY: Prometheus Books, 2012.
- Caspar, Max. *Kepler*. New York: Dover, 1993.
- Clark, David H. Parkinson, John H. Stephenson, F. Richard. "An Astronomical Re-Appraisal of the Star of Bethlehem—A Nova in 5 BC", *Quarterly Journal of the Royal Astronomical Society* 18 (1977): 443-449.
- Clarke, Arthur C. "The Star", *Infinity Science Fiction* 1, 1 (1955): 120-127.
- Clow, Barbara Hand. *Chiron: Rainbow Bridge between the Inner and Outer Planets*. St. Paul: Llewellyn, 1987.
- Cullen, Christopher. "Can We Find the Star of Bethlehem in Far Eastern Records?" *Quarterly Journal of the Royal Astronomical Society* 20 (1979): 153-159.
- Dandamaev, M. A. *Persien unter den erster Achämenidian (6. Jh. v. Chr.)*. Wiesbaden: Reichert, 1976.
- Davies, William D. Allison, Dale C. *A Critical and Exegetical Commentary on the Gospel According to Saint Matthew: Introduction and Commentary*. Edinburgh: T. & T. Clark, 1988-1997.
- de Jong, Albert. *Traditions of the Magi: Zoroastrianism in Greek and Latin Literature*. Leiden: Brill, 1997.
- dei Rossi, Azariah ben Moses. Weinberg, Joanna. *The Light of the Eyes*. New Haven: Yale University Press, 2001.
- Dibelius, Martin. *Botschaft und Geschichte: Gesammelte Aufsätze*. Tübingen: Mohr, 1953-1956.
- Dieterich, Albrecht. "Die Weisen aus dem Morgenlande", *Zeitschrift für die Neutestamentliche Wissenschaft* 3, 1 (1902): 1-14.

- Dione, R. L. *God Drives a Flying Saucer*. New York: Bantam Books, 1968.
- Dodge, Bayard. *The Fihrist of al-Nadim: A Tenth-Century Survey of Muslim Culture*. New York: Columbia University Press, 1970.
- Downing, Barry H. *The Bible and Flying Saucers*. New York: Avon, 1970.
- Elwell, Douglas A. *Planet X, the Sign of the Son of Man, and the End of the Age*. Crane, MS: Defender Publishing Group, 2011.
- Faracovi, Ornella Pompeo. *Gli oroscopi di Cristo*. Venezia: Marsilio, 1999.
- Ferrari, Leo. "Astronomy and Augustine's Break with the Manichees", *Revue des Études Augustiniennes* 19 (1973): 263-276.
- Ferrari d'Occhieppo, Konradin. "The Star of Bethlehem", *Quarterly Journal of the Royal Astronomical Society* 19 (1978): 517-520.
- "Neue Argumente zu Aufgang und Stillstand des Sterns in der Magierperikope Matthaüs 2, 1-12", *Sitzungsberichte der Österreichische Akademie der Wissenschaften, Abteilung II. Mathematische, Physikalische und Technische Wissenschaften*, 206 (1997): 317-344.
- Freitag, Ruth S. *The Star of Bethlehem: A List of References*. Washington: Library of Congress, 1979.
- Frisch, Christian. *Joannis Kepleri Astronomi Opera Omnia*. Francofurti a.M. et Erlanga: Heyder & Zimmer, 1858-1871.
- Gardiner, Martin. *Urantia: The Great Cult Mystery*. Amherst, NY: Prometheus Books, 1995.
- Giberson, Karl. Collins, Francis. *The Language of Science and Faith: Straight Answers to Genuine Questions*. Downers Grove, IL: Intervarsity Press, 2011.
- Gignoux, Philippe. "L'Inscription de Kartir à Sar Mashad", *Journal Asiatique* 256 (1968): 387-418.
- Gingerich, Owen. Hoskin, Michael. Hughes, David. Birdsall, J. Neville. "Review Symposium: The Star of Bethlehem", *Journal for the History of Astronomy* 33, 4, (2002): 386-394.
- Goldstein, Bernard R. Pingree, David E. *Levi ben Gerson's Prognostication for the Conjunction of 1345*. Transactions of

- the American Philosophical Society, vol. 80, pt. 6. Philadelphia: American Philosophical Society, 1990.
- Goodacre, Mark. *The Case Against Q: Studies in Markan Priority and the Synoptic Problem*. Harrisburg, PA: Trinity Press International, 2002.
- Gundel, Wilhem. Schott, Siegfried. *Dekane und Dekansternbilder: Ein Beitrag zur Geschichte der Sternbilder der Kulturvölker*. Glückstadt und Hamburg: J.J. Augustin, 1936.
- Heiberg, Johan Ludvig. *Claudii Ptolemaei opera quae exstant omnia I: Syntaxis mathematica*. Lipsiae : Teubner, 1898.
- Ho Peng Yoke. "Ancient and Medieval Observations of Comets and Novae in Chinese Sources", *Vistas in Astronomy*, vol. 5 (1962): 127-255.
- Horsley, Richard. *The Liberation of Christmas: The Infancy Narratives in Social Context*. New York: Crossroad, 1989.
- Hughes, David W. *The Star of Bethlehem: An Astronomer's Confirmation*. New York: Walker, 1979.
- Humphreys, Colin. "The Star of Bethlehem—a Comet in 5 BC—and the Date of the Birth of the Christ", *Quarterly Journal of the Royal Astronomical Society*, 32 (1991): 389-407.
- Hunger, Hermann. *Astrological Reports to Assyrian Kings*. Helsinki: Helsinki University Press, 1992.
- Hunger, Hermann. Parpola, Simo. "Bedeckungen des Planeten Jupiter durch den Mond", *Archiv für Orientforschung*, 29/30 (1983/4): 46-49.
- Hunger, Hermann. Stephenson, F. Richard. Walker, C. B. F. et al. *Halley's Comet in History*. London: British Museum Publications, 1985.
- Instone-Brewer, David. "Balaam-Laban as the Key to the Old Testament Quotations of Matthew 2", in *Built Upon the Rock: Studies in the Gospel of Matthew*, Daniel M. Gurtner and John Nolland, eds. (Grand Rapids, MI : William B. Eerdmans Pub., 2008), pp. 207-227.
- Jenkins, Rod M. "The Star of Bethlehem as the Comet of AD 66", *Journal of the British Astronomical Association* 114, 6 (June 1999): 336-343.
- Keller, Werner. *The Bible as History*. New York: Barnes & Noble, 1995.

- Kennedy, Edward S. "Comets in Islamic Astronomy and Astrology", *Journal of Near Eastern Studies* 16, 1 (1957): 44-51.
- Kennedy, Edward S. Pingree, David. *The Astrological History of Masha'allah*. Cambridge, MA: Harvard University Press, 1971.
- Kepple, George Robert. Sanner, Glen W. *The Night Sky Observer's Guide*. Richmond, VA: Willmann-Bell, 1998.
- Kronk, Gary W. *Cometography: A Catalog of Comets*. Cambridge: Cambridge University Press, 1999.
- Kidger, Mark R. *The Star of Bethlehem: An Astronomer's View*. Princeton: Princeton University Press, 1999.
- Koch, Dieter. *Der Stern von Bethlehem*. Frankfurt: Verl. der Häretischen Blätter, 2006.
- Koch-Westenholz, Ulla. "The Astrological Commentary *Šumma Sin ina tamurtišu* Table 1" in *La Sciences des Cieux: Sages, Mages, Astrologues*, Rika Gyselen, ed. (Bures-sur-Yvette: Groupe pour l'étude de la civilisation du Moyen-Orient, 1999), pp. 149-166.
- Kühnöl, Christian Gottlieb. *Evangelium Matthaei*. Leipzig: Barth, 1807.
- Landau, Brent Christopher. "The Sages and the Star-Child: An Introduction to the *Revelation of the Magi*, an Ancient Christian Apocryphon", Th.D. Thesis: Harvard University, 2008.
- MacDonald, George. "The Pseudo-Autonomous Coinage of Antioch", *The Numismatic Chronicle* 4 (1904): 105-135.
- Martin, Ernest L. *The Star that Astonished the World*. Portland: ASK Publications, 1991.
- McDowell, Al. *Uncommon Knowledge: New Science on Gravity, Light, the Origin of Life, and the Mind of Man*. Bloomington, IN: AuthorHouse, 2009.
- McGrew, John. McFall, Richard. "A Scientific Inquiry into the Validity of Astrology", *Journal of Scientific Exploration* 4, 1 (1990): 75-83.
- Miller, Robert J. *The Complete Gospels: Annotated Scholars Version*. San Francisco: Harper, 1992.
- Mobberley, Martin. *It Came from Outer Space Wearing an RAF Blazer!: A Fan's Biography of Sir Patrick Moore*. New York: Springer, 2013.

- Molnar, Michael R. *The Star of Bethlehem: The Legacy of the Magi*. New Brunswick: Rutgers University Press, 1999.
- Moore, Patrick. *The Star of Bethlehem*. Bath: Canopus Publ., Ltd., 2001.
- Moreland, J. P. Craig, William Lane. *Foundations for a Christian Worldview*. Downers Grove, IL: InterVarsity Press, 2003.
- Mussies, Gerard. "Some Astrological Presuppositions of Matthew 2: Oriental, Classical and Rabbinical Parallels", in *Aspects of Religious Contact and Conflict in the Ancient World*, ed. Peter Willem van der Horst (Utrecht: Faculteit der Godgeleerdheid Universiteit Utrecht, 1995), pp. 25-44.
- Nanninga, Rob. "The Astrotest: A Tough Match for Astrologers," *Correlation* 15, 2 (1996): 14-20.
- Neugebauer, Otto. *A History of Ancient Mathematical Astronomy*. Berlin: Springer-Verlag, 1975.
- Newman, Robert. "The Star of Bethlehem: A Natural-Supernatural Hybrid?" *Interdisciplinary Biblical Research Institute* 2001, pp. 1-16.
- Nigosian, Solomon. *The Zoroastrian Faith: Tradition and Modern Research*. Montreal: McGill-Queen's University Press, 1993.
- North, John David. *Horoscopes and History*. London: Warburg Institute, University of London, 1986.
- Olson, Donald W. "Who First Saw the Zodiacal Light?" *Sky & Telescope* 77, 2 (Feb 1989): 146-148.
- O'Reilly, Bill. Dugard, Martin. *Killing Jesus: A History*. New York: Henry Holt and Company, 2013.
- Paschalis, Michael. *Virgil's Aeneid: Semantic Relations and Proper Names*. Oxford: Oxford University press, 1997.
- Pearce, Jonathan. *The Nativity: A Critical Examination*. Fareham: Onus Books, 2012.
- Pingree, David. "Astronomy and Astrology in India and Iran", *Isis* 54, 2 (1963): 229-246.
- "Historical Horoscopes", *Journal of the American Oriental Society* 82, 4 (Oct-Dec 1962): 487-502.
- Phil Plait, "Starry, Starry Night", in *There's Probably No God: The Atheist's Guide to Christmas*, ed. Adriane Smith (London: Friday Project, 2009), pp. 59-67.

- Poole, Michael. *The 'New' Atheism: 10 Arguments that Don't Hold Water*. Oxford: Lion Hudson, 2009.
- Price, Robert M. *Night of the Living Savior*. Cranford, NJ: American Atheist Press, 2011.
- Ramsey, John T. "Mithridates, the Banner of Ch'ih-yu, and the Comet Coin", *Harvard Studies in Classical Philology* 99 (1999): 197-253.
- Ramsey, John T. Licht, A. Lewis. *The Comet of 44 B.C. and Caesar's Funeral Games*. Atlanta: Scholars Press, 1997.
- Ratzinger, Joseph. *Jesus of Nazareth: The Infancy Narratives*. New York: Image Books, 2012.
- Roberts, Courtney. *The Star of the Magi: The Mystery that Heralded the Coming of Christ*. Franklin Lakes, NJ: New Page Books, 2007.
- Rochberg, Francesca. *Babylonian Horoscopes*. Transactions of the American Philosophical Society vol. 88, pt. 1. Philadelphia: American Philosophical Society, 1998.
- Rosenberg, Roy A. "The 'Star of the Messiah' Reconsidered", *Biblica* 53, 1 (1972): 105-109.
- Sagan, Carl. Druyan, Ann. *Comet*. New York: Random House, 1985.
- Schoedel, William R. *Ignatius of Antioch: A Commentary on the Letters of Ignatius of Antioch*. Philadelphia: Fortress Press, 1985.
- Schonfield, Hugh J. *The Original New Testament*. London: Firethorn, 1985.
- Segal, Robert A. *In Quest of the Hero*. Princeton: Princeton University Press, 1990.
- Sigismondi, Costantino. "Mira Ceti and the Star of Bethlehem", *Quodlibet: Online Journal of Christian Theology and Philosophy* 4, 1 (Winter 2002).
- Sim, David. "The Magi: Gentiles or Jews?" *Hervormde Theologische Studies* 55, 4 (1999): 990-996.
- "Reconstructing the Social and Religious Milieu of Matthew: Methods, Sources, and Possible Results", in *Matthew, James, and Didache*, eds. Huub van de Sandt, Jürgen Zangenburg (Atlanta: Society of Biblical Literature, 2008), pp. 13-32.
- Sinnott, Robert. "Thoughts on the Star of Bethlehem", *Sky & Telescope* 36 (1968): 384-386.

- Stasiuk, Garry T. "The Star of Bethlehem Reconsidered: A Mythological Approach." *Planetarian* 10, 1 (1980): 16–17.
- Stern, Sacha. *Calendars in Antiquity: Empires, States, and Societies*. Oxford: Oxford University Press, 2012.
- Strauss, David Friedrich. *Das Leben Jesu kritisch bearbeitet*. Tübingen: C.F. Osiander, 1835.
- Teres, Gustav. *The Bible and Astronomy: The Magi and the Star in the Gospel*. Oslo: Solum Forlag, 2002.
- Tester, S. J. *A History of Western Astrology*. Woodbridge, Suffolk: Boydell Press, 1987.
- Thomas, Paul. *Flying Saucers through the Ages*. trans. Gavin Gibbons. London: Neville Spearman, 1965.
- Thompson, R. Campbell. *Reports of the Magicians and Astrologers of Nineveh and Babylon in the British Museum*. London: Luzac, 1900.
- Thorndike, Lynn. *History of Magic and Experimental Science*. New York: The Macmillan Company, 1923–58.
- Tipler, Frank. "The Star of Bethlehem: a Type Ia/Ic Supernova in the Andromeda Galaxy?" *The Observatory* 125 (2005): 168–174.
- *The Physics of Christianity*. New York: Doubleday, 2007.
- Toomer, G. J. *Ptolemy's Almagest*. Princeton: Princeton University Press, 1998.
- Tschudin, Max. "Das Horoskop von Jesus-Christus—ein Versuch", *Astrologie Heute* 52 (Dec/Jan 1994/5): 8–11.
- Upham, Francis William. *Star of our Lord: or, Christ Jesus, King of all worlds, both of time or space. With thoughts on inspiration, and the astronomic doubt as to Christianity*. New York: Nelson & Phillips, 1873.
- Van Tassel, George. *When Stars Look Down*. Los Angeles: Kruckberg Press, 1976.
- Vines, Michael E. *The Problem of Markan Genre: The Gospel of Mark and the Jewish Novel*. Leiden: Brill, 2002.
- Voigt, Heinrich. *Die Geschichte Jesu und die Astrologie: Eine religionsgeschichtliche und chronologische Untersuchung zu der Erzählung von den Weisen aus dem Morgenlande*. Leipzig: Hinrich, 1911.
- Wallace-Hadrill, Andrew. *Suetonius: The Scholar and his Caesars*. New Haven: Yale University Press, 1984.

- Warmington, B. H. *Suetonius: Nero*. Bristol: Bristol Classical Press, 1977.
- Weiss, J.-C. "2000 Jahre Jesus Christ", *Astrologie Heute* 52 (Dec/Jan 1994/5): 12-16.
- Yarshater, Ehsan. *Encyclopaedia Iranica*. London: Routledge, 1982-2004.
- Zaehner, R. C. *Zurvan: A Zoroastrian Dilemma*. Oxford: Clarendon Press, 1955.
- *The Dawn and Twilight of Zoroastrianism*. New York: Putnam, 1961.
- Zerubavel, Eviatar. *The Seven Day Circle: The History and Meaning of the Week*. New York: Free Press, 1985.

